Lesson 4 of 5
Visualizing and Writing

Intro to R workshop, LU Skill School

Instructor:

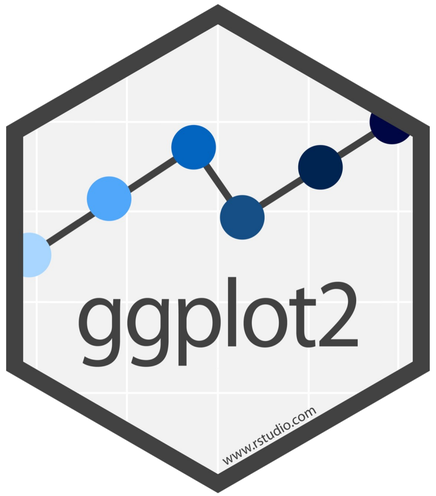
Christopher Swader
LU Sociology (Assoc. Prof)

Slides adapted from and inspired by:
Irina Vartanova, Institute for Future Studies, Stockholm

Today’s agenda

- Visualizing with ggplot2, plotly, gganimate
- Rmarkdown

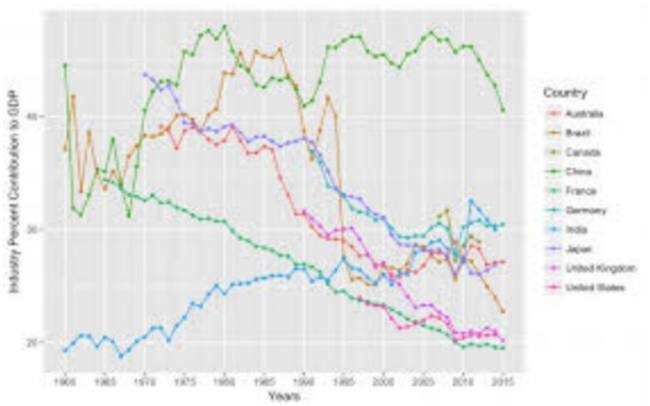
- Use the link to download the files we will be using today: https://github.com/ChristopherSwader/R_introduction
- Download the Day 4 folder.

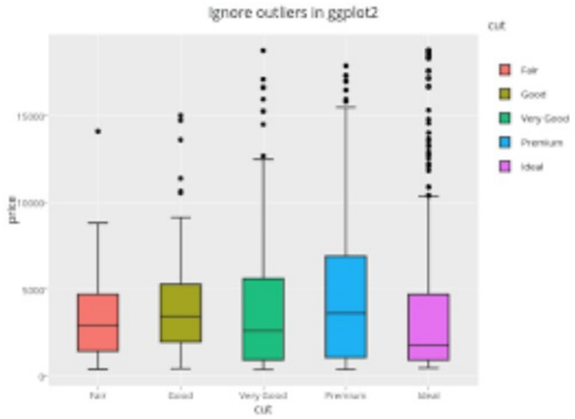


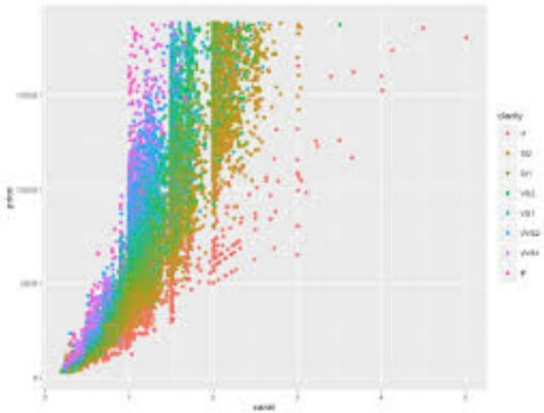
Basics of Visualisation in R’s ggplot2 package

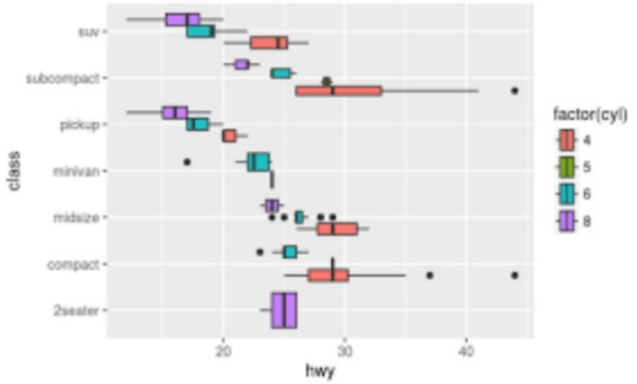
ggplot2: the grammar of graphics

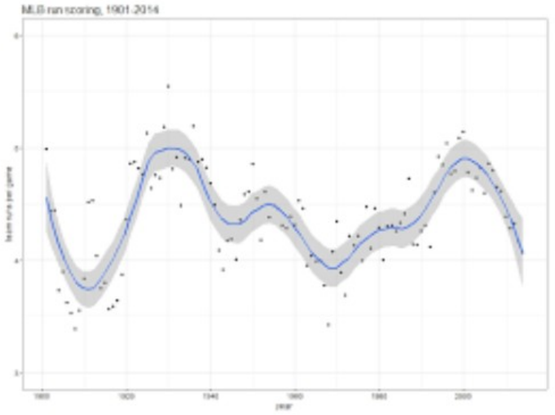
A unified system for variety of graphs

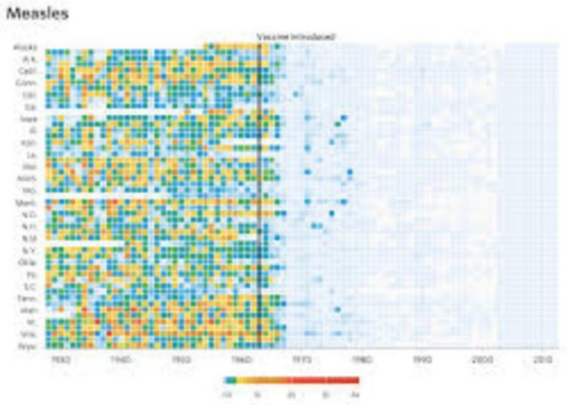


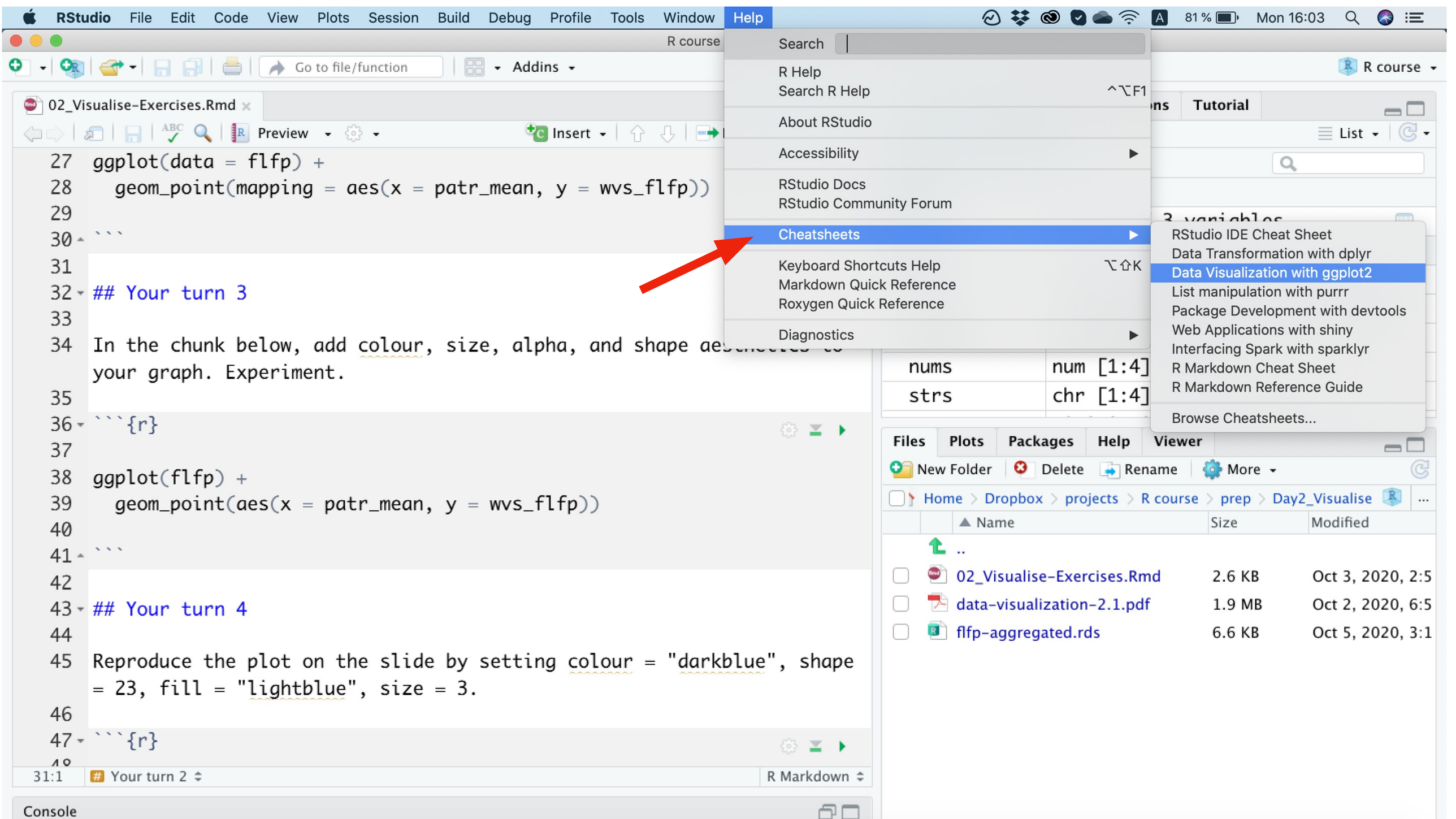








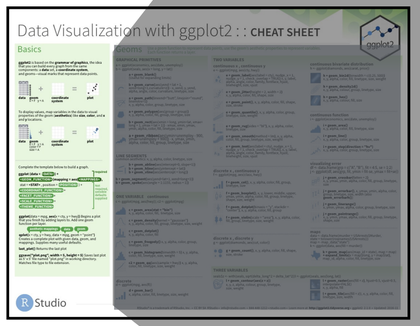



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Package Development with devtools
Web Applications with shiny
Interfacing Spark with sparklyr
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Browse Cheatsheets...
R course
Go to file/function
Addins
02_Visualise-Exercises.Rmd
Preview
Insert
Tutorial
List
nums num [1:4]
strs chr [1:4]
Files Plots Packages Help Viewer
New Folder Delete Rename More
Home Dropbox projects R course prep Day2_Visualise
Name Size Modified
02_Visualise-Exercises.Rmd 2.6 KB Oct 3, 2020, 2:5
data-visualization-2.1.pdf 1.9 MB Oct 2, 2020, 6:5
flfp-aggregated.rds 6.6 KB Oct 5, 2020, 3:1
27 ggplot(data = flfp) +
28 geom_point(mapping = aes(x = patr_mean, y = wvs_flfp))
29
30 ```
31
32 ## Your turn 3
33
34 In the chunk below, add colour, size, alpha, and shape aesthetics to your graph. Experiment.
35
36 ```{r}
37
38 ggplot(flfp) +
39 geom_point(aes(x = patr_mean, y = wvs_flfp))
40
41 ```
42
43 ## Your turn 4
44
45 Reproduce the plot on the slide by setting colour = "darkblue", shape = 23, fill = "lightblue", size = 3.
46
47 ```{r}
31:1 Your turn 2 R Markdown
Console

geom_functions

Different bases for visualization

(Each requires a mapping argument.)



Geoms

Use a geom function to represent data points, use the geom's aesthetic properties to represent variables. Each function returns a layer.

ggplot2

GRAPHICAL PRIMITIVES

a <- ggplot(economics, aes(date, unemploy))
b <- ggplot(seals, aes(x = long, y = lat))

a + geom_blank()
(Useful for expanding limits)

b + geom_curve(aes(yend = lat + 1, xend=long+1,curvature=z)**)** - x, xend, y, yend, alpha, angle, color, curvature, linetype, size

a + geom_path(lineend="butt", linejoin="round", linemitre=1**)**
x, y, alpha, color, group, linetype, size

a + geom_polygon(aes(group = group)**)**
x, y, alpha, color, fill, group, linetype, size

b + geom_rect(aes(xmin = long, ymin=lat, xmax= long + 1, ymax = lat + 1)**)** - xmax, xmin, ymax, ymin, alpha, color, fill, linetype, size

a + geom_ribbon(aes(ymin=unemploy - 900, ymax=unemploy + 900)**)** - x, ymax, ymin, alpha, color, fill, group, linetype, size

LINE SEGMENTS

common aesthetics: x, y, alpha, color, linetype, size

b + geom_abline(aes(intercept=0, slope=1)**)**
b + geom_hline(aes(yintercept = lat)**)**
b + geom_vline(aes(xintercept = long)**)**

b + geom_segment(aes(yend=lat+1, xend=long+1)**)**
b + geom_spoke(aes(angle = 1:1155, radius = 1)**)**

ONE VARIABLE continuous

c <- ggplot(mpg, aes(hwy)); c2 <- ggplot(mpg)

c + geom_area(stat = "bin"))
x, y, alpha, color, fill, linetype, size

c + geom_density(kernel = "gaussian"**)**
x, y, alpha, color, fill, group, linetype, size, weight

c + geom_dotplot()
x, y, alpha, color, fill

c + geom_freqpoly() x, y, alpha, color, group, linetype, size

c + geom_histogram(binwidth = 5**)** x, y, alpha, color, fill, linetype, size, weight

c2 + geom_qq(aes(sample = hwy)**)** x, y, alpha, color, fill, linetype, size, weight

discrete

d <- ggplot(mpg, aes(fl))

d + geom_bar()
x, alpha, color, fill, linetype, size, weight

TWO VARIABLES

continuous x , continuous y

e <- ggplot(mpg, aes(cty, hwy))

e + geom_label(aes(label = cty), nudge_x = 1, nudge_y = 1, check_overlap = TRUE**)** x, y, label, alpha, angle, color, family, fontface, hjust, lineheight, size, vjust

e + geom_jitter(height = 2, width = 2**)**
x, y, alpha, color, fill, shape, size

e + geom_point(), x, y, alpha, color, fill, shape, size, stroke

e + geom_quantile(), x, y, alpha, color, group, linetype, size, weight

e + geom_rug(sides = "bl"**)**, x, y, alpha, color, linetype, size

e + geom_smooth(method = lm**)**, x, y, alpha, color, fill, group, linetype, size, weight

e + geom_text(aes(label = cty), nudge_x = 1, nudge_y = 1, check_overlap = TRUE**)**, x, y, label, alpha, angle, color, family, fontface, hjust, lineheight, size, vjust

discrete x , continuous y

f <- ggplot(mpg, aes(class, hwy))

f + geom_col(), x, y, alpha, color, fill, group, linetype, size

f + geom_boxplot(), x, y, lower, middle, upper, ymax, ymin, alpha, color, fill, group, linetype, shape, size, weight

f + geom_dotplot(binaxis = "y", stackdir = "center"**)**, x, y, alpha, color, fill, group

f + geom_violin(scale = "area"**)**, x, y, alpha, color, fill, group, linetype, size, weight

discrete x , discrete y

g <- ggplot(diamonds, aes(cut, color))

g + geom_count(), x, y, alpha, color, fill, shape, size, stroke

continuous bivariate distribution

h <- ggplot(diamonds, aes(carat, price))

h + geom_bin2d(binwidth = c(0.25, 500)**)**
x, y, alpha, color, fill, linetype, size, weight

h + geom_density2d()
x, y, alpha, colour, group, linetype, size

h + geom_hex()
x, y, alpha, colour, fill, size

continuous function

i <- ggplot(economics, aes(date, unemploy))

i + geom_area()
x, y, alpha, color, fill, linetype, size

i + geom_line()
x, y, alpha, color, group, linetype, size

i + geom_step(direction = "hv")
x, y, alpha, color, group, linetype, size

visualizing error

df <- data.frame(grp = c("A", "B"), fit = 4:5, se = 1:2)
j <- ggplot(df, aes(grp, fit, ymin = fit-se, ymax = fit+se))

j + geom_crossbar(fatten = 2**)**
x, y, ymax, ymin, alpha, color, fill, group, linetype, size

j + geom_errorbar(), x, ymax, ymin, alpha, color, group, linetype, size, width (also **geom_errorbarh()**)

j + geom_linerange()
x, ymin, ymax, alpha, color, group, linetype, size

j + geom_pointrange()
x, y, ymin, ymax, alpha, color, fill, group, linetype, shape, size

maps

data <- data.frame(murder = USArrests$Murder, state = tolower(rownames(USArrests)))
map <- map_data("state")
k <- ggplot(data, aes(fill = murder))

k + geom_map(aes(map_id = state), map = map**)**
+ expand_limits(x = map$long, y = map$lat**)**,
map_id, alpha, color, fill, linetype, size

THREE VARIABLES

seals$z <- with(seals, sqrt(delta_long^2 + delta_lat^2))l <- ggplot(seals, aes(long, lat))

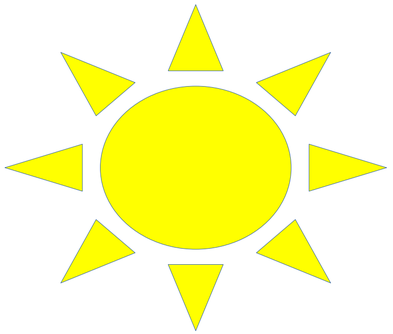
l + geom_contour(aes(z = z))
x, y, z, alpha, colour, group, linetype, size, weight

l + geom_raster(aes(fill = z), hjust=0.5, vjust=0.5, interpolate=FALSE**)**
x, y, alpha, fill

l + geom_tile(aes(fill = z)**)**, x, y, alpha, color, fill, linetype, size, width

Graph Gallery!

https://r-graph-gallery.com/



To make a graph

1. Pick a **data** set

[template]

```
ggplot(data = <DATA>) +
  <GEOM_FUNCTION>(mapping = aes(<MAPPINGS>))
```

2. Choose a **geom** to display cases

3. **Map** aesthetic properties to variables

Histograms and Bar Charts

Histogram

data

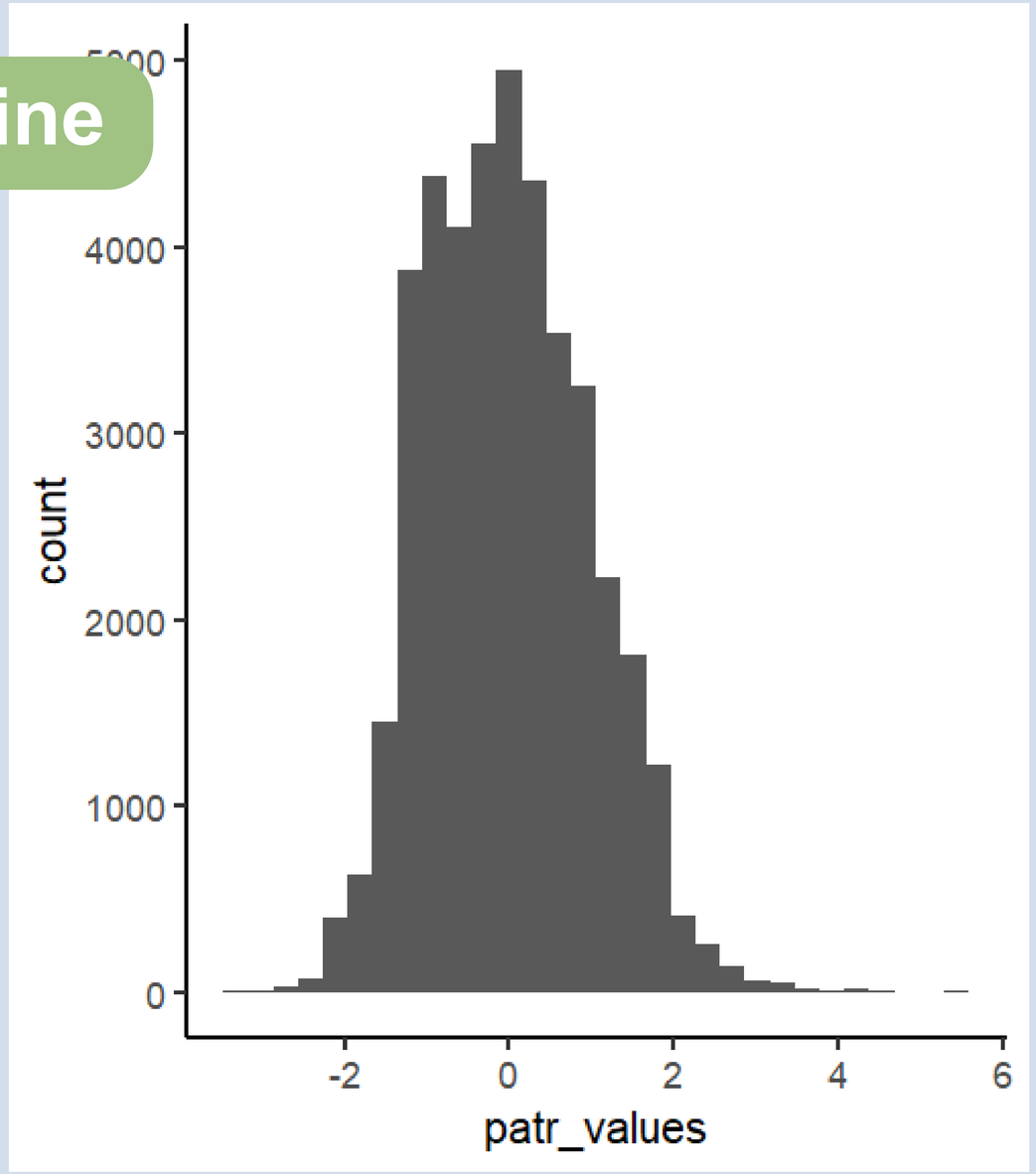
Aesthetic mappings

+ before new line

```
ggplot(flfp, aes(patr_values)) +
  geom_histogram()
```

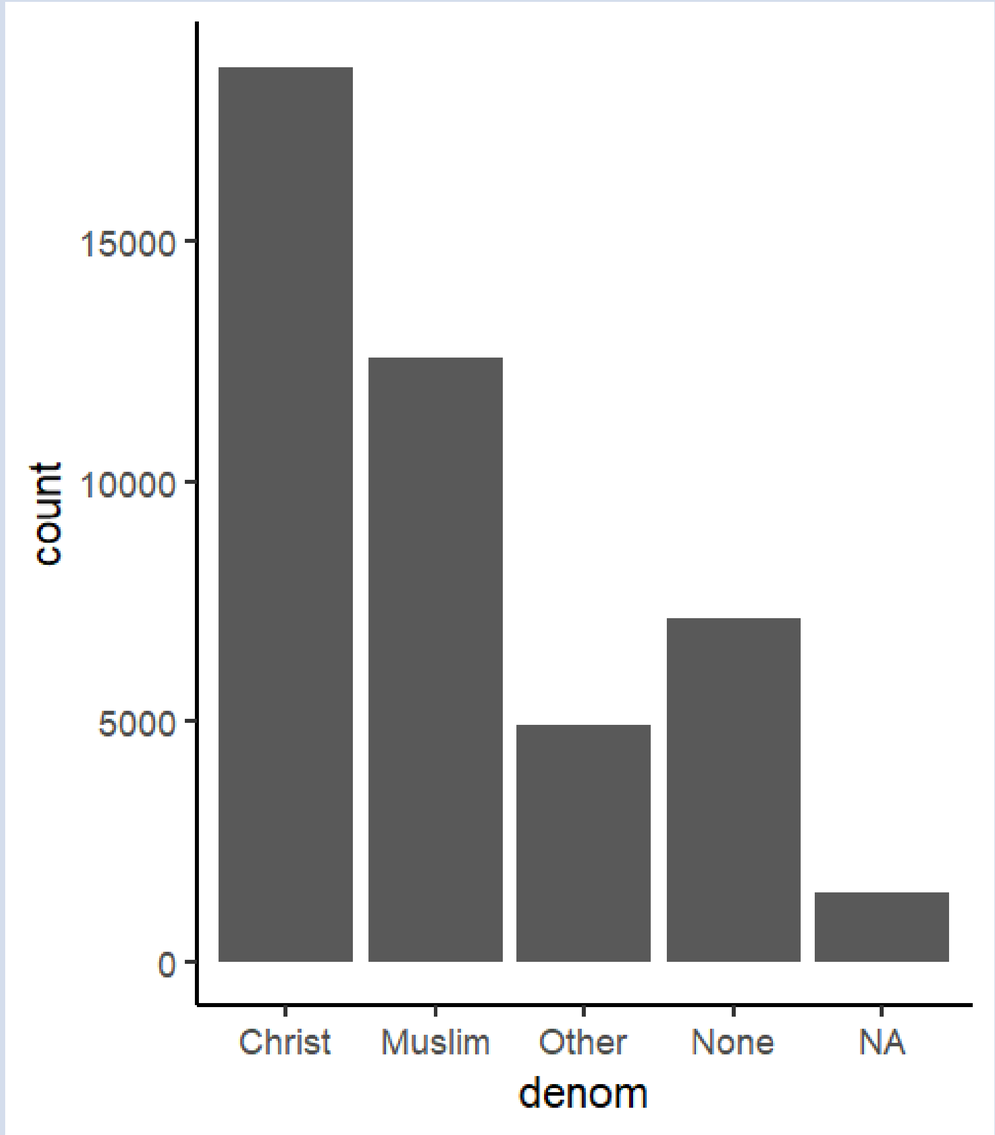
type of layer

The geom (style, shape) of the graph

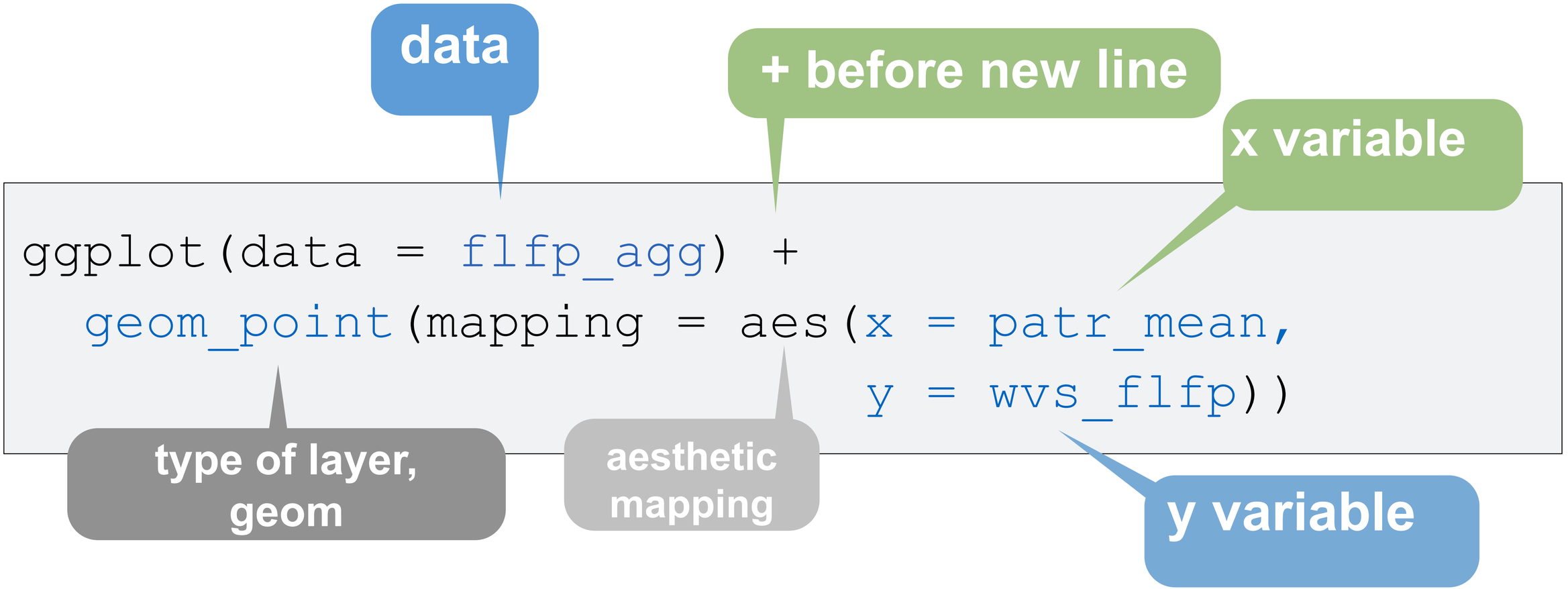


Bar chart - frequencies

```
ggplot(flfp, aes(denom)) +
  geom_bar()
```

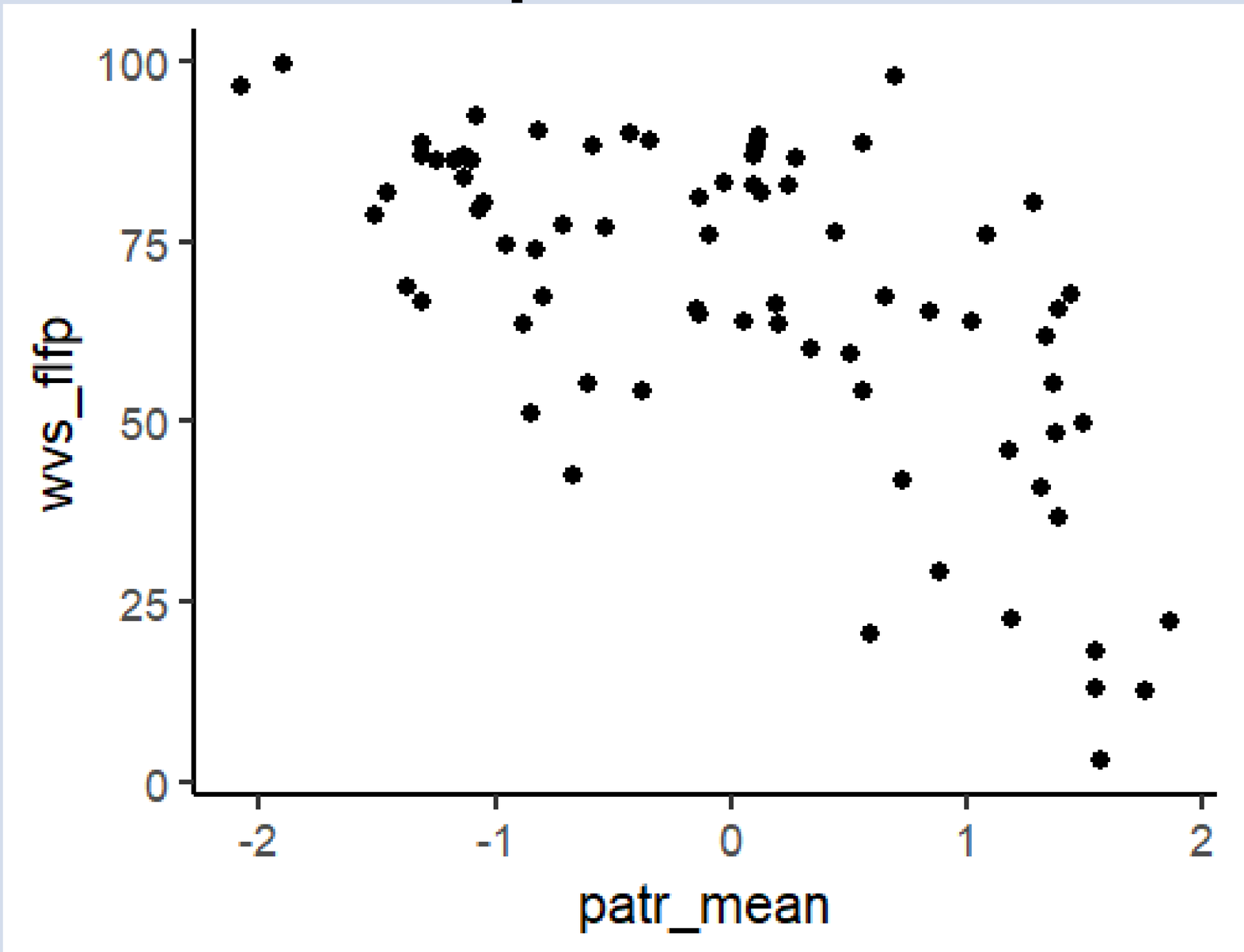


Scatterplots


data
+ before new line
x variable
ggplot(data = flfp_agg) +
geom_point(mapping = aes(x = patr_mean,
y = wvs_flfp))
type of layer, geom
aesthetic mapping
y variable

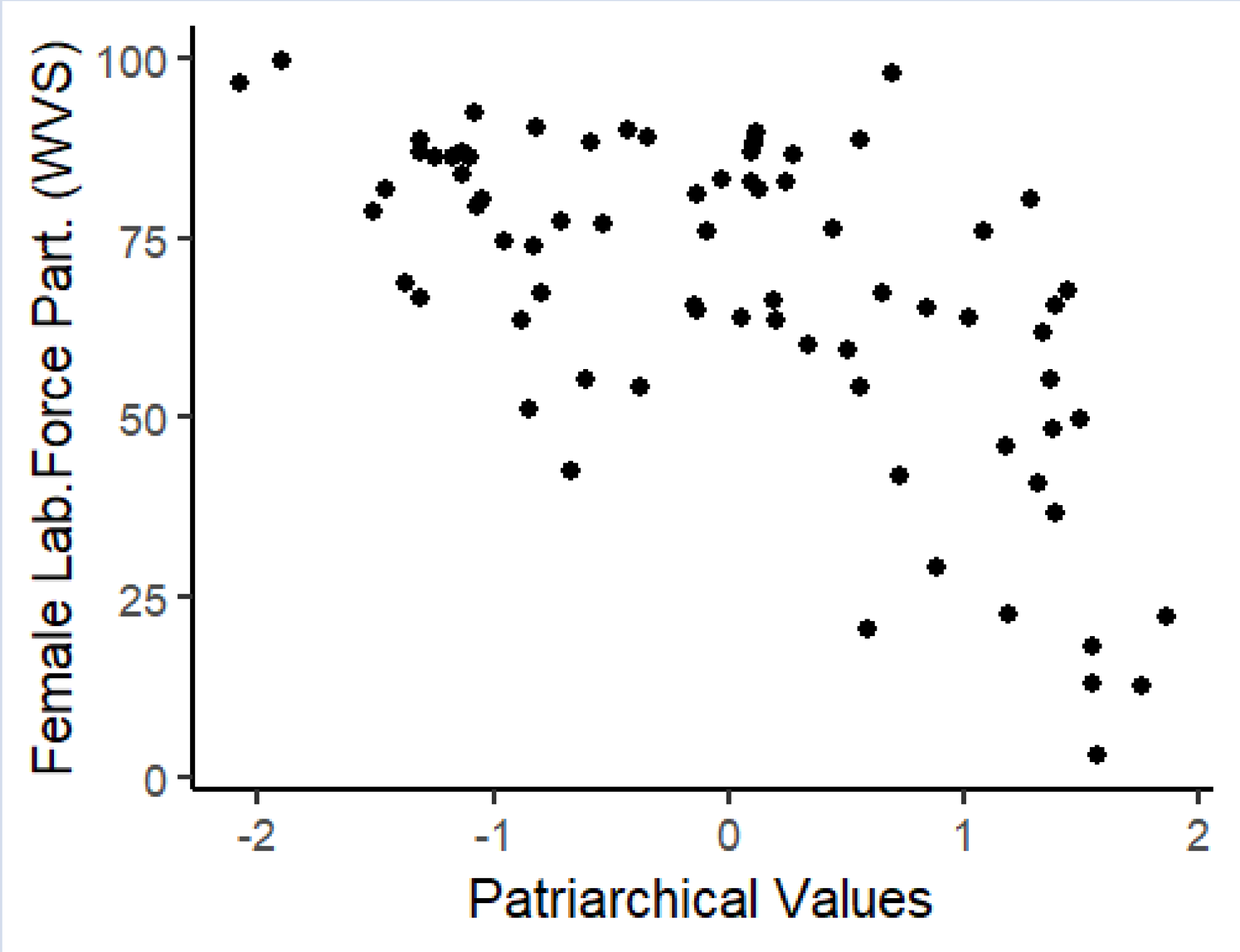
Basic Scatterplot

```
ggplot(flfp_agg, aes(x=patr_mean, y=wvs_flfp)) +
  geom_point()
```



Fix axis labels

```
ggplot(flfp_agg, aes(x=patr_mean,
y=wvs_flfp)) +
  geom_point()+
  labs(y= "Female Lab.Force Part.
(WVS)", x = "Patriarchical Values")
```



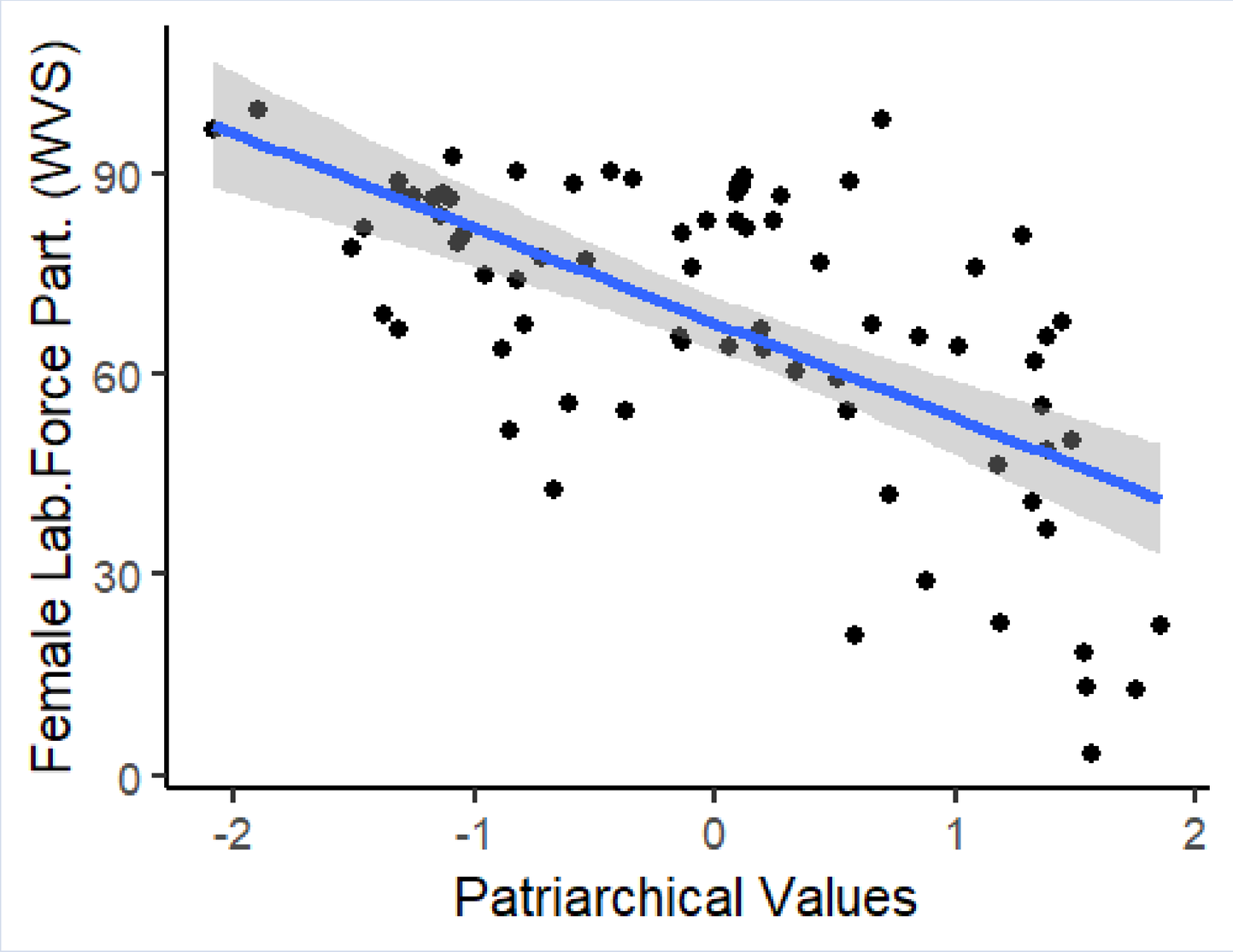
Add a regression line
(with confidence interval)

```
ggplot(flfp_agg, aes(x=patr_mean,
y=wvs_flfp)) +

  geom_point()+

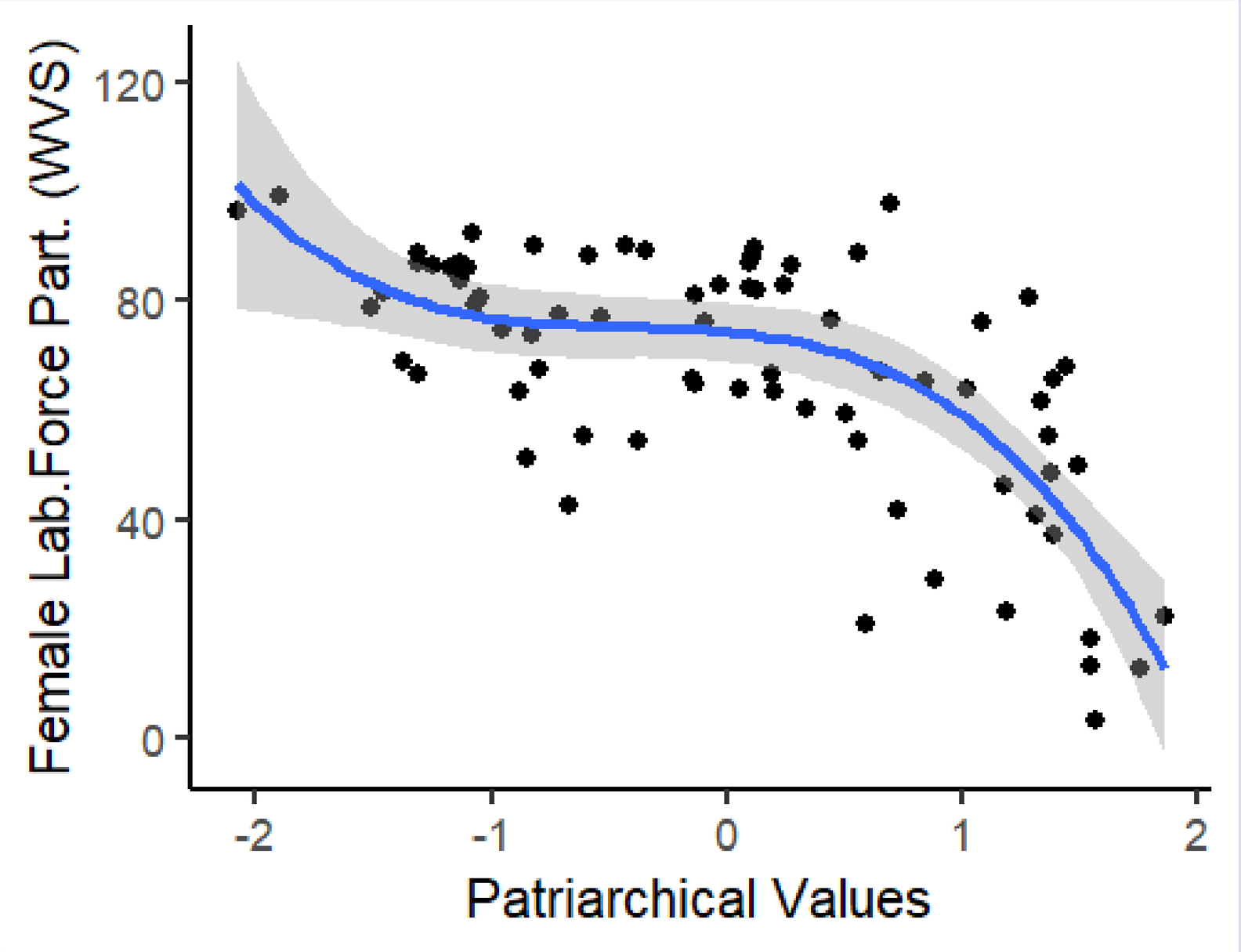
    labs(y= "Female Lab.Force Part.
(WVS)", x = "Patriarchical Values")
+

  geom_smooth(method = "lm", se
= T)
```



Add a curvier regression line

```
ggplot(flfp_agg, aes(x=patr_mean, y=wvs_flfp)) +
  geom_point()+
  labs(y= "Female Lab.Force Part. (WVS)", x = "Patriarchical Values") +
  geom_smooth(method = "lm", se = T, formula = y ~ poly(x, 3))
```



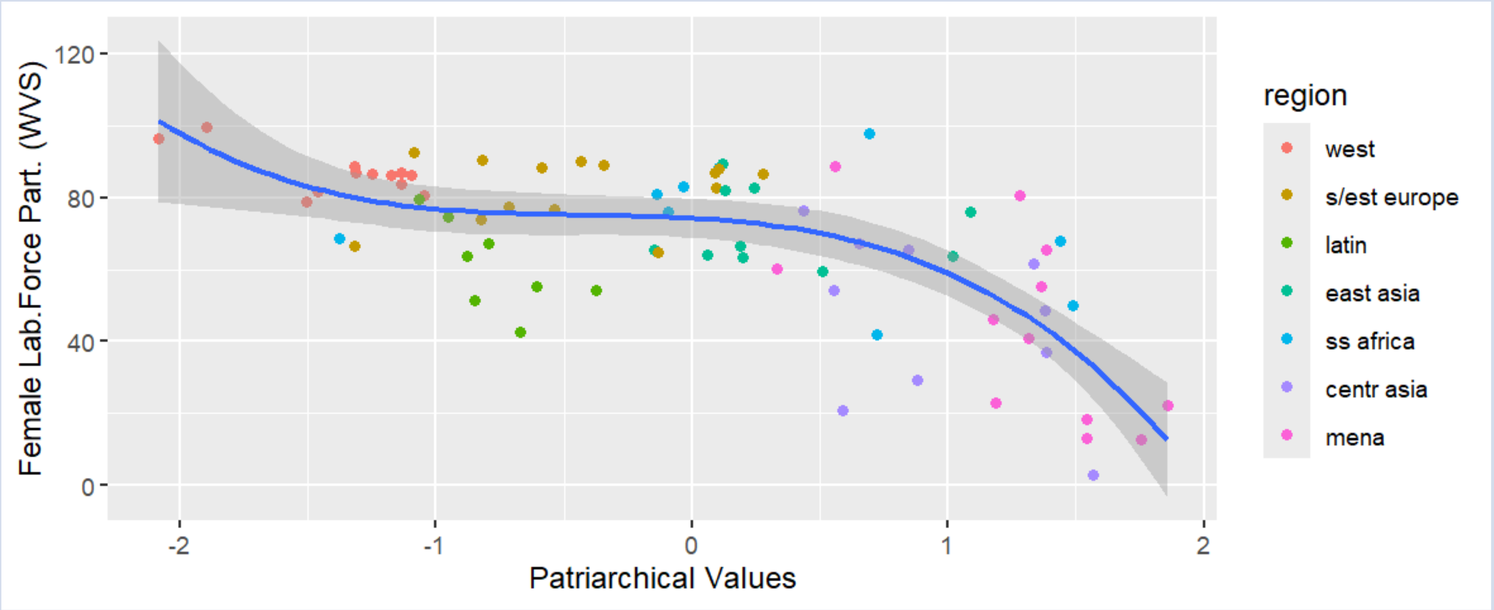
Highlight Region Differences

```
ggplot(flfp_agg, aes(x=patr_mean, y=wvs_flfp)) +

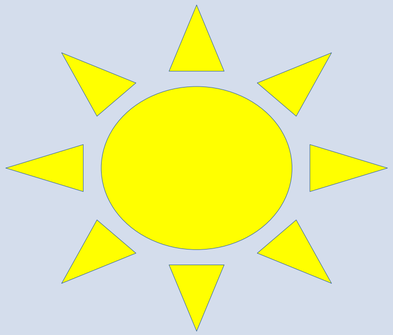
  geom_point(aes(colour = region)) +

  labs(y= "Female Lab.Force Part. (WVS)", x = "Patriarchical Values") +

  geom_smooth(method = "lm", se = T, formula = y ~ poly(x, 3))
```

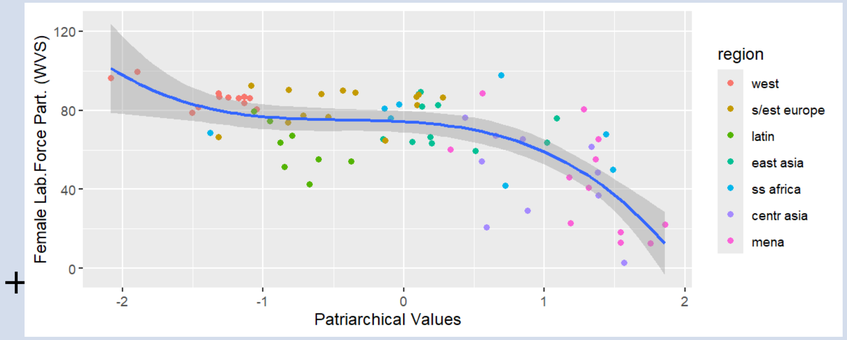


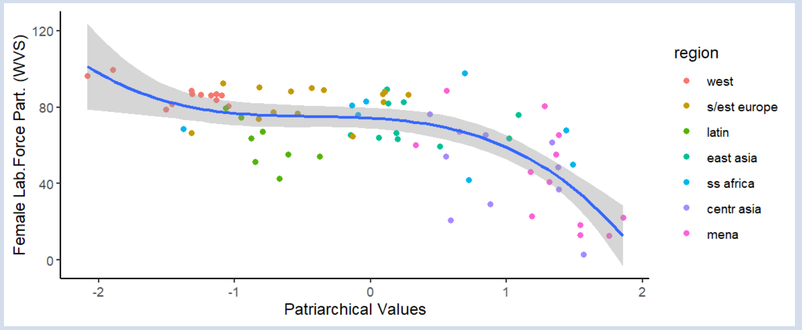
We had to add this aesthetic to geom_point! If we would add it to ggplot() on top, it would impact EVERYTHING, including the regression line, thus splitting it.



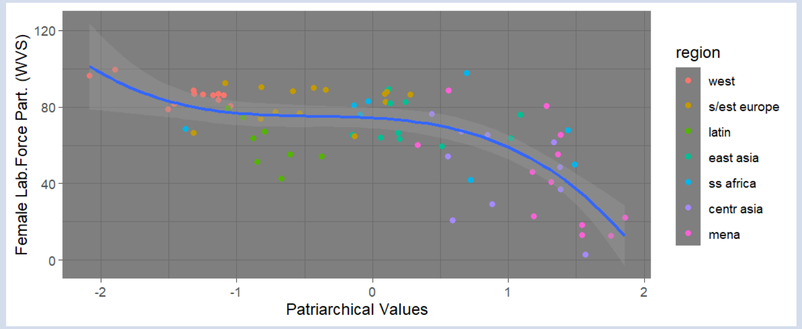
Change stylistic *theme*

```
ggplot(flfp_agg, aes(x=patr_mean, y=wvs_flfp)) +
  geom_point(aes(colour = region))+
  theme_classic() +
  labs(y= "Female Lab.Force Part. (WVS)",
       x = "Patriarchical Values") +
  geom_smooth(method = "lm", se = T, formula =
y ~ poly(x, 3))
```

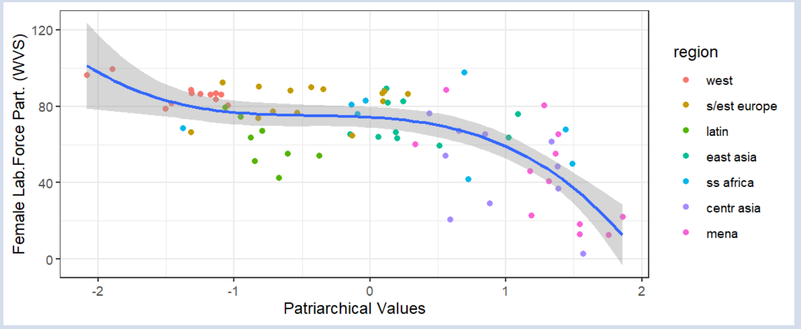




theme_classic()



theme_dark()



theme_bw()

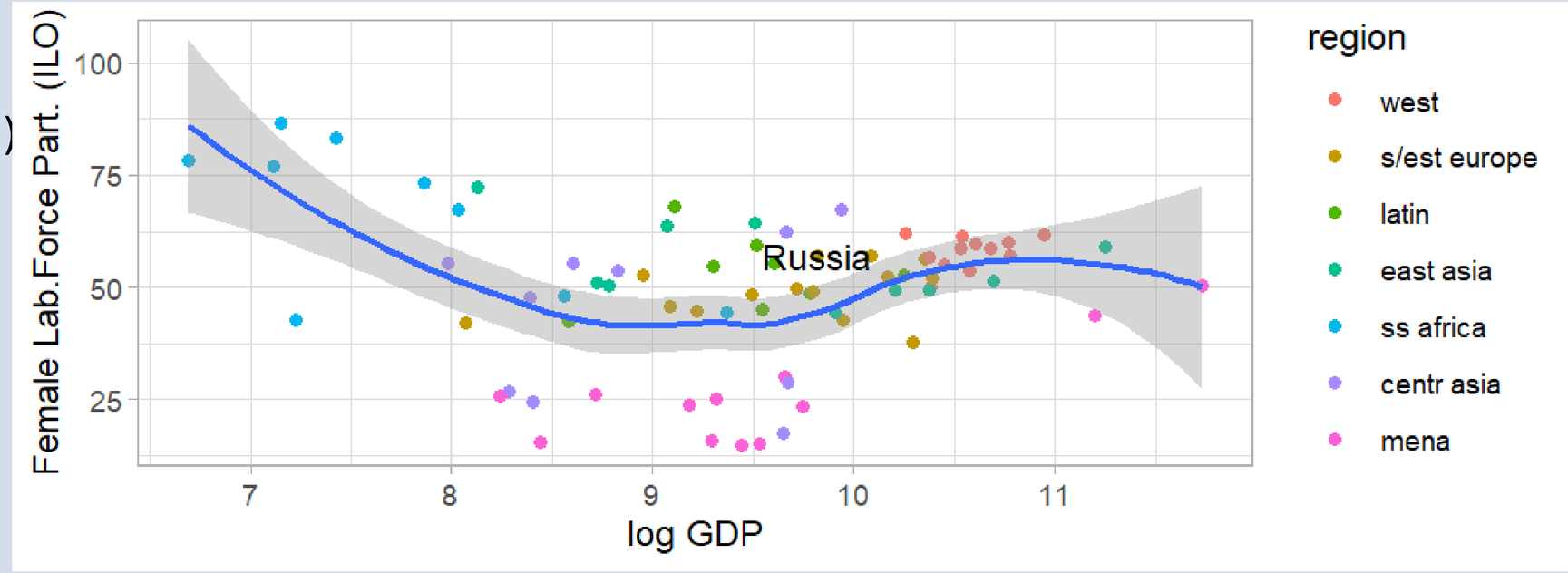
Lots of others can be found online. There are also ways to specify all of these manually

Your Turn

- Use ggplot with flfp_agg to make a scatterplot with log_gdp on the x-axis and ilo_flfp on the y-axis.
- Add an unspecified smoothed line (no method), with a confidence interval.
- Split by region, and
- use the theme_light().
- Adapt your axis labels.
- Add a special label to the to a country name of your choosing (*cntry*) using geom_text().

Highlight Region Differences

```
ggplot(flfp_agg, aes(x=log_gdp, y=ilo_flfp)) +
  geom_point(aes(colour = region)) +
  theme_light() +
  labs(y= "Female Lab.Force Part. (ILO)", x = "log GDP") +
  geom_smooth(se = T)+
  geom_text(data=filter(flfp_agg, cntry=="Russia"), aes(label=cntry))
```



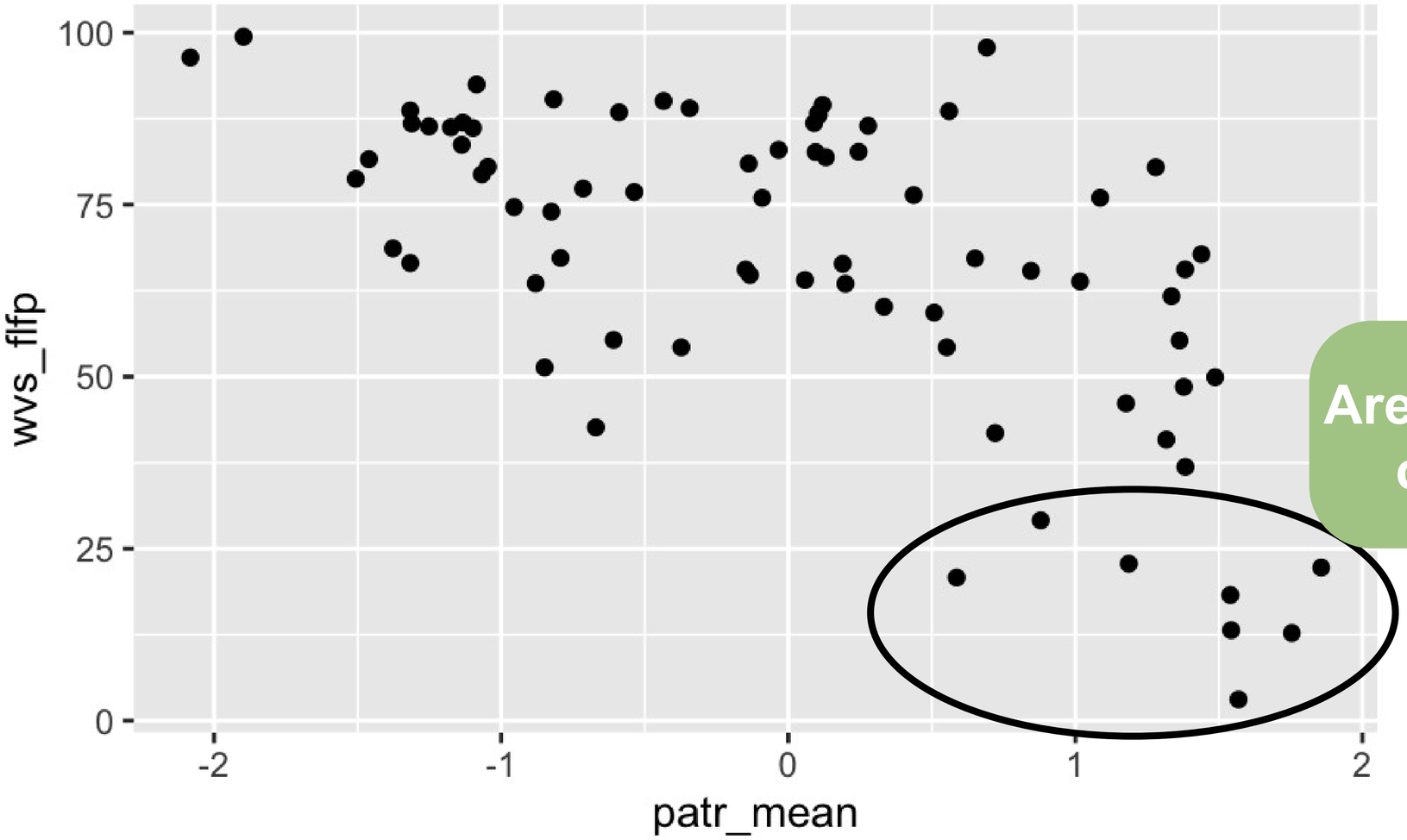
Solution is to use a combination of geom_text, filter() to limit the data under consideration, and aes(label=cntry) to add the label(s).

Layering

Using this same method, we can adapt ANYTHING and everything about this plot. e.g. the font, where the label appears in relation to the point. We can connect the two with a line, offset them so they do not overlap. Change the legend...etc etc .etc

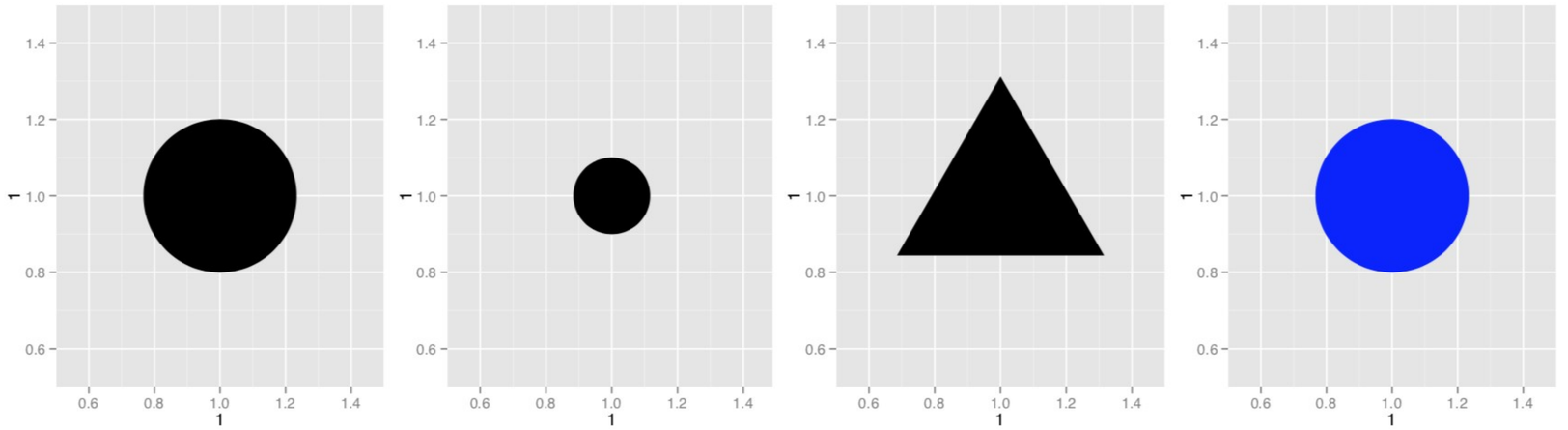
Aesthetic Mappings

(linking variables to visual properties on the plots)




```
ggplot(flfp_agg) +
   geom_point(aes(x = patr_mean, y = wvs_flfp))
```

Aesthetics

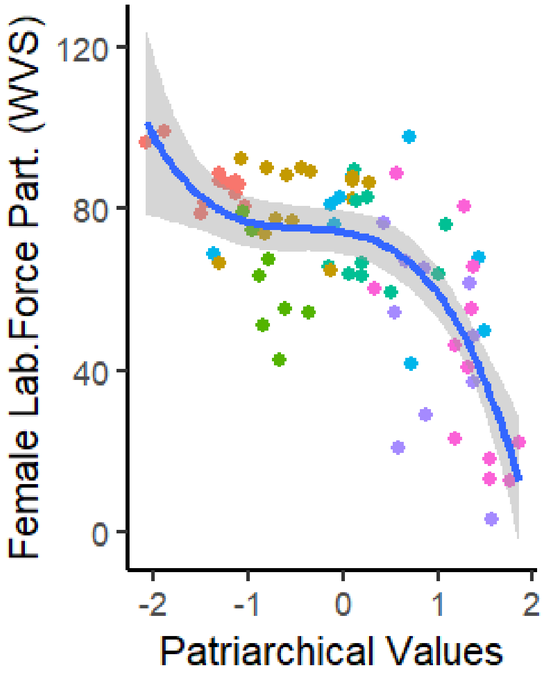


Aesthetics

aesthetic property

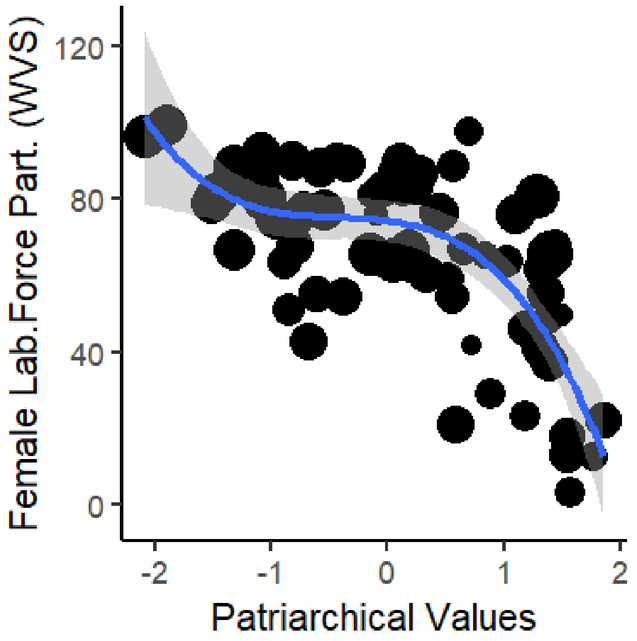
variable to map it to

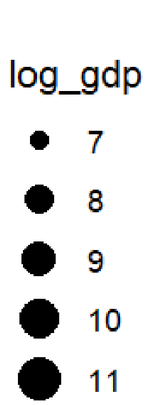
```
ggplot(flfp_agg) + geom_point(aes(x = patr_mean, y = wvs_flfp, color = region))
ggplot(flfp_agg) + geom_point(aes(x = patr_mean, y = wvs_flfp, size = log_gdp))
ggplot(flfp_agg) + geom_point(aes(x = patr_mean, y = wvs_flfp, shape = region))
ggplot(flfp_agg) + geom_point(aes(x = patr_mean, y = wvs_flfp, alpha = log_gdp))
```





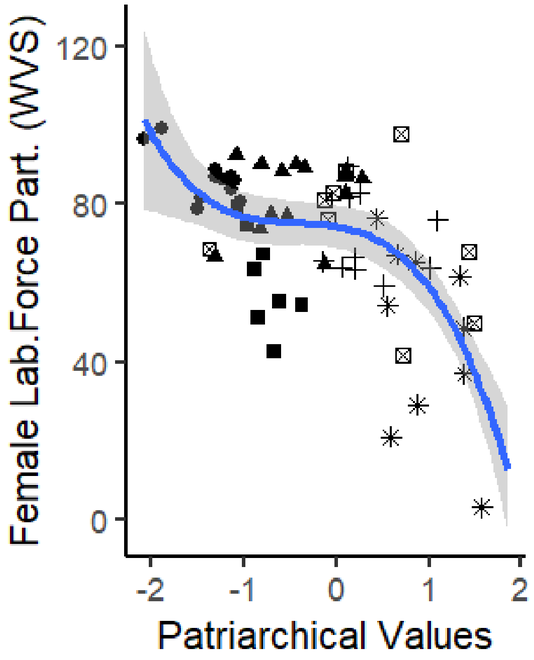
color





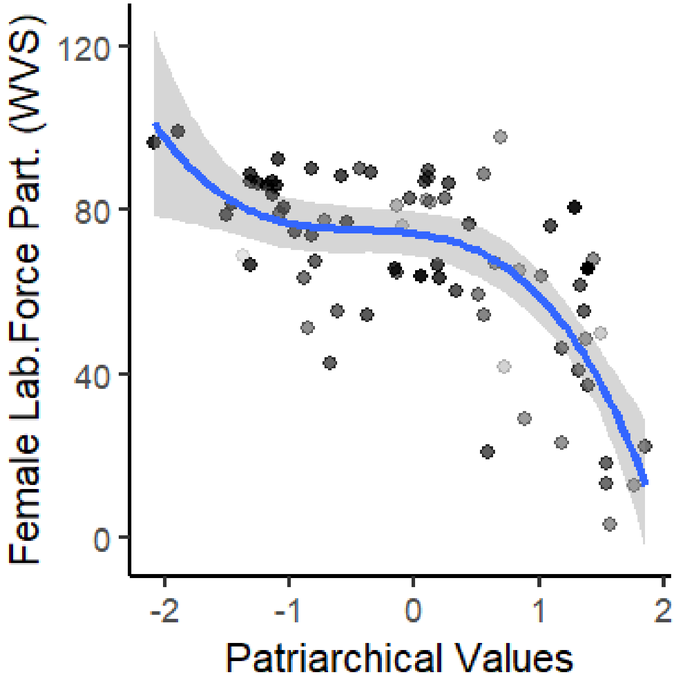
Legends added automatically

size





shape



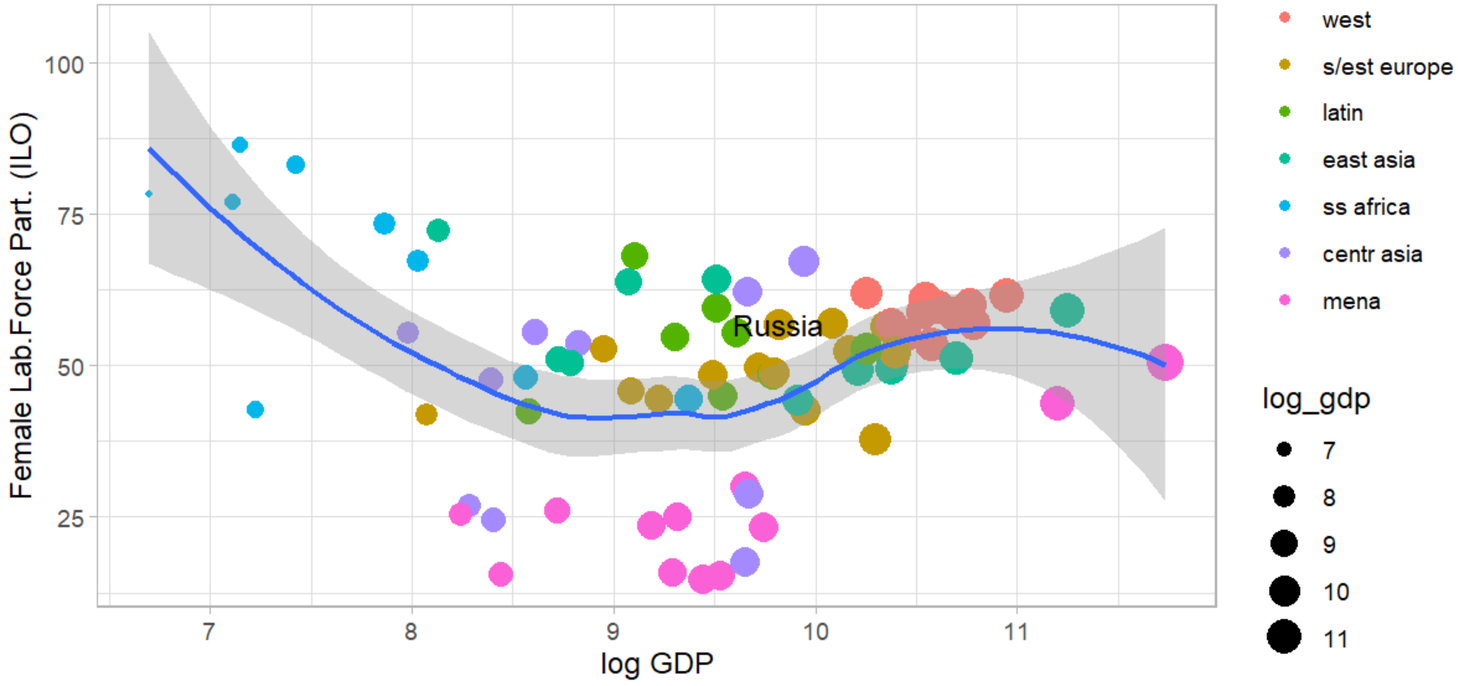


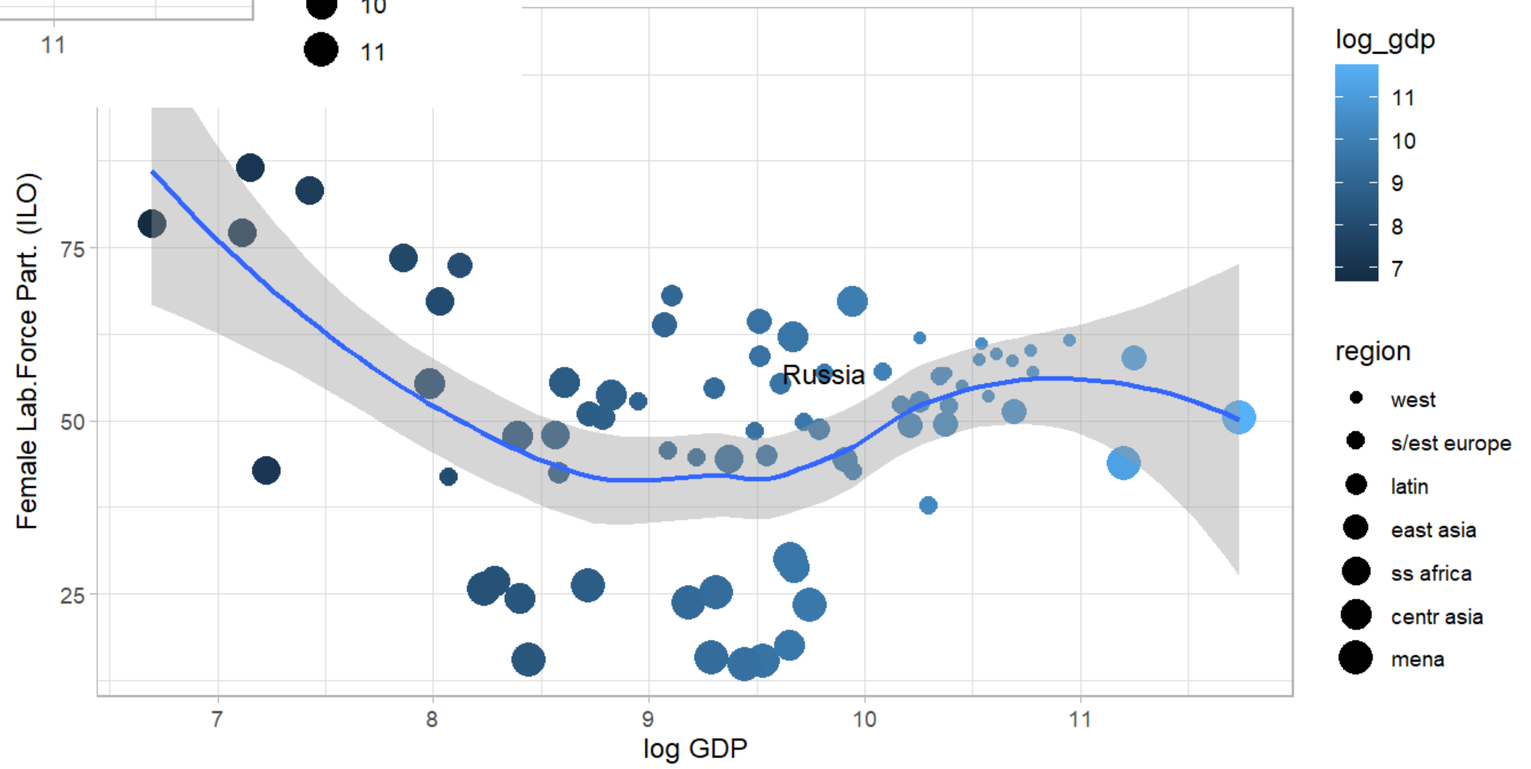
alpha

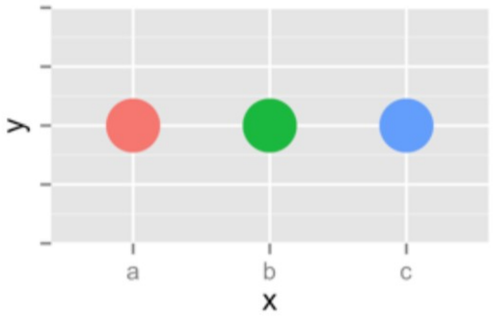
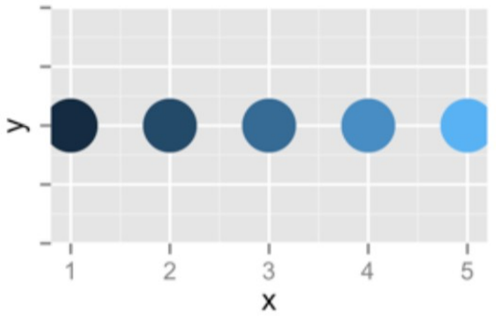
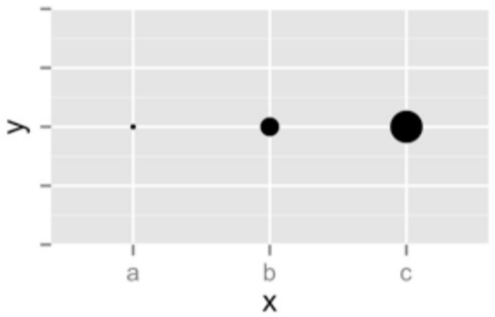
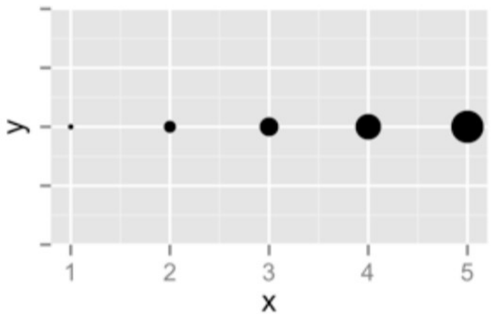
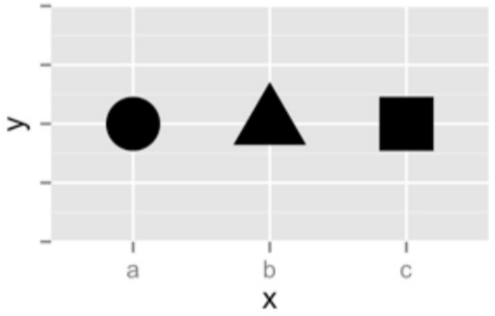
***size* and *alpha* more appropriate for non-categorical variables**

Your turn

- Add colour, size, alpha, and shape aesthetics to your graph from your last individual exercise. Experiment.
- What happens when you map aesthetics to nominal vs. continuous variables?
- What happens when you use more than one aesthetic?

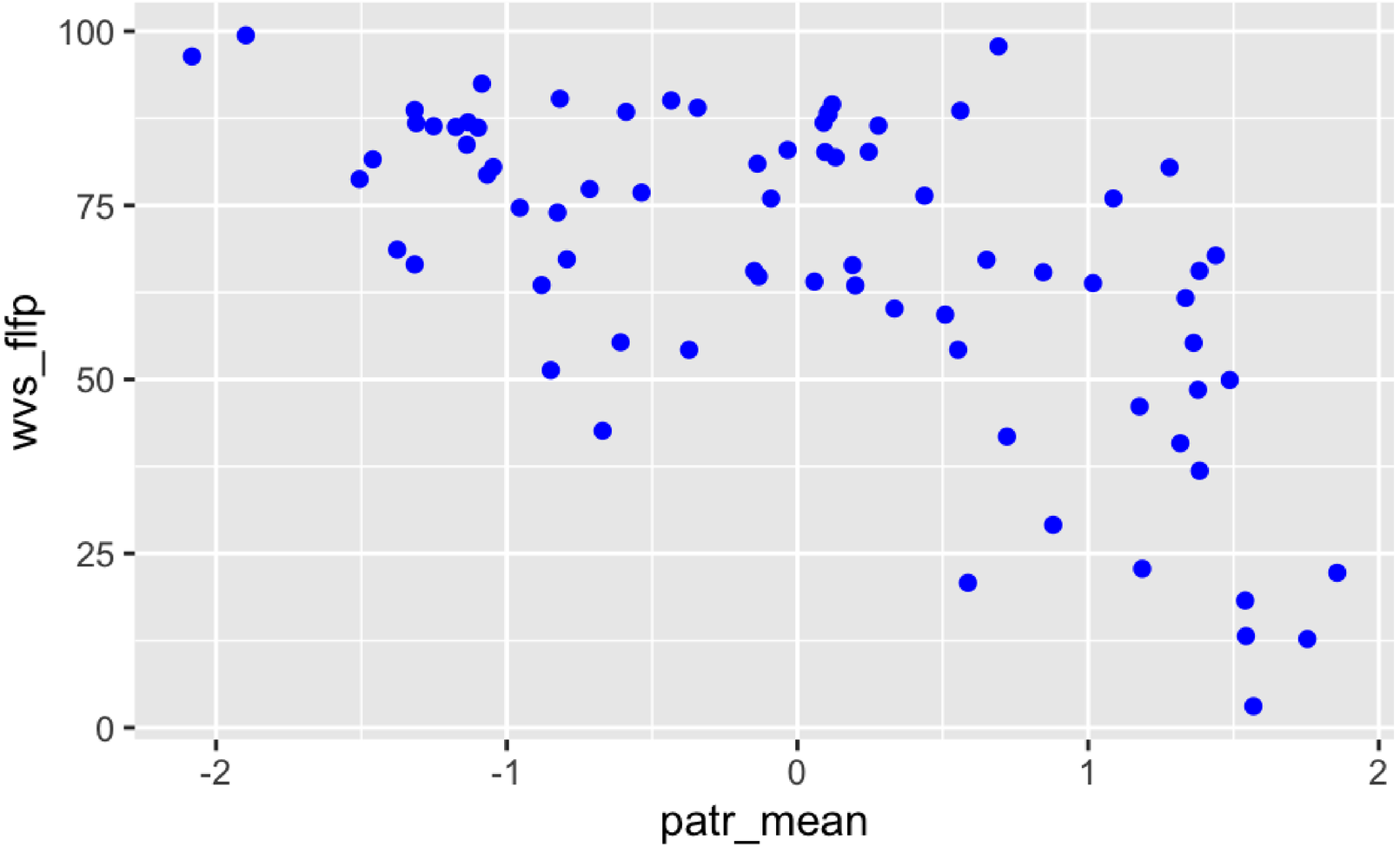

Female Lab.Force Part. (ILO)
100
75
50
25
7
8
9
10
11
log GDP
Russia
west
s/est europe
latin
east asia
ss africa
centr asia
mena
log_gdp
7
8
9
10
11


Female Lab.Force Part. (ILO)
75
50
25
7
8
9
10
11
log GDP
Russia
log_gdp
11
10
9
8
7
region
west
s/est europe
latin
east asia
ss africa
centr asia
mena

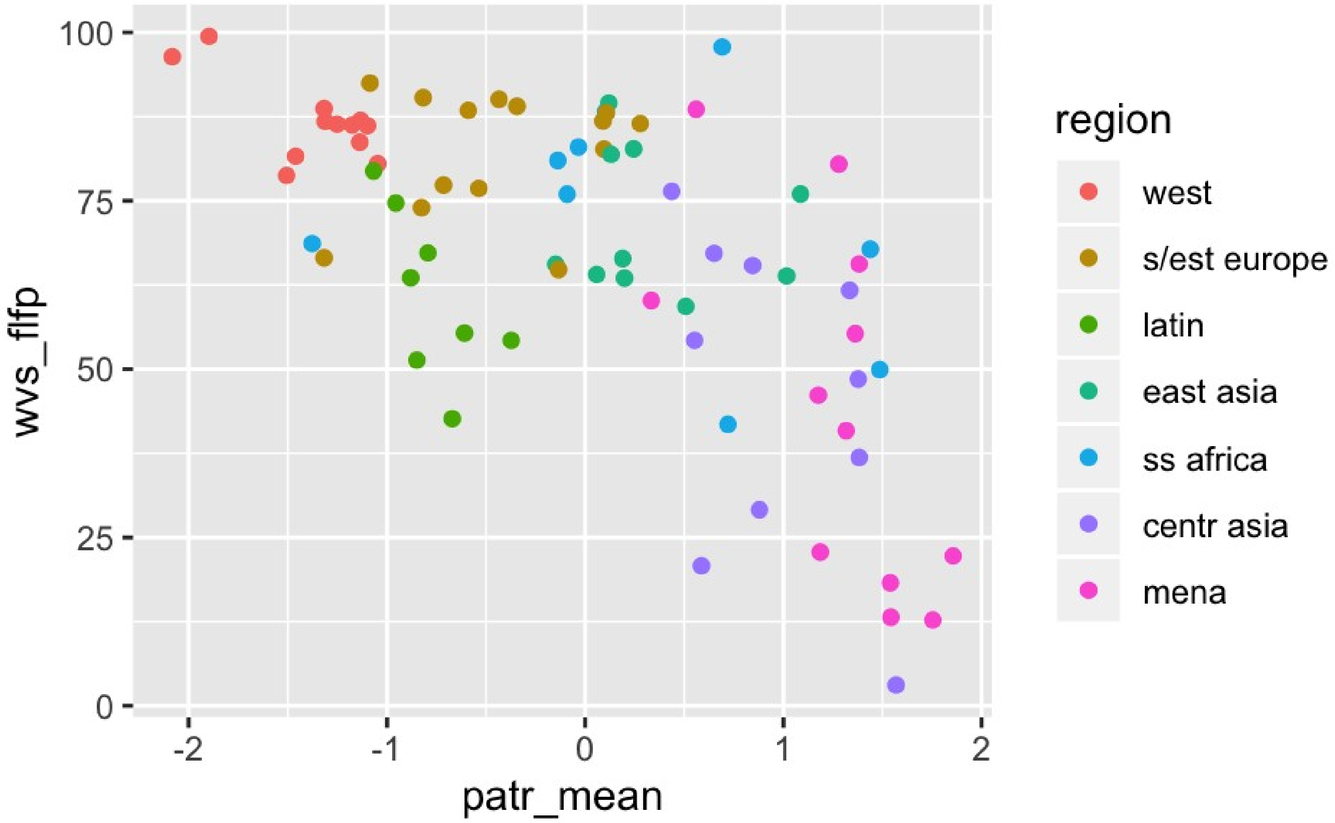
	Discrete	Continuous
Color		
Size		
Shape		

set vs. map

How do you make this plot?



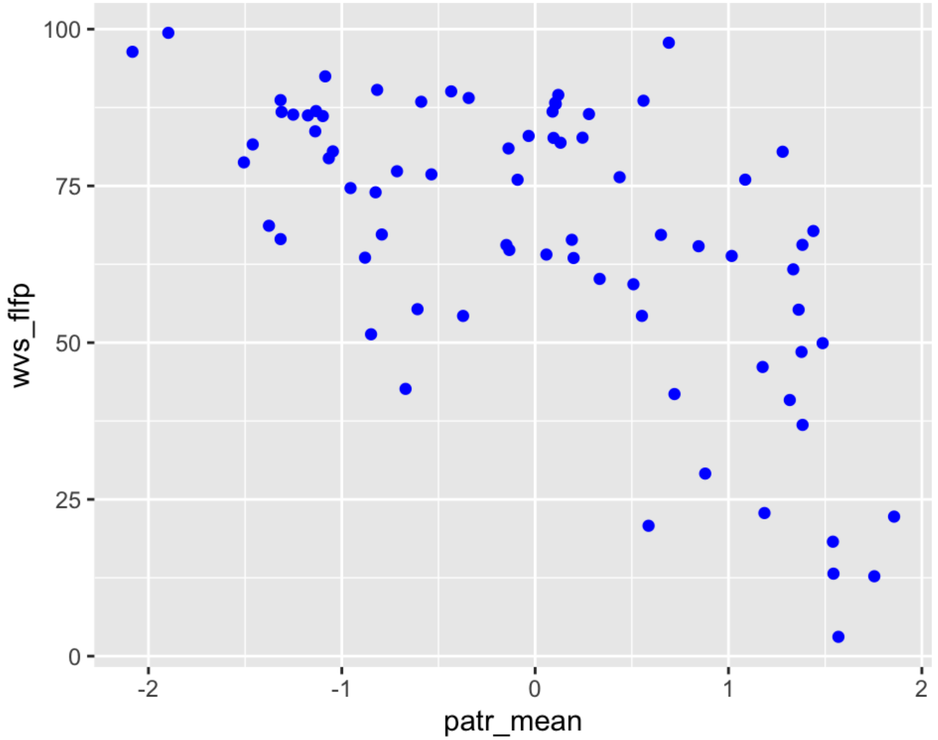
Map variable



Inside of aes(): maps an aesthetic to a variable

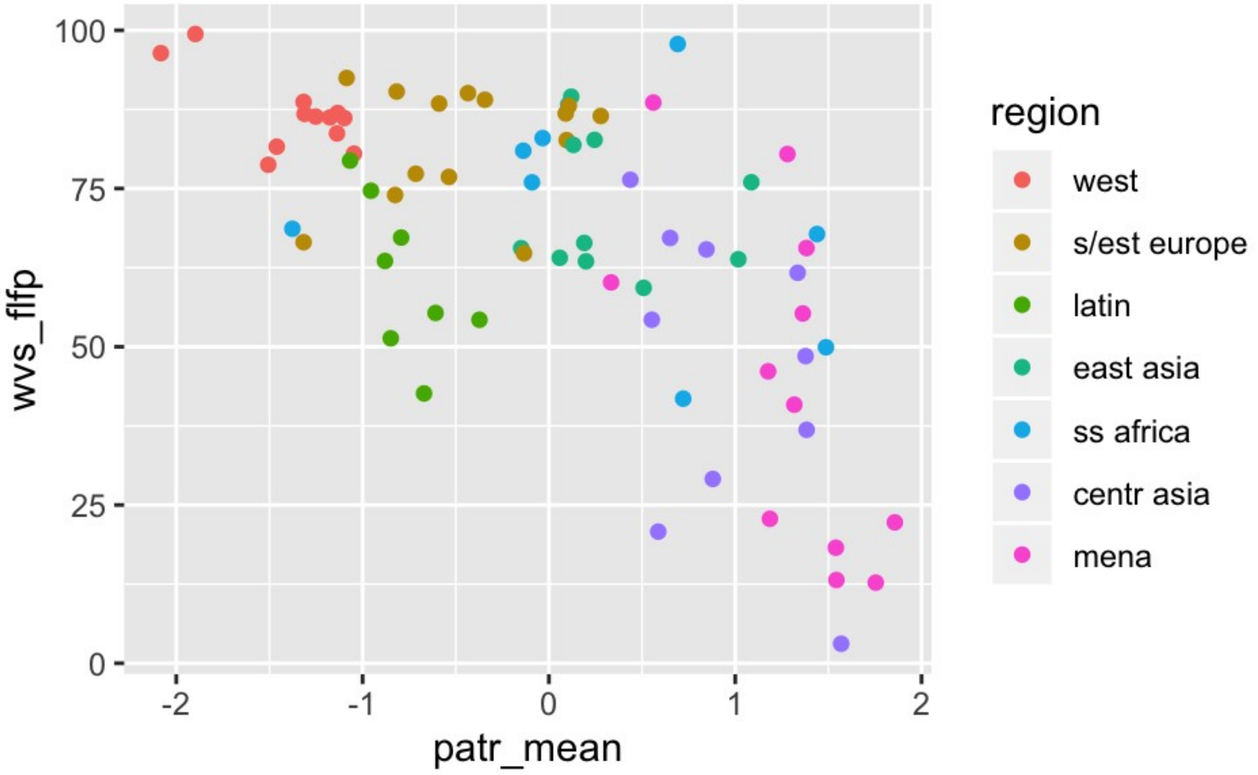
```
ggplot(flfp_agg) +
  geom_point(aes(x = patr_mean, y = wvs_flfp, color = region))
```

Set values

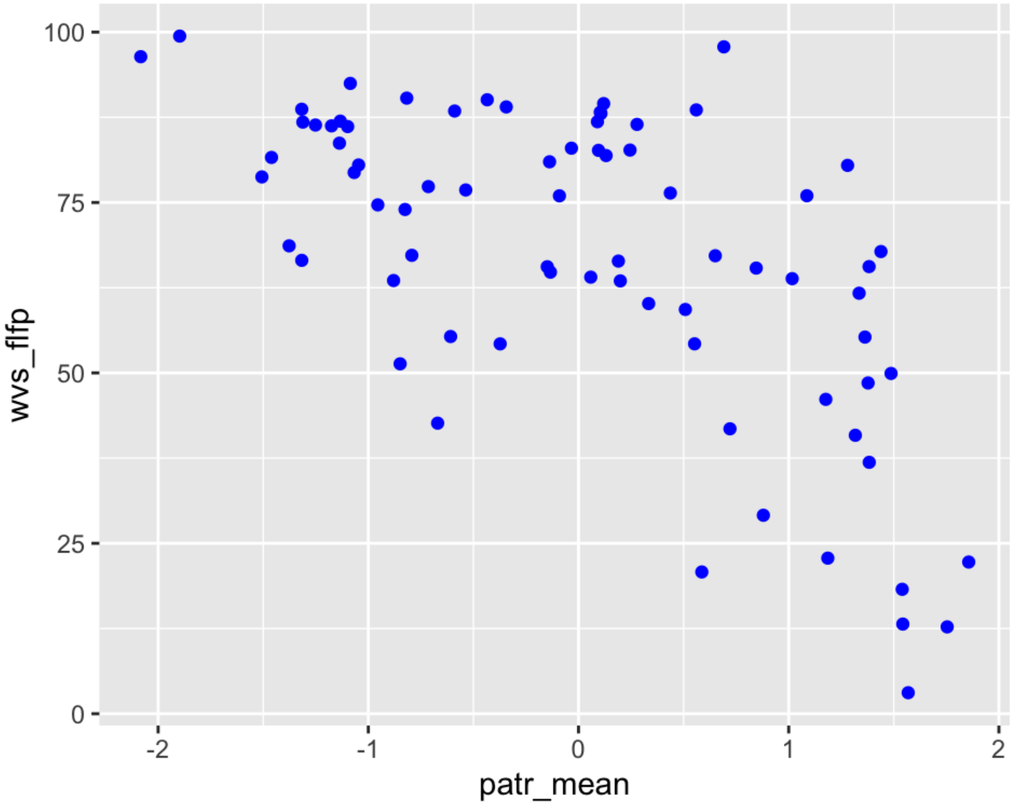


Outside of aes():
sets an aesthetic to a value

```
ggplot(flfp_agg) +
  geom_point(mapping = aes(x = patr_mean, y = wvs_flfp),
                           colour ="blue")
```




```
ggplot(flfp_agg) +
geom_point(aes(x = patr_mean,
               y = wvs_flfp,
               colour = region))
```

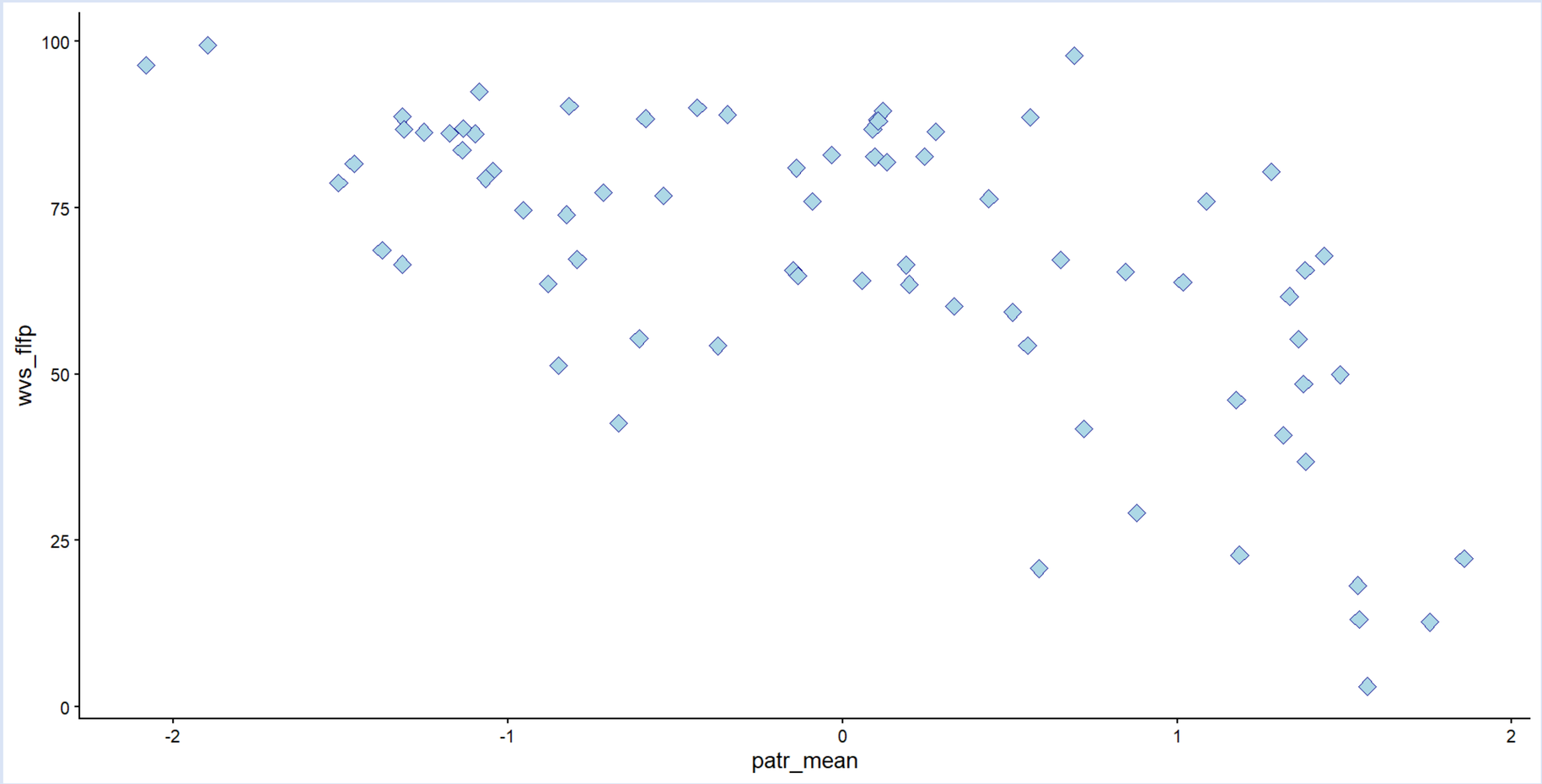



```
ggplot(flfp_agg) +
 geom_point(aes(x = patr_mean,
                y = wvs_flfp),
                colour ="blue")
```

Quiz

Reproduce the plot by setting:

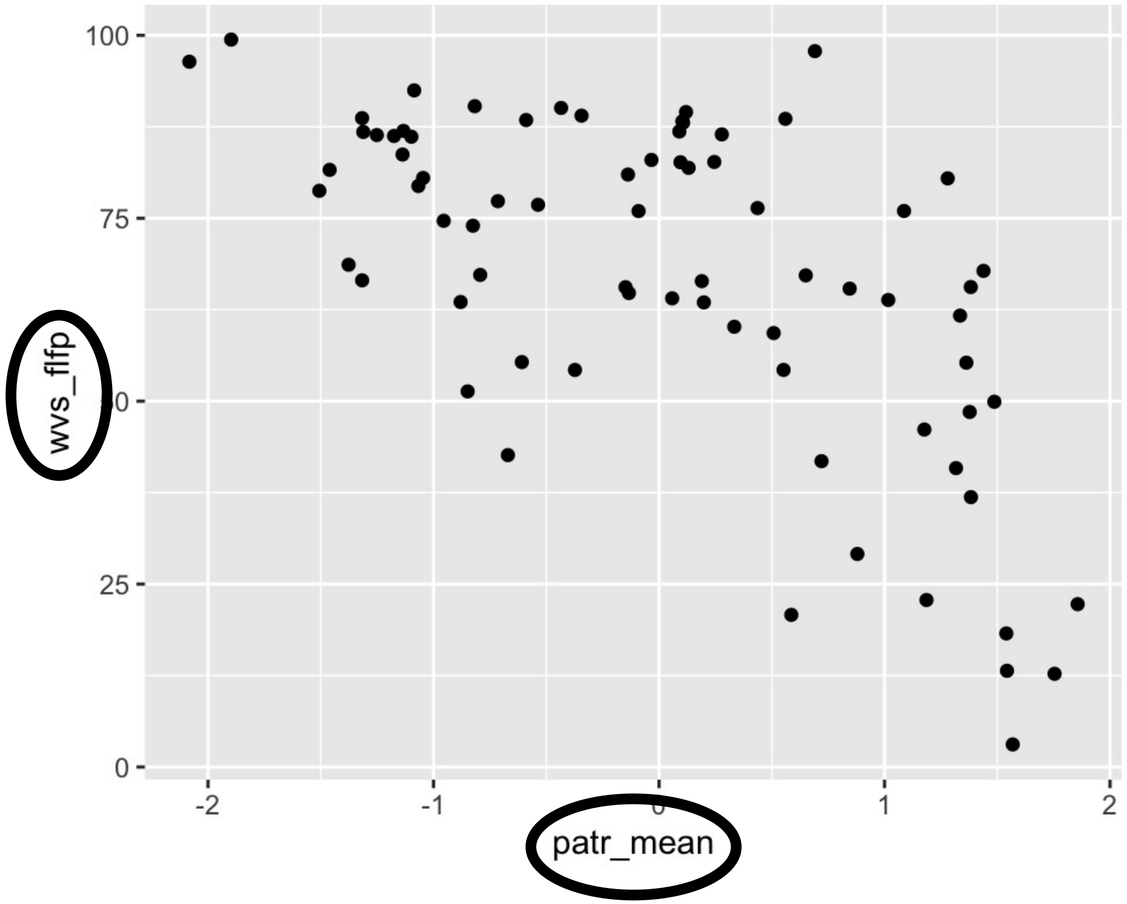
- colour = "darkblue"
- shape = 23
- fill = "lightblue"
- size = 3

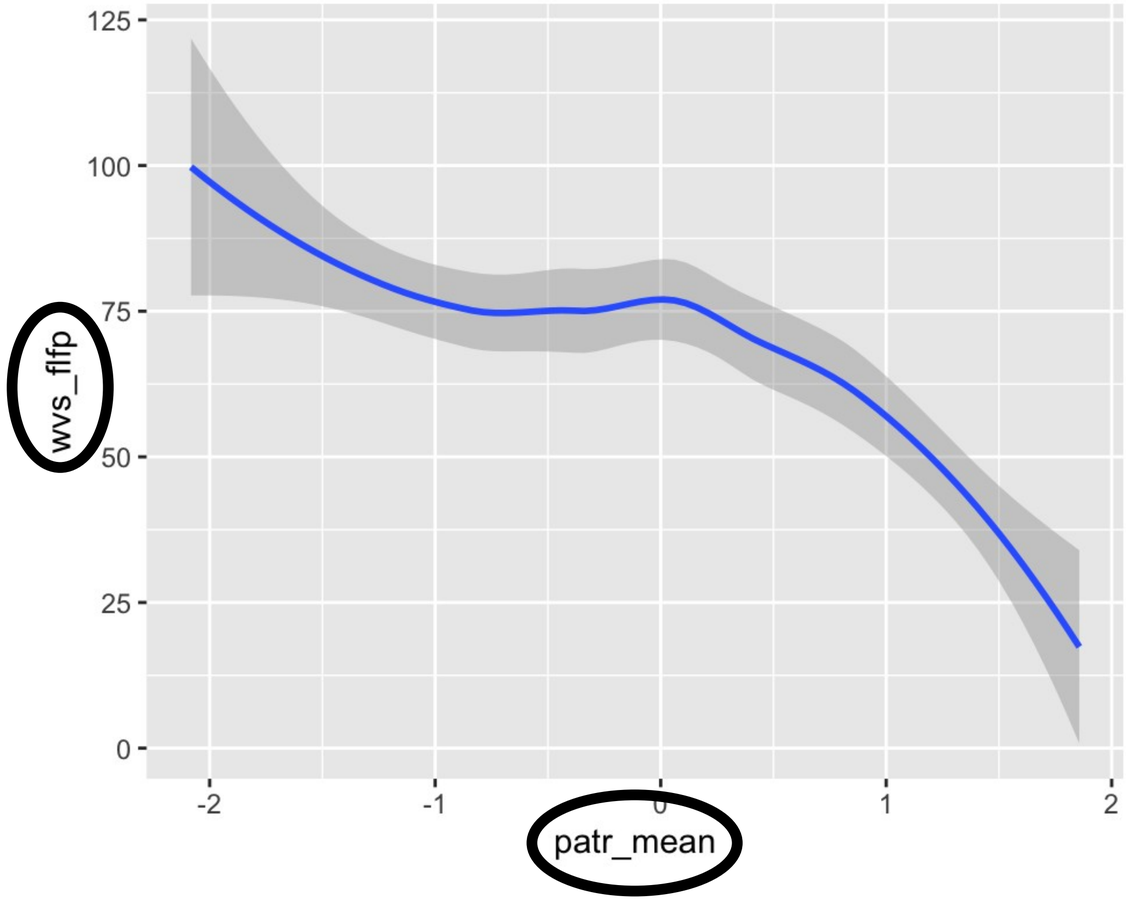


Geoms

How are these plots similar?

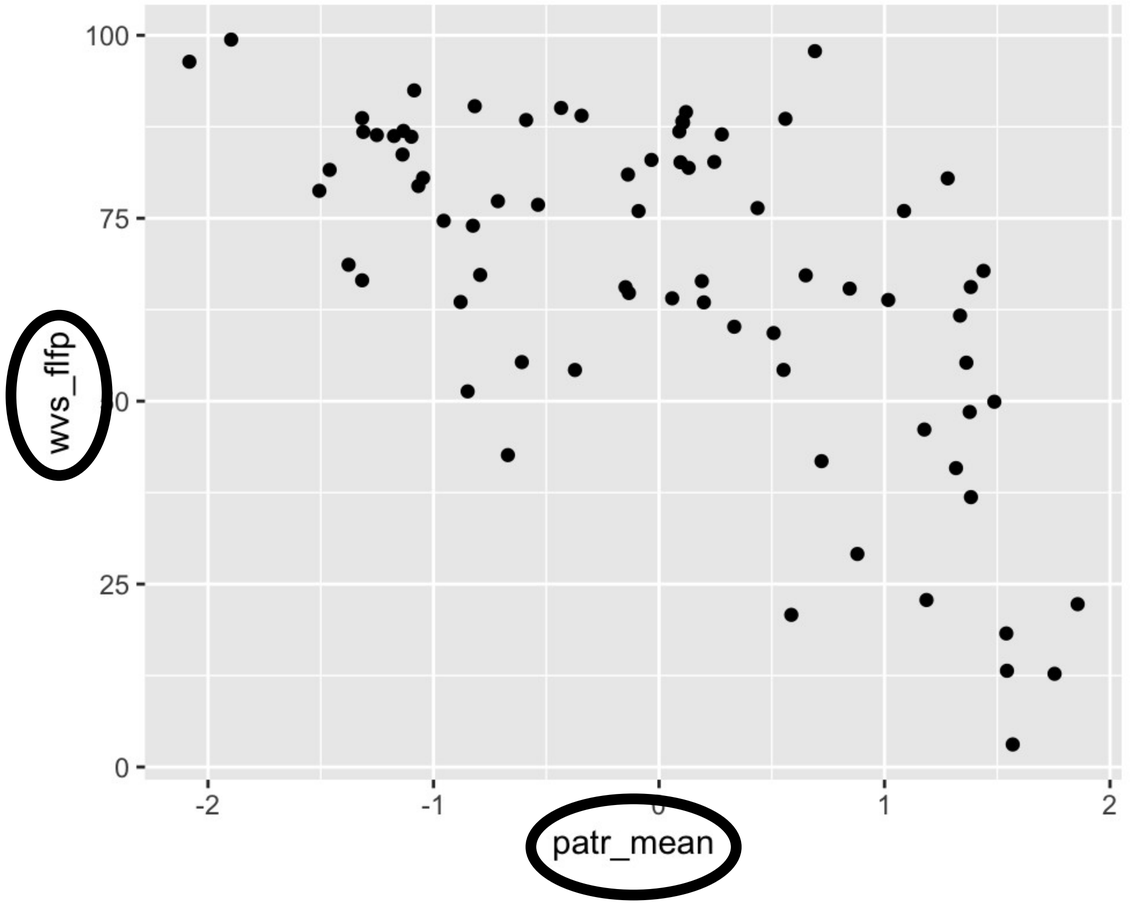
Same: x var, y var, data

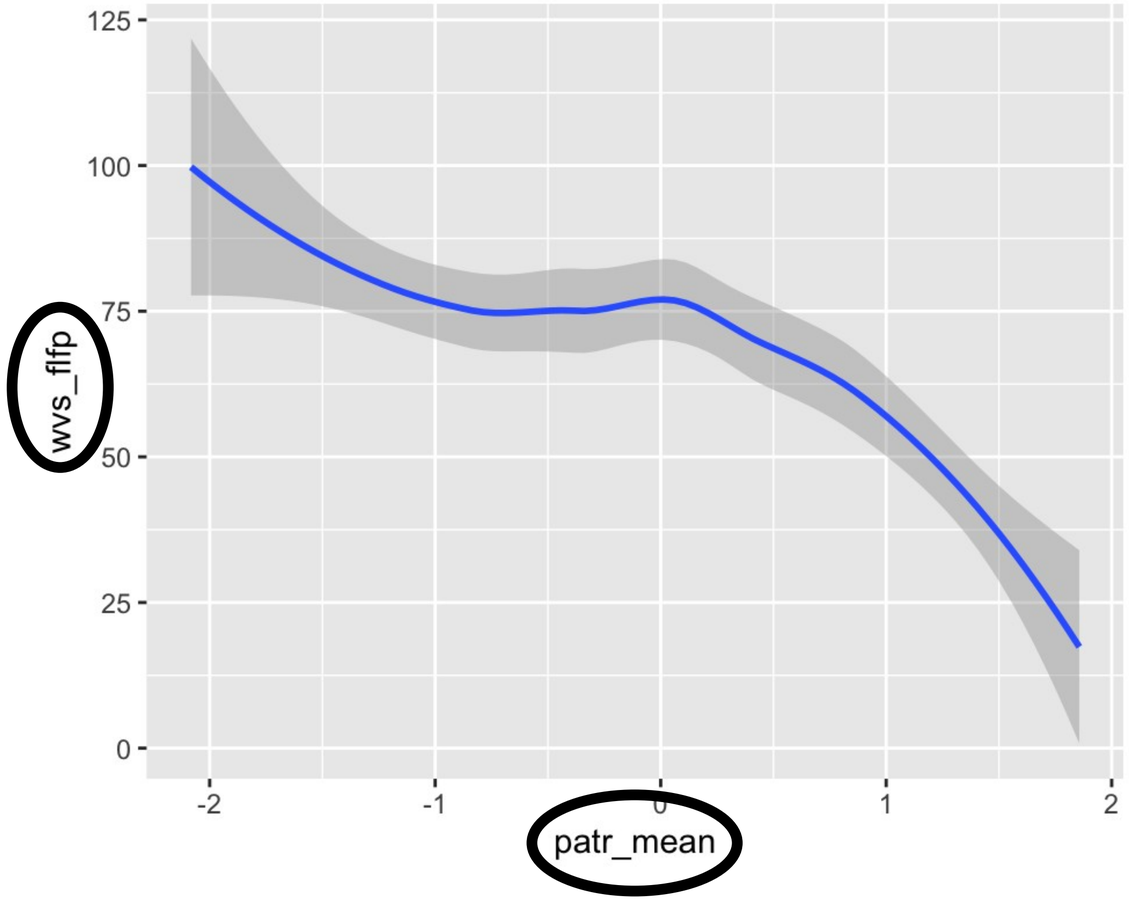




How are these plots different?

Different: geometric shape/object (geom function)



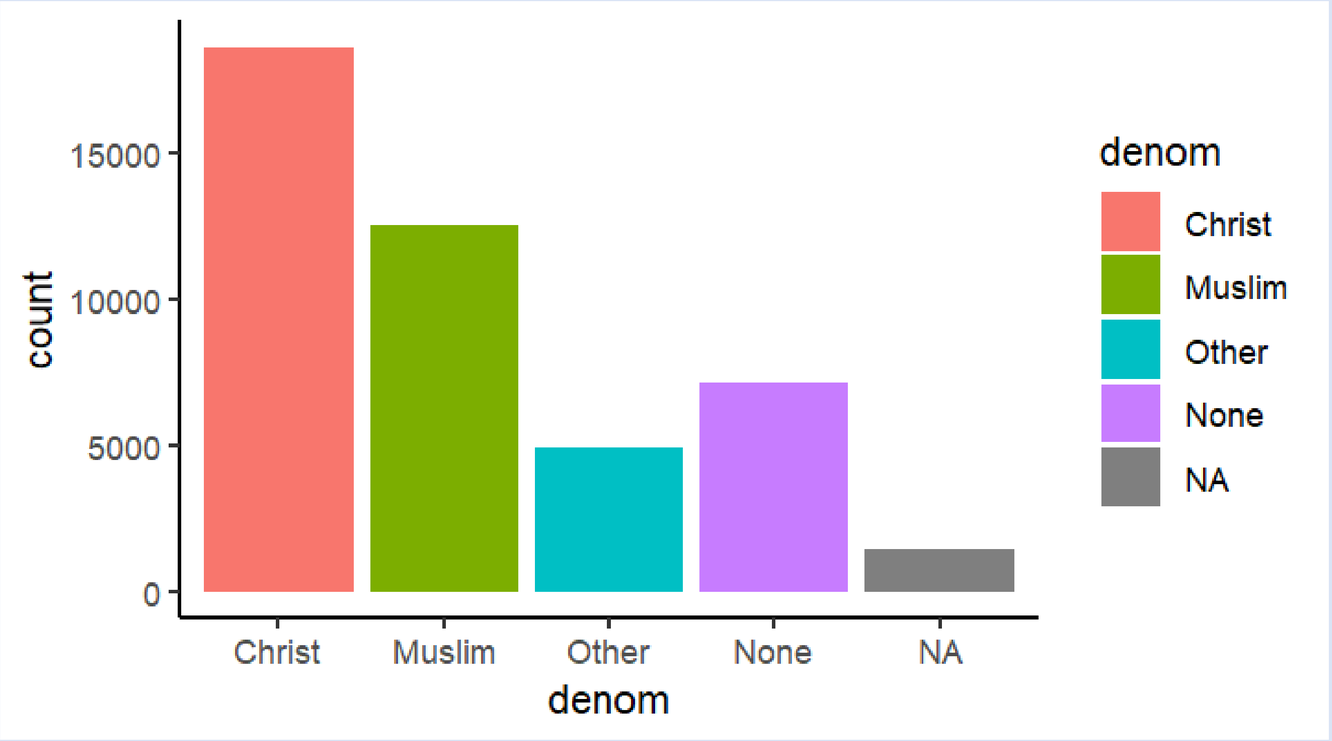


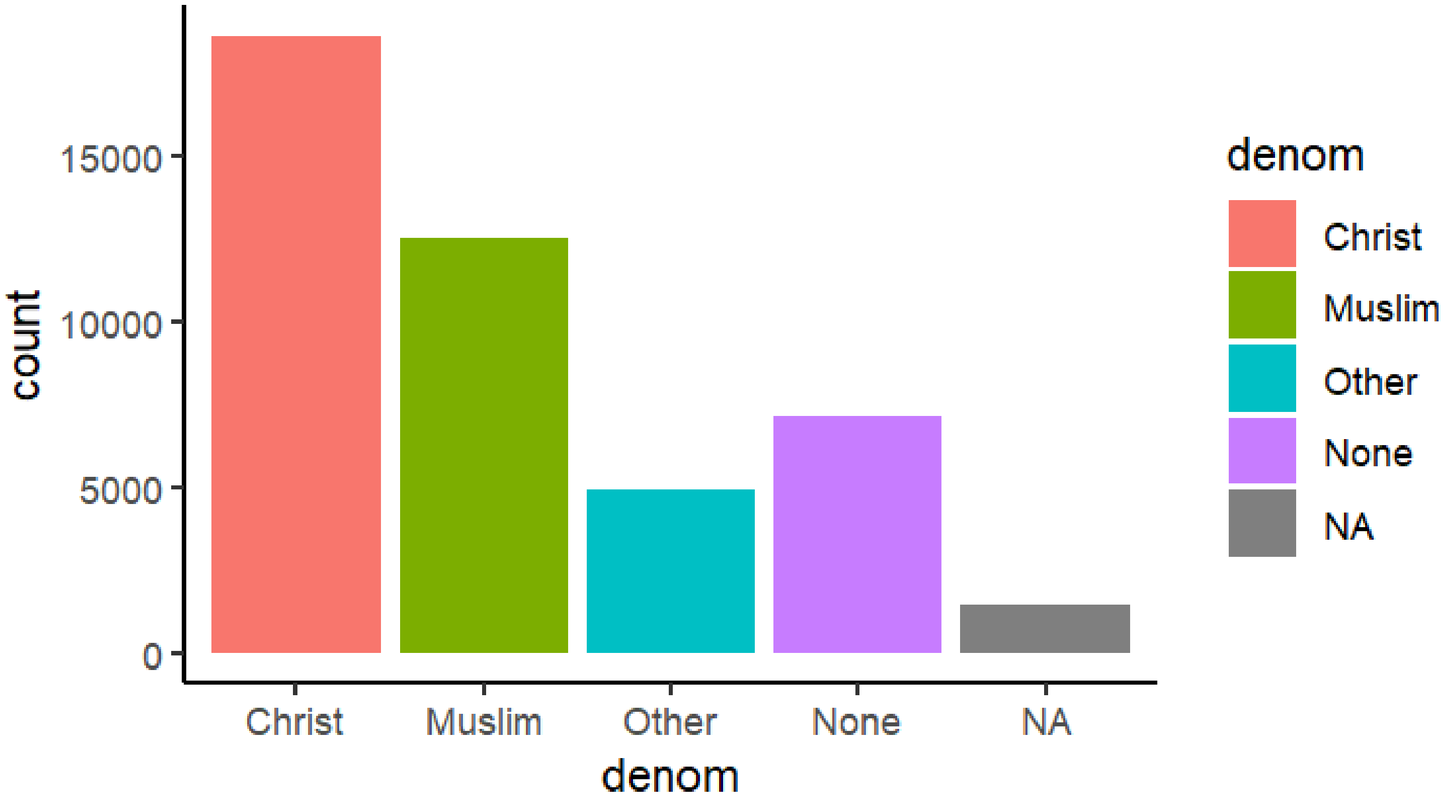
geoms

```
ggplot(data = <DATA>) +
  <GEOM_FUNCTION>(mapping = aes(<MAPPINGS>))
```

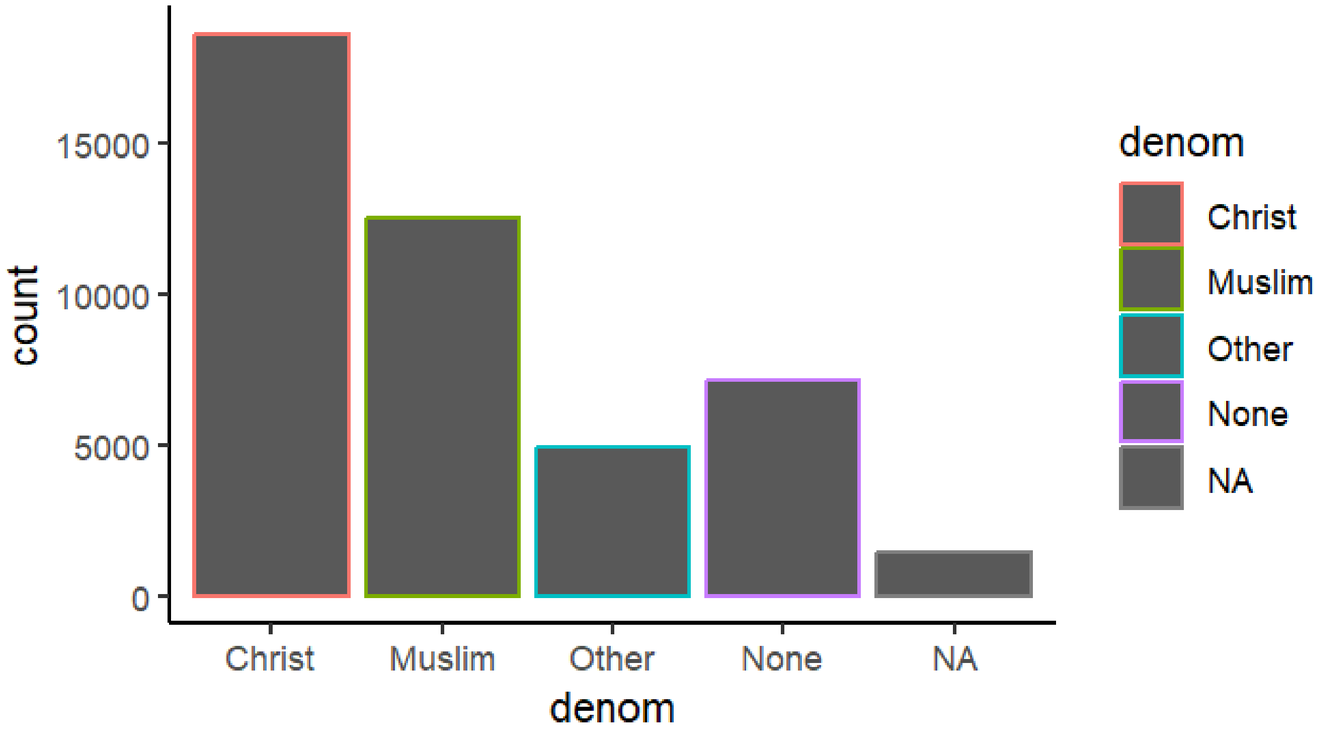
Quiz

- Make the bar-plot of **denom** from the flfp df which you made earlier today, coloured by denom, like below.



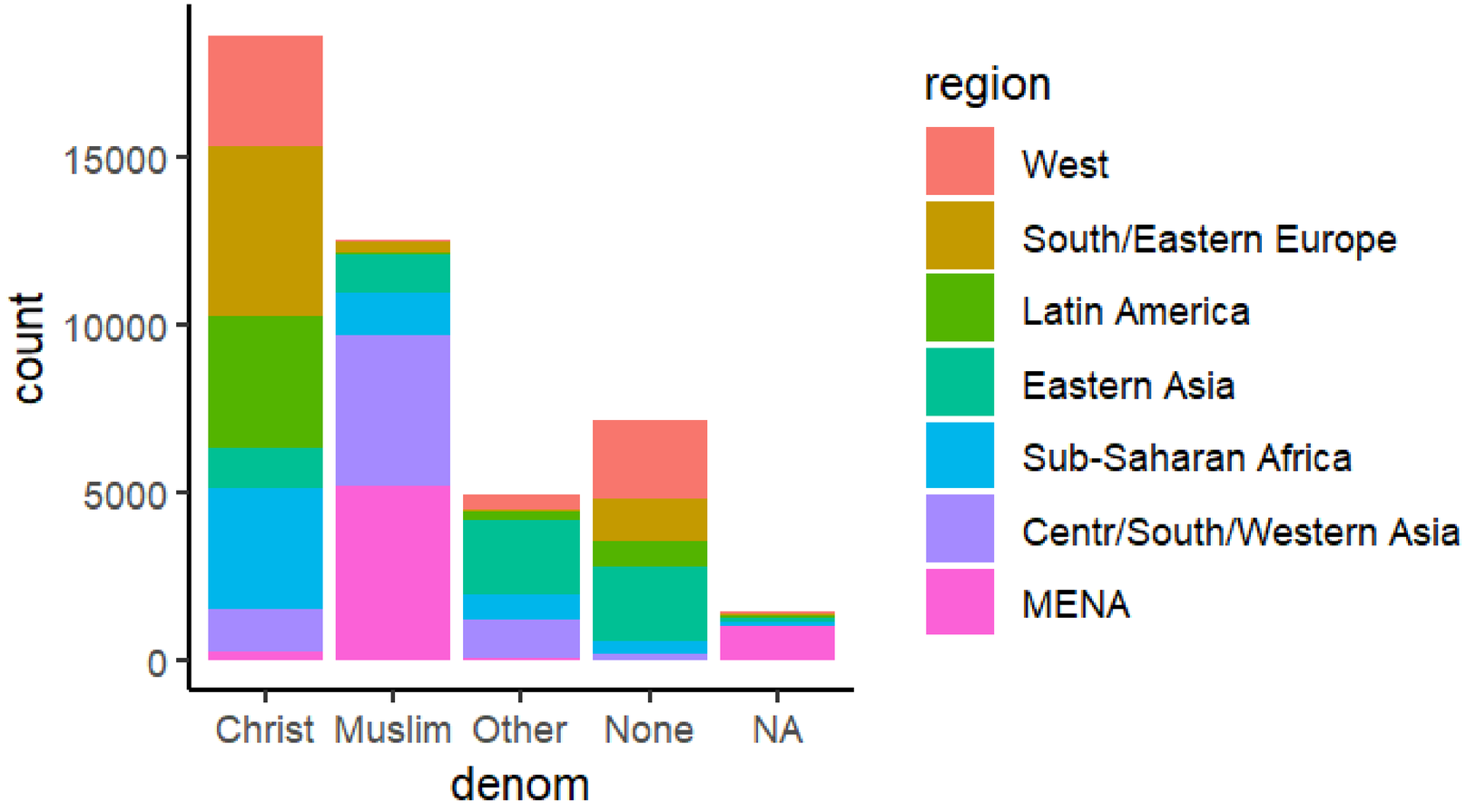



```
ggplot(flfp, aes(denom)) +
  geom_bar(aes(fill=denom))+
  theme_classic()
```




```
ggplot(flfp, aes(denom)) +
  geom_bar(aes(color=denom))+
  theme_classic()
```

How to get this graph?

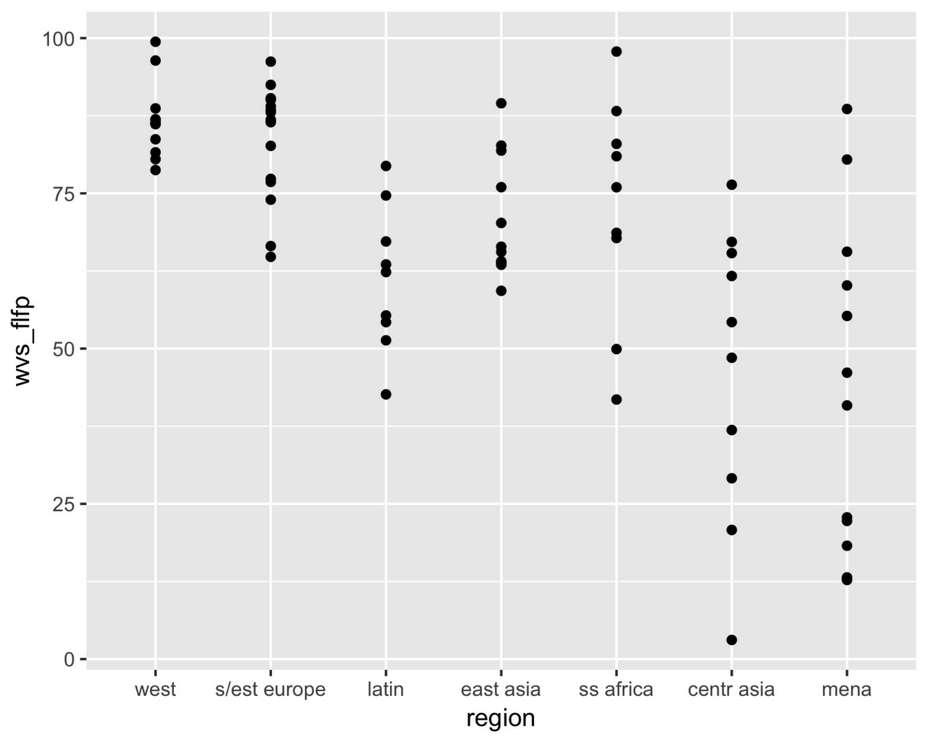


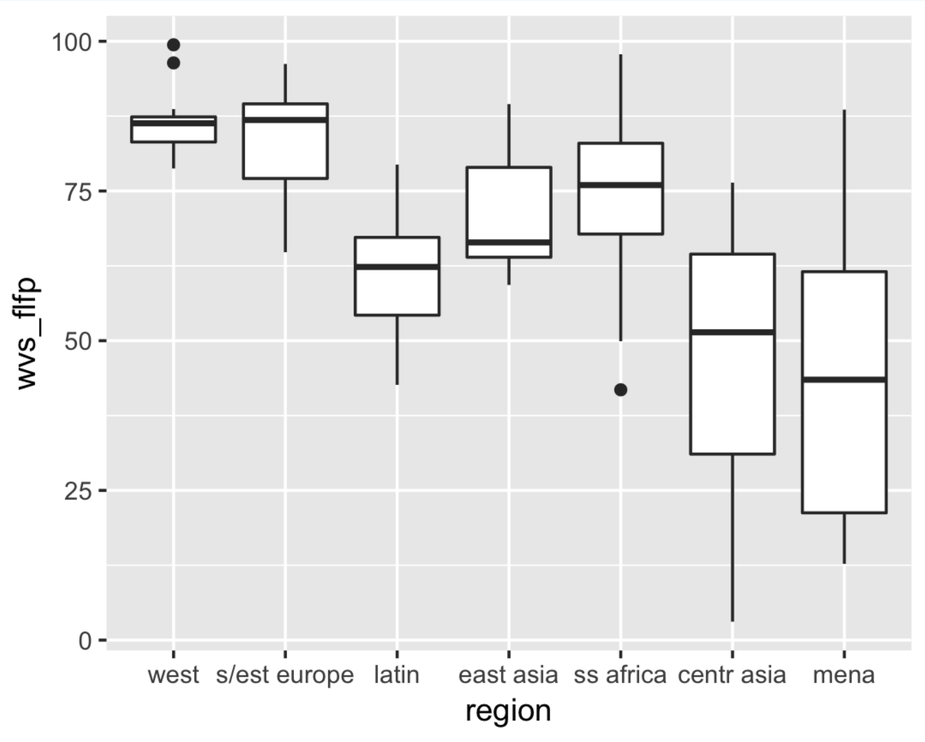

```
ggplot(flfp, aes(x=denom)) +
  geom_bar(aes(fill=region))+
  theme_classic()
```

Boxplots
(aka whisker plots)

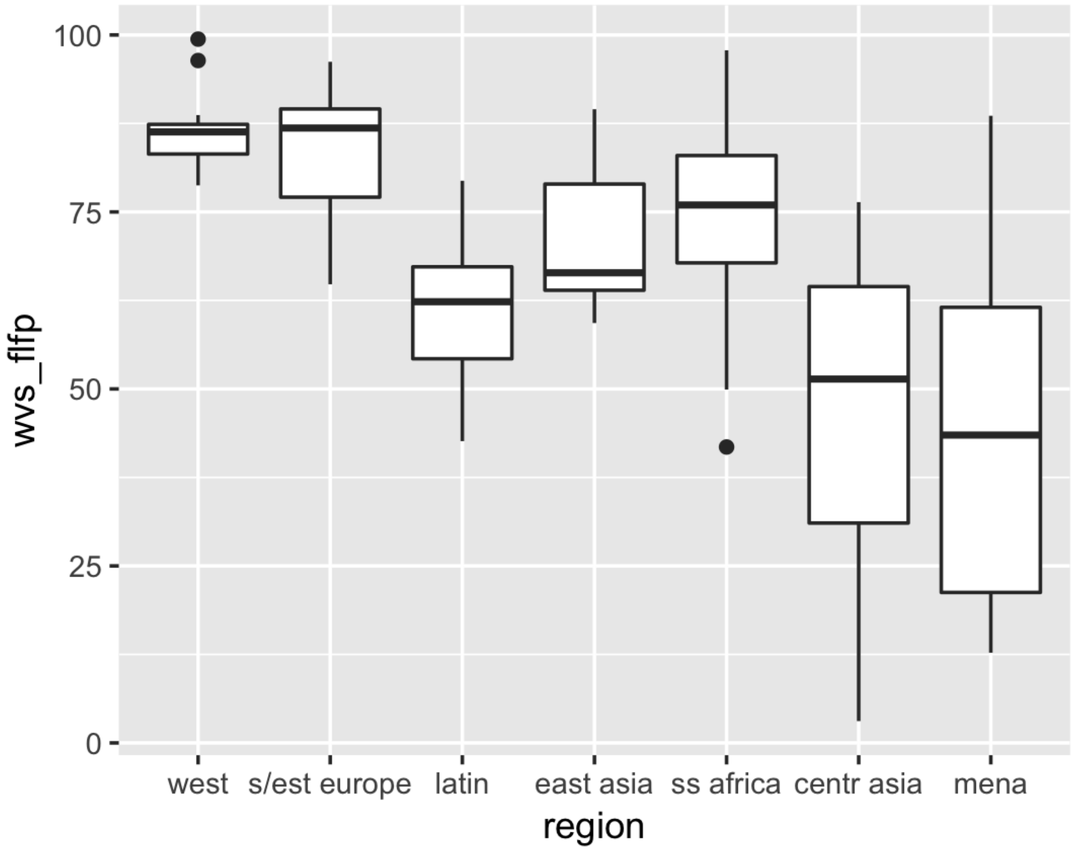
Your turn

- Use the cheatsheet/ google and decide how to replace the scatterplot on the left with a box-plot






```
ggplot(flfp_agg) +
geom_point(aes(x = region, y = wvs_flfp))+
theme_bw()
```




```
ggplot(flfp_agg) +
geom_boxplot(aes(x = region, y = wvs_flfp))+
theme_bw()
```

"Help" pages

To open the documentation for a function, type

```
?geom_histogram
```

?

function name (no parentheses)

geom_freqpoly {ggplot2} R Documentation

Histograms and frequency polygons

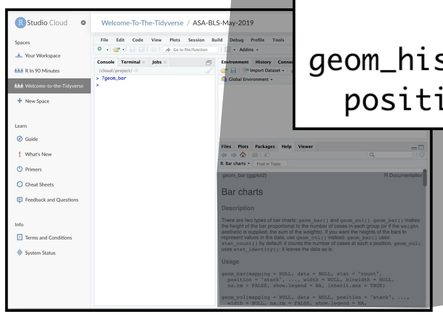
Description

Visualise the distribution of a single continuous variable by dividing the x axis into bins and counting the number of observations in each bin. Histograms (`geom_histogram()`) display the counts with bars; frequency polygons (`geom_freqpoly()`) display the counts with lines. Frequency polygons are more suitable when you want to compare the distribution across the levels of a categorical variable.

Usage

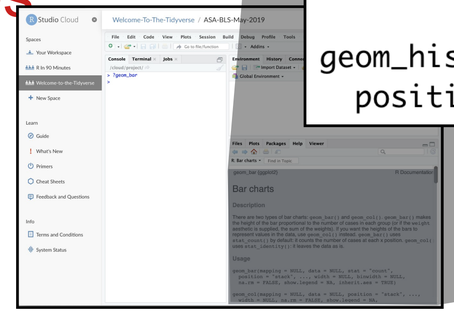
```
geom_freqpoly(mapping = NULL, data = NULL, stat = "bin",
  position = "identity", ..., na.rm = FALSE, show.legend = NA,
  inherit.aes = TRUE)

geom_histogram(mapping = NULL, data = NULL, stat = "bin",
  position = "stack", ..., binwidth = NULL, bins = NULL,
```



Tips

- **scan** page for relevant info, especially the arguments
- **ignore** things that don't make sense and you don’t seem to need
- **try** out the examples



geom_freqpoly {ggplot2} R Documentation

Histograms and frequency polygons

Description

Visualise the distribution of a single continuous variable by dividing the x axis into bins and counting the number of observations in each bin. Histograms (`geom_histogram()`) display the counts with bars; frequency polygons (`geom_freqpoly()`) display the counts with lines. Frequency polygons are more suitable when you want to compare the distribution across the levels of a categorical variable.

Usage

```
geom_freqpoly(mapping = NULL, data = NULL, stat = "bin",
  position = "identity", ..., na.rm = FALSE, show.legend = NA,
  inherit.aes = TRUE)

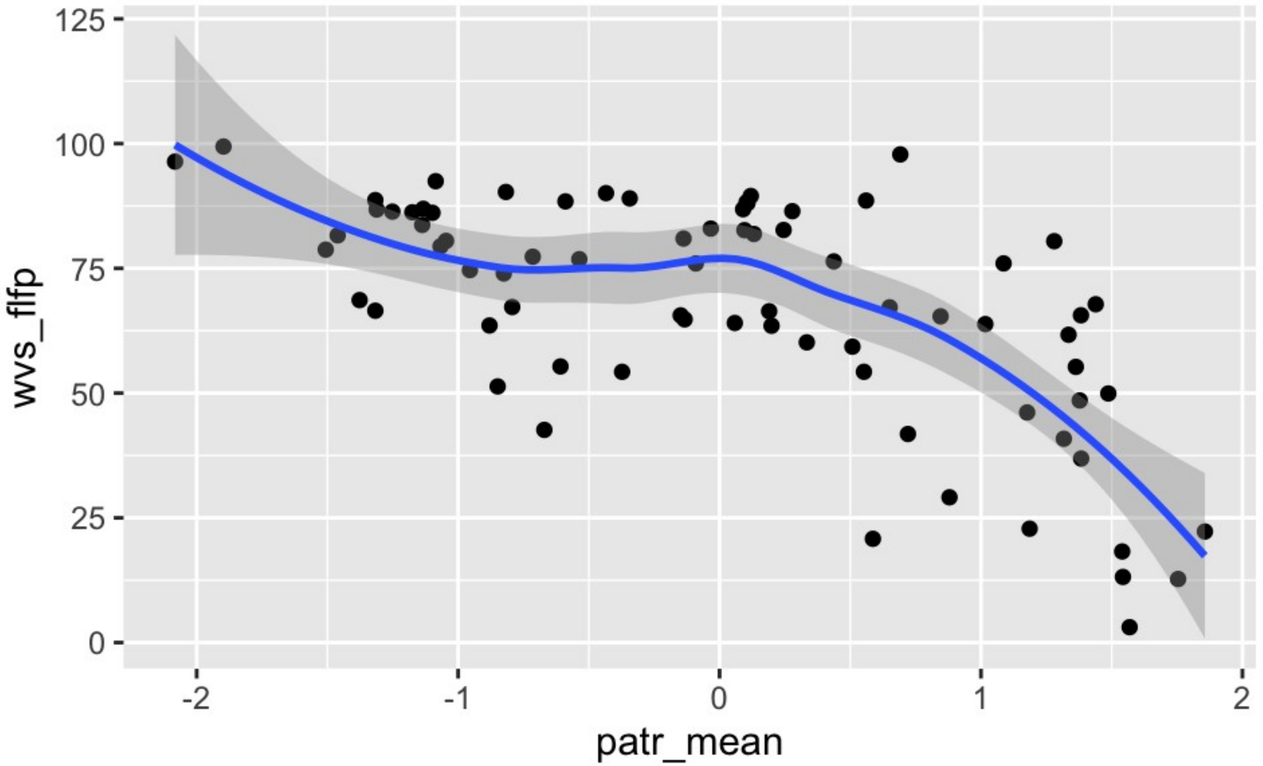
geom_histogram(mapping = NULL, data = NULL, stat = "bin",
  position = "stack", ..., binwidth = NULL, bins = NULL,
```

Your turn

- Try to predict what this code will do. Then run it.

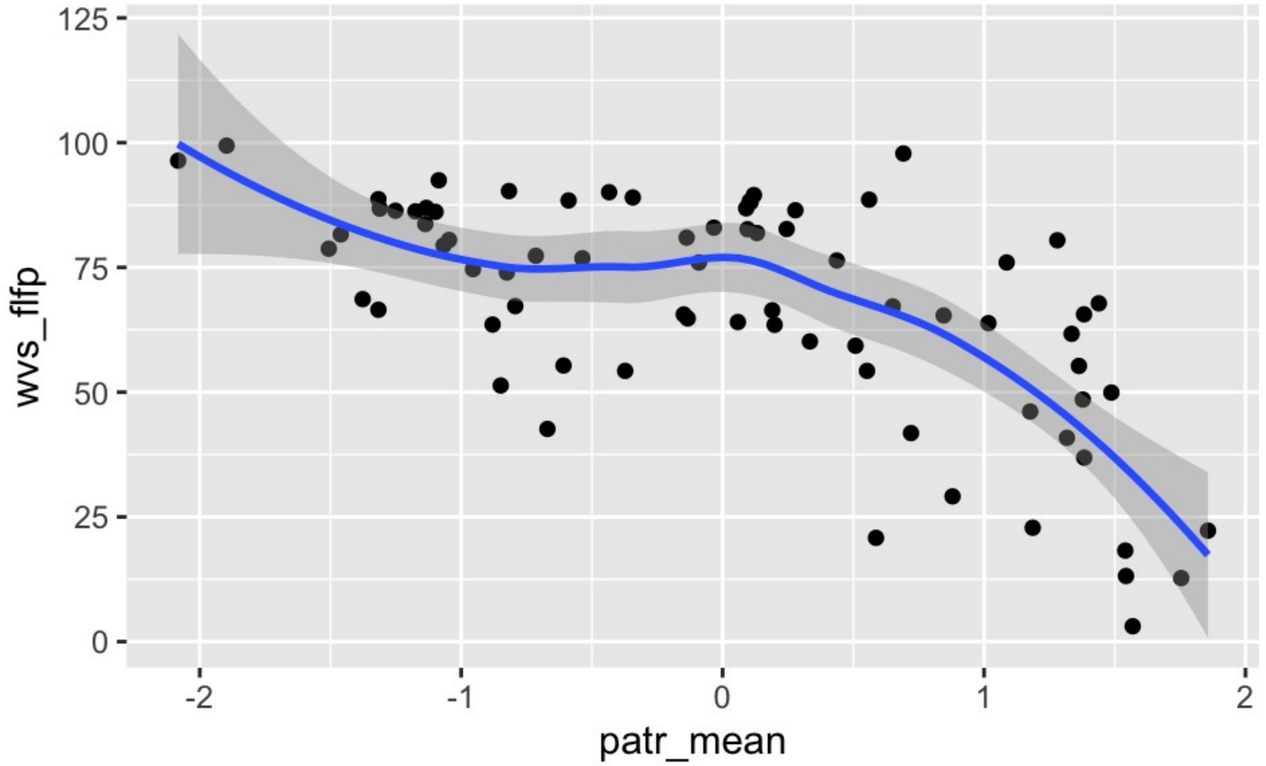
```
ggplot(flfp_agg) +
  geom_point(aes(x = patr_mean, y = wvs_flfp))+
  geom_smooth(aes(x = patr_mean, y = wvs_flfp))+
  theme_bw()
```

Multiple geoms on the same plot




```
ggplot(flfp_agg) +
geom_point(aes(x = patr_mean, y = wvs_flfp)) +
geom_smooth(aes(x = patr_mean, y = wvs_flfp))
```

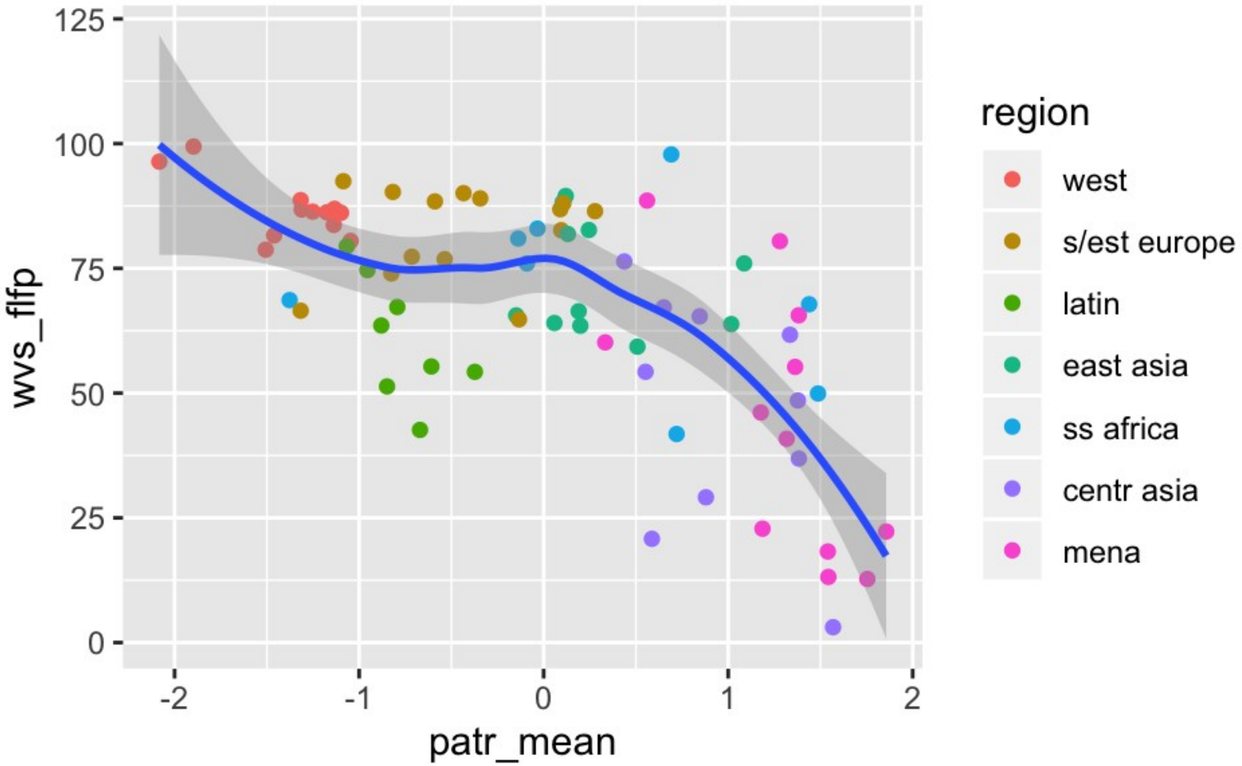
Global mapping



Mappings (and data) that appear in ggplot() will apply globally to every layer

```
ggplot(flfp, aes(x = patr_mean, y = wvs_flfp)) +
      geom_point() +
      geom_smooth()
```

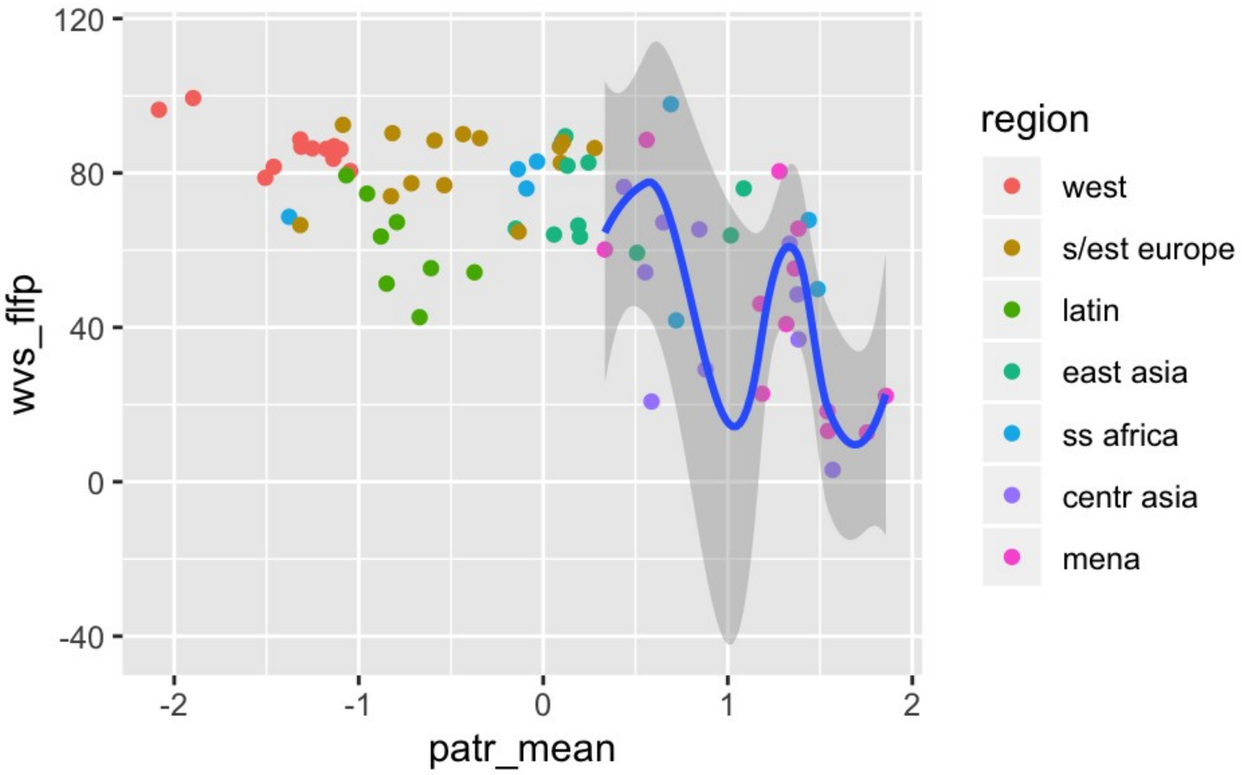
Local mapping



Mappings (and data) that appear in a geom_ function will add to or override the global mappings for that layer only

```
ggplot(flfp_agg, aes(x = patr_mean, y = wvs_flfp)) +
  geom_point(aes(colour = region)) +
  geom_smooth()+
  theme_bw()
```

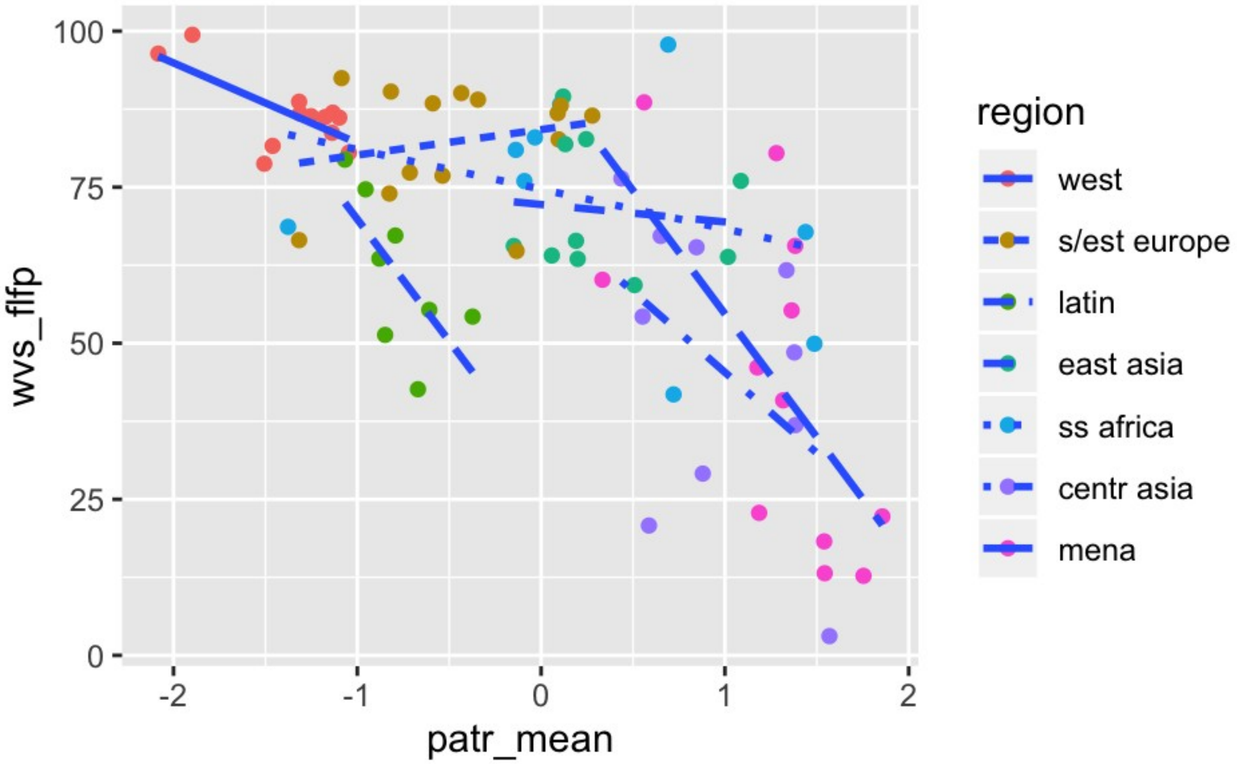
Local mapping



Data can also be set locally by geom

```
ggplot(flfp_agg, aes(x = patr_mean, y = wvs_flfp)) +
  geom_point(aes(colour = region)) +
  geom_smooth(data = filter(flfp_agg, region == "mena"))+
  theme_bw()
```

Different geoms use different aesthetics




```
ggplot(flfp_agg, aes(x = patr_mean, y = wvs_flfp)) +
  geom_point(aes(colour = region)) +
  geom_smooth(aes(linetype = region),se = FALSE, method = "lm")+
  theme_bw()
```

Different geoms use different aesthetics

TWO VARIABLES

continuous x , continuous y

e <- ggplot(mpg, aes(cty, hwy))



e + geom_label(aes(label = cty), nudge_x = 1, nudge_y = 1, check_overlap = TRUE**)** x, y, label, alpha, angle, color, family, fontface, hjust, lineheight, size, vjust



e + geom_jitter(height = 2, width = 2**)**
x, y, alpha, color, fill, shape, size



e + geom_point(), x, y, alpha, color, fill, shape, size, stroke



e + geom_quantile(), x, y, alpha, color, group, linetype, size, weight



e + geom_rug(sides = "bl"**)**, x, y, alpha, color, linetype, size



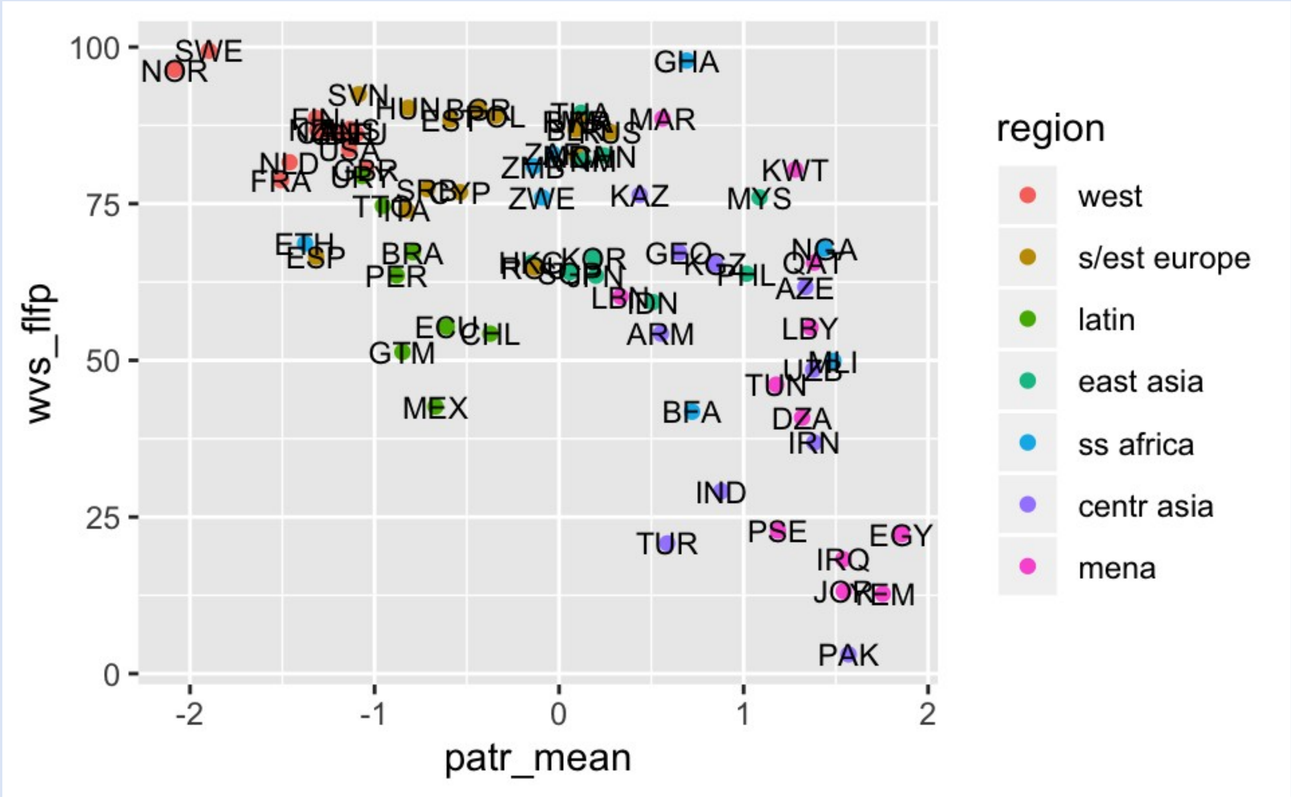
e + geom_smooth(method = lm**)**, x, y, alpha, color, fill, group, linetype, size, weight

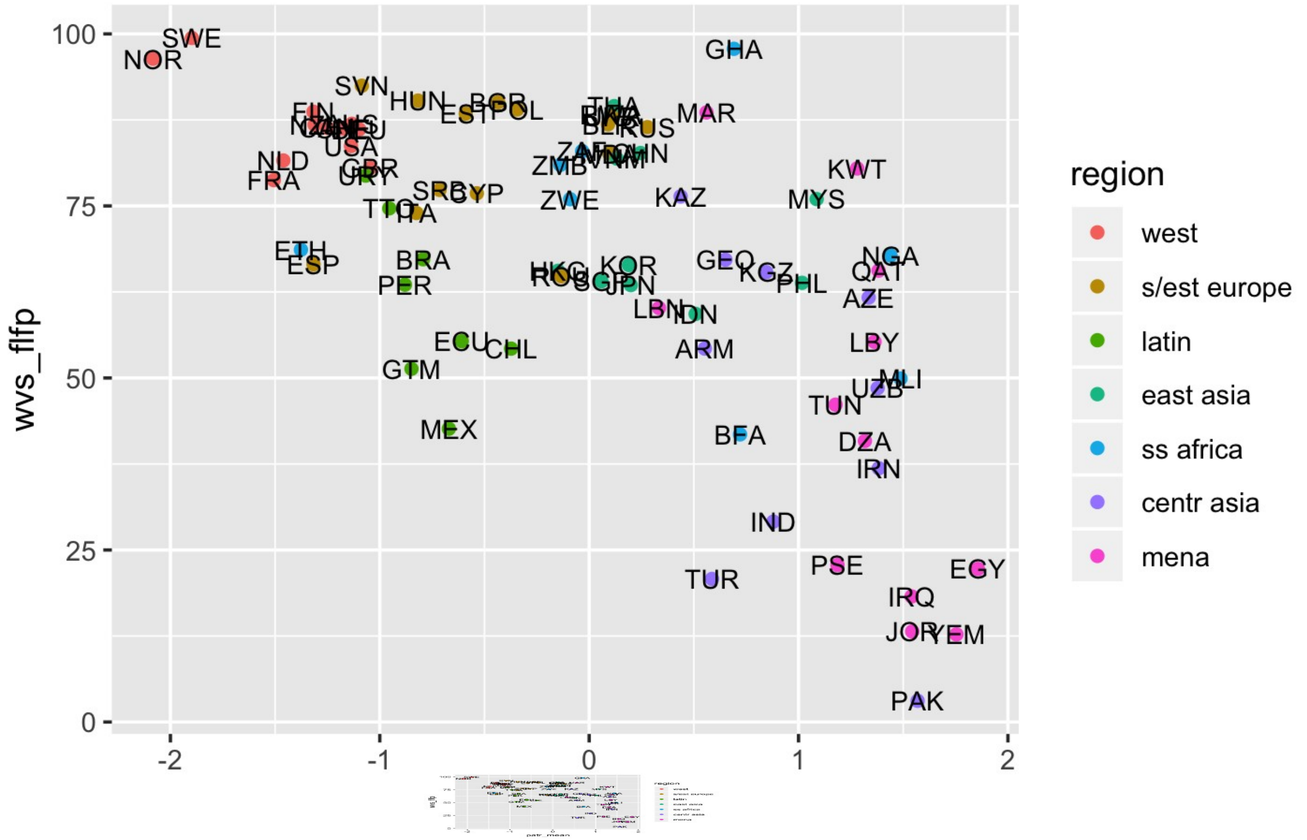


e + geom_text(aes(label = cty), nudge_x = 1, nudge_y = 1, check_overlap = TRUE**)**, x, y, label, alpha, angle, color, family, fontface, hjust, lineheight, size, vjust

Quiz

- Make the scatterplot below with **labels** representing country codes. Use the cheatsheet.



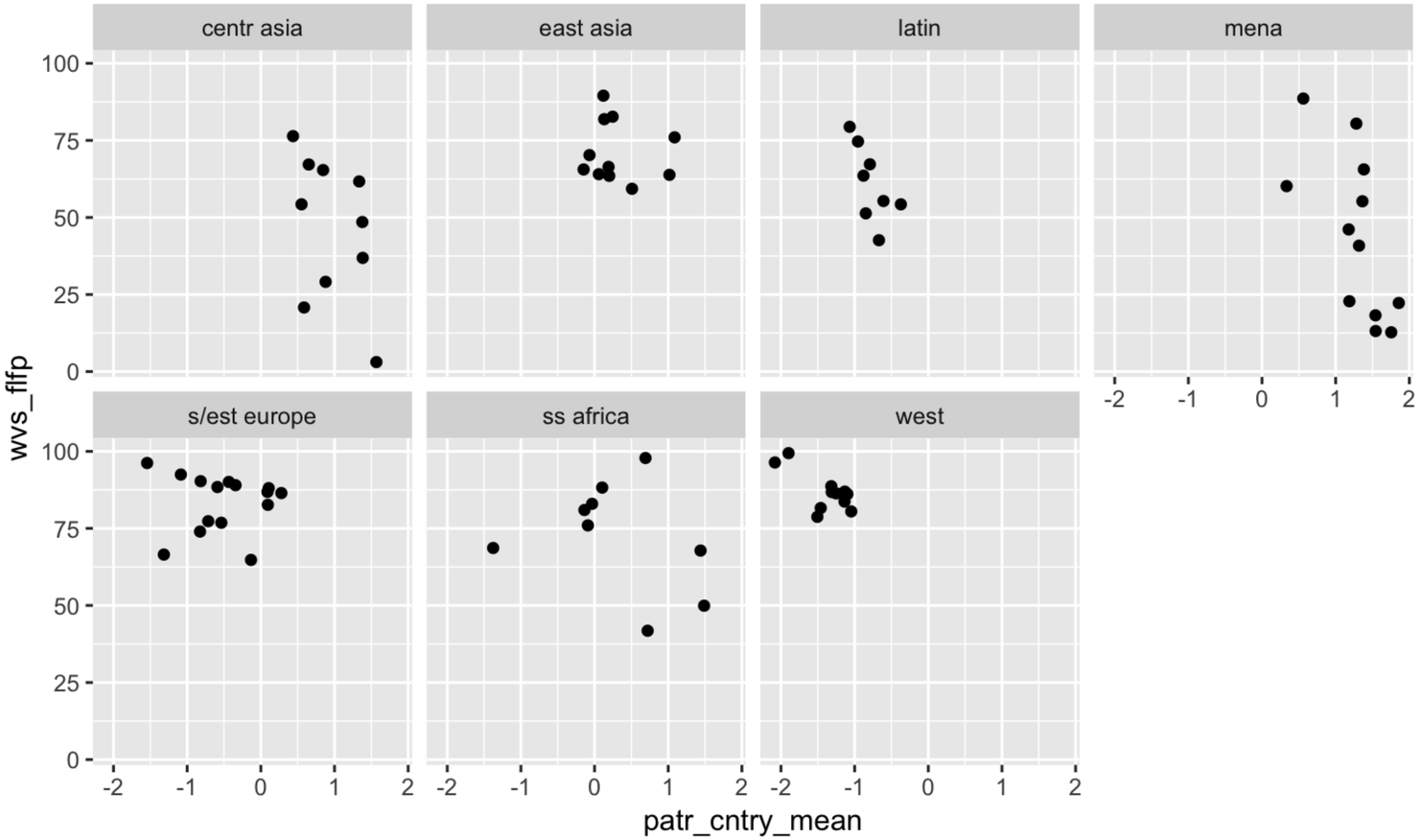


Of course, one can also adjust the labels so they do not cover the points.

```
ggplot(flfp, aes(x = patr_mean, y = wvs_flfp)) +
geom_point(aes(colour = region)) +
geom_text(aes(label = code))
```

What else?

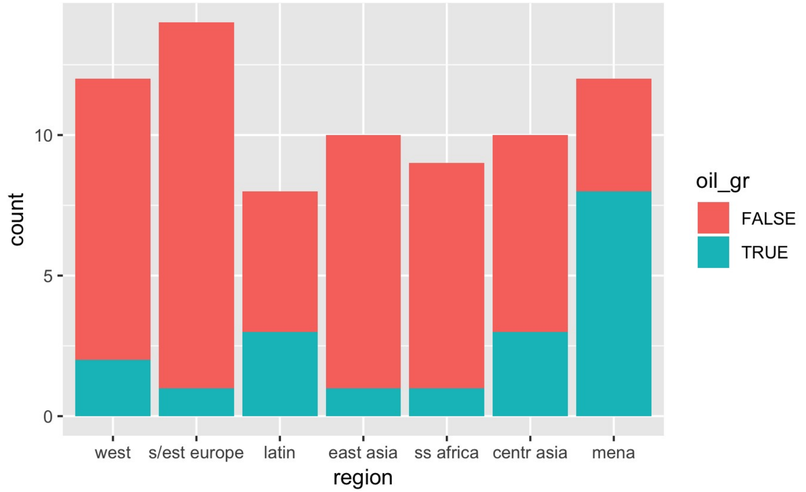
Facets

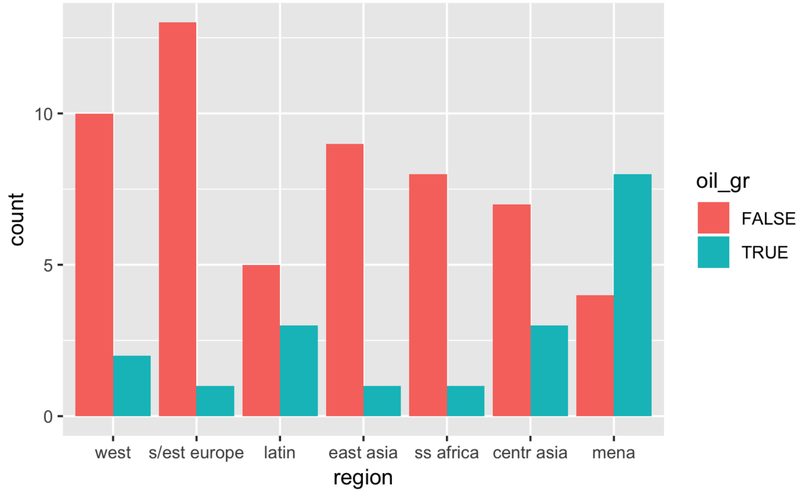



```
ggplot(flfp_agg) +
  geom_point(aes(x = patr_mean, y = wvs_flfp)) +
  facet_wrap(vars(region))+
  theme_bw()
```

Position adjustments

How overlapping objects are arranged






```
ggplot(flfp_agg) +
  geom_bar(aes(x = region,
               fill = oil_gr),
           position = "stack") +
  theme_bw()
```

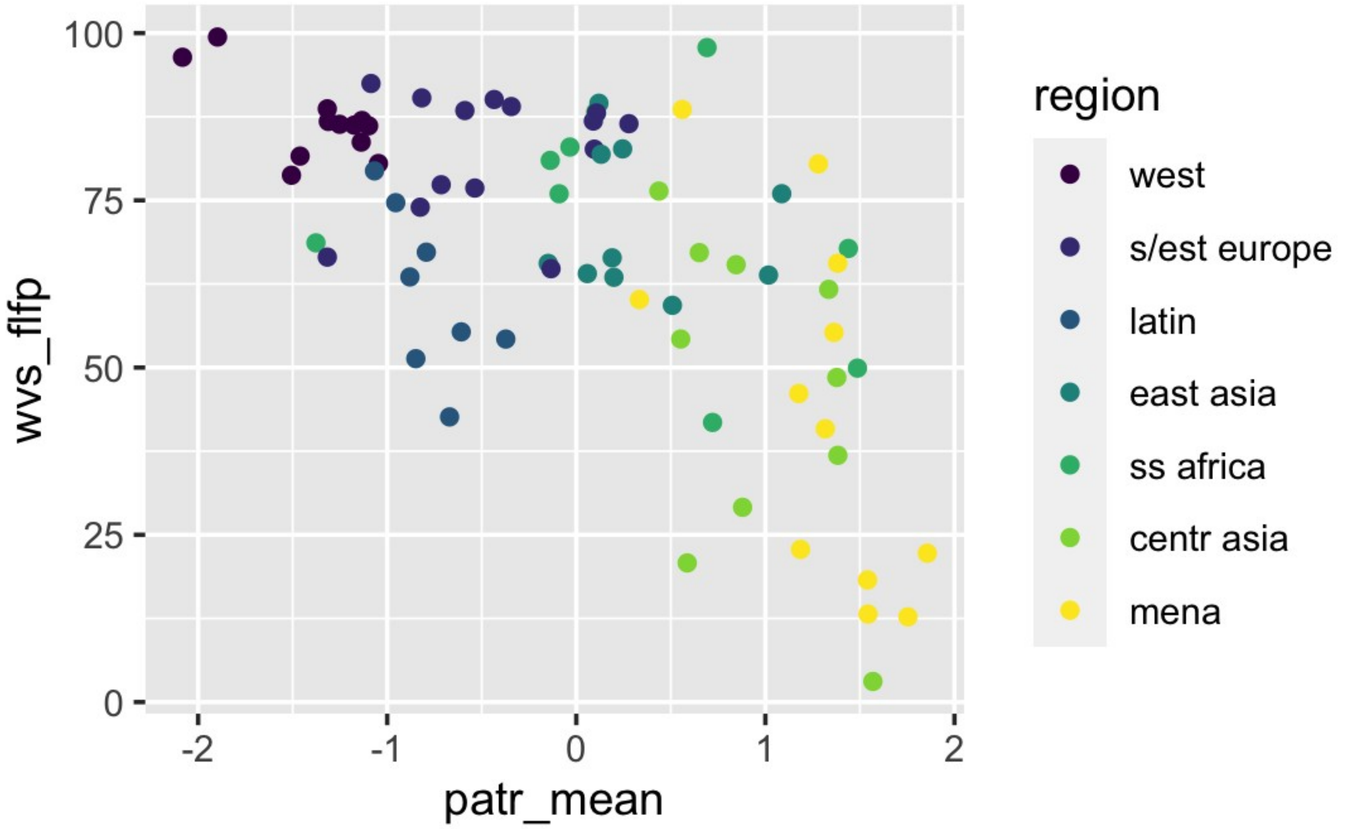
Default position

```
ggplot(flfp) +
   geom_bar(aes(x = region,
                fill = oil_gr),
            position = "dodge")
```

Dodged position

Scales

Customise colour scales, other mappings




```
ggplot(flfp_agg) +
  geom_point(aes(x = patr_mean, y = wvs_flfp, colour = region)) +
  scale_color_viridis_d()+
  theme_bw()
```

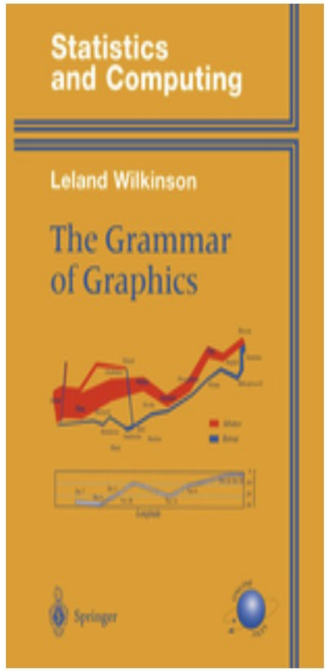
Use the template to make any plot of your choice

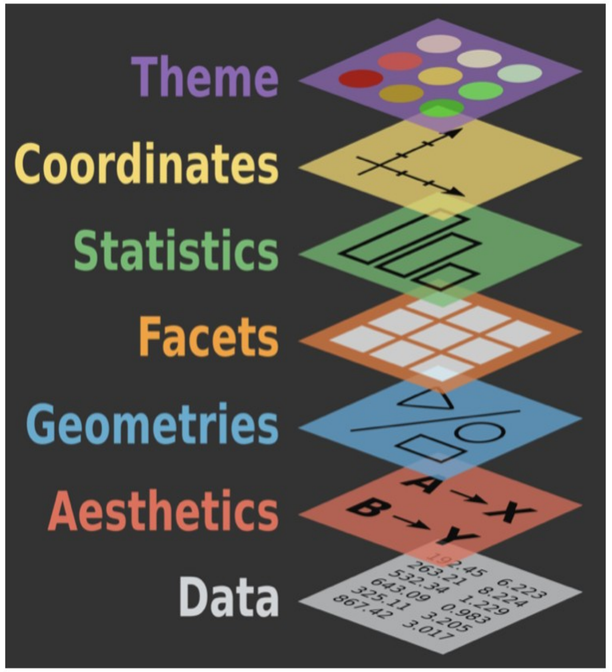
Complete the template below to build a graph.

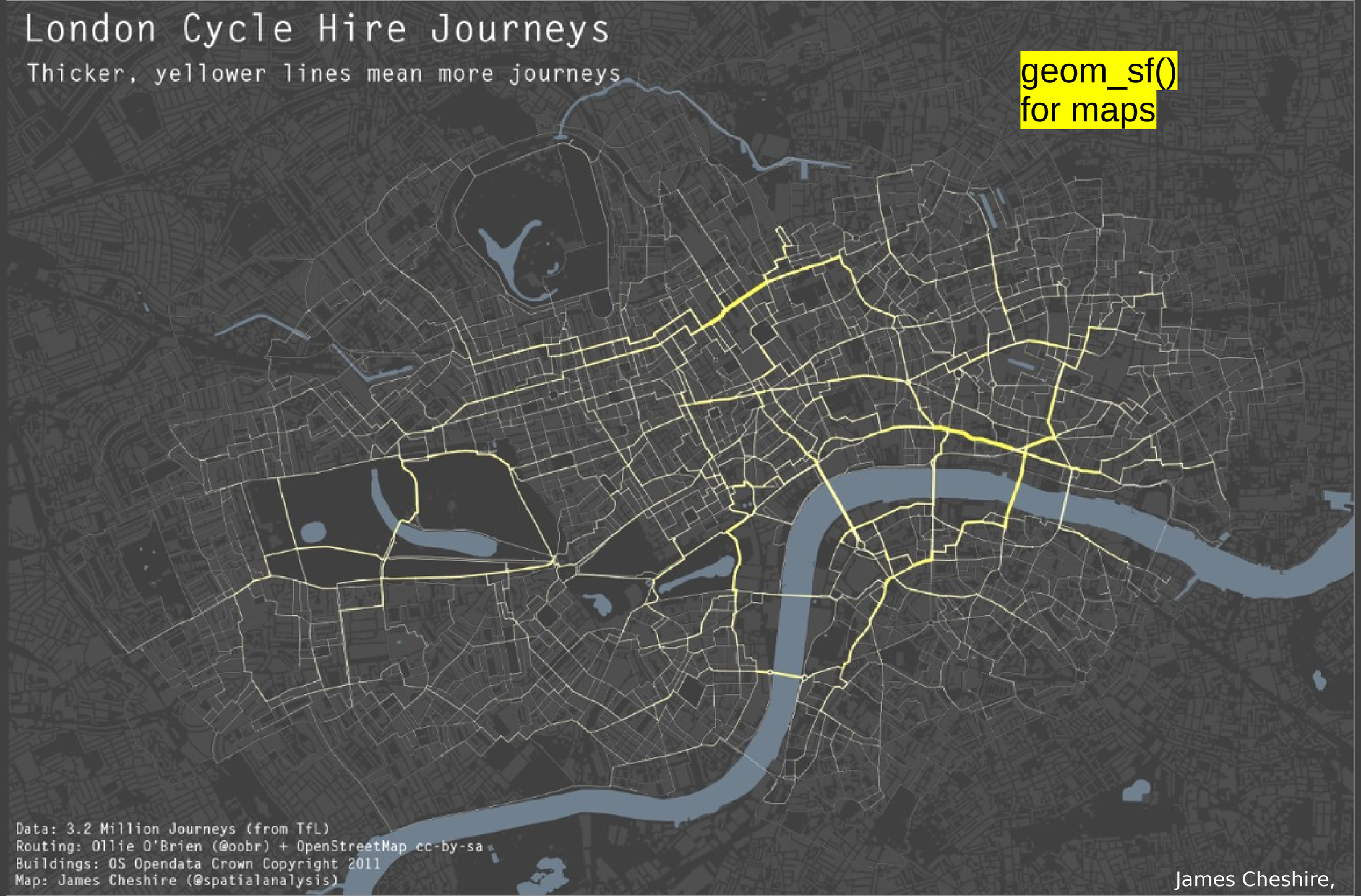
```
ggplot (data = <DATA>) +
<GEOM_FUNCTION>(mapping = aes(<MAPPINGS>),
 stat = <STAT> , position = <POSITION> ) +
<COORDINATE_FUNCTION>+
<FACET_FUNCTION>+
<SCALE_FUNCTION>+
<THEME_FUNCTION>
```

required

Not required, sensible defaults supplied







James Cheshire,

https://ggplot2.tidyverse.org/reference/index.html

Reference

Plot basics

All ggplot2 plots with a call to `ggplot()`, supplying default data and aesthethic mappings, specified by `aes()`. You then add layers, scales, coords and facets with `+`. To save a plot to disk, use `ggsave()`.

`ggplot()`	Create a new ggplot
`aes()`	Construct aesthetic mappings
`` `+`(<gg>) `%+%` ``	Add components to a plot
`ggsave()`	Save a ggplot (or other grid object) with sensible defaults
`qplot()` `quickplot()`	Quick plot

Layer: geoms

A layer combines data, aesthetic mapping, a geom (geometric object), a stat (statistical transformation), and a position

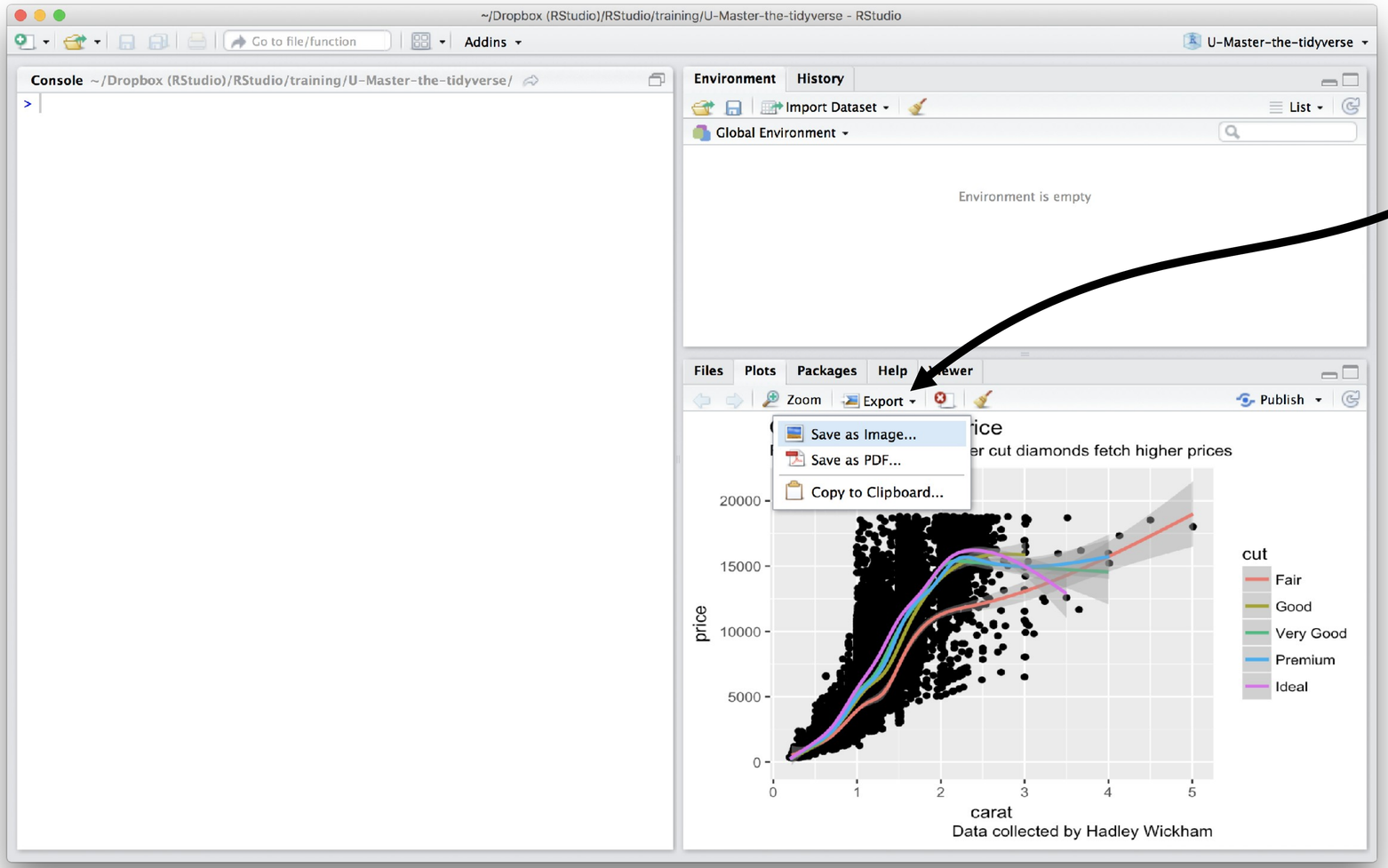
Contents

- Plot basics
- Layer: geoms
- Layer: stats
- Layer: position adjustment
- Layer: annotations
- Aesthetics
- Scales
- Guides: axes and legends
- Facetting
- Facetting: labels
- Coordinate systems
- Themes
- Programming with ggplot2
- Extending ggplot2
- Vector helpers
- Data

Saving graphs

Manually saving plots

Save plots manually with the export menu



Saving plots

ggsave() saves the last plot.
Uses size on screen:

```
ggsave("my-plot.pdf")
ggsave("my-plot.png")
```

Specify size in inches

```
ggsave("my-plot.pdf", width = 6, height = 6)
```

Your Turn

- Use the ggsave() function to save your last plot in a pdf file called "01_patr-values-vs-flfp-scatter.pdf".

```
ggsave("01_patr-values-vs-flfp-scatter.pdf")
```

I generally have way better results using ”zoom” and then saving manually compared to this function

Occasionally, I have had to do a screen shot to get the right look.

We can also make graphics interactive

Plotly demonstration
(in R script)

Your turn

- Run the code below.

```
pl <- ggplot(flfp_agg, aes(x = patr_mean,
                           y = wvs_flfp,
                          colour = region,
                          label = cntry)) +
                           geom_point()


library(plotly)
ggplotly(pl)
```

gganimate demonstration

- Line 293 in Day 4 code

Your turn

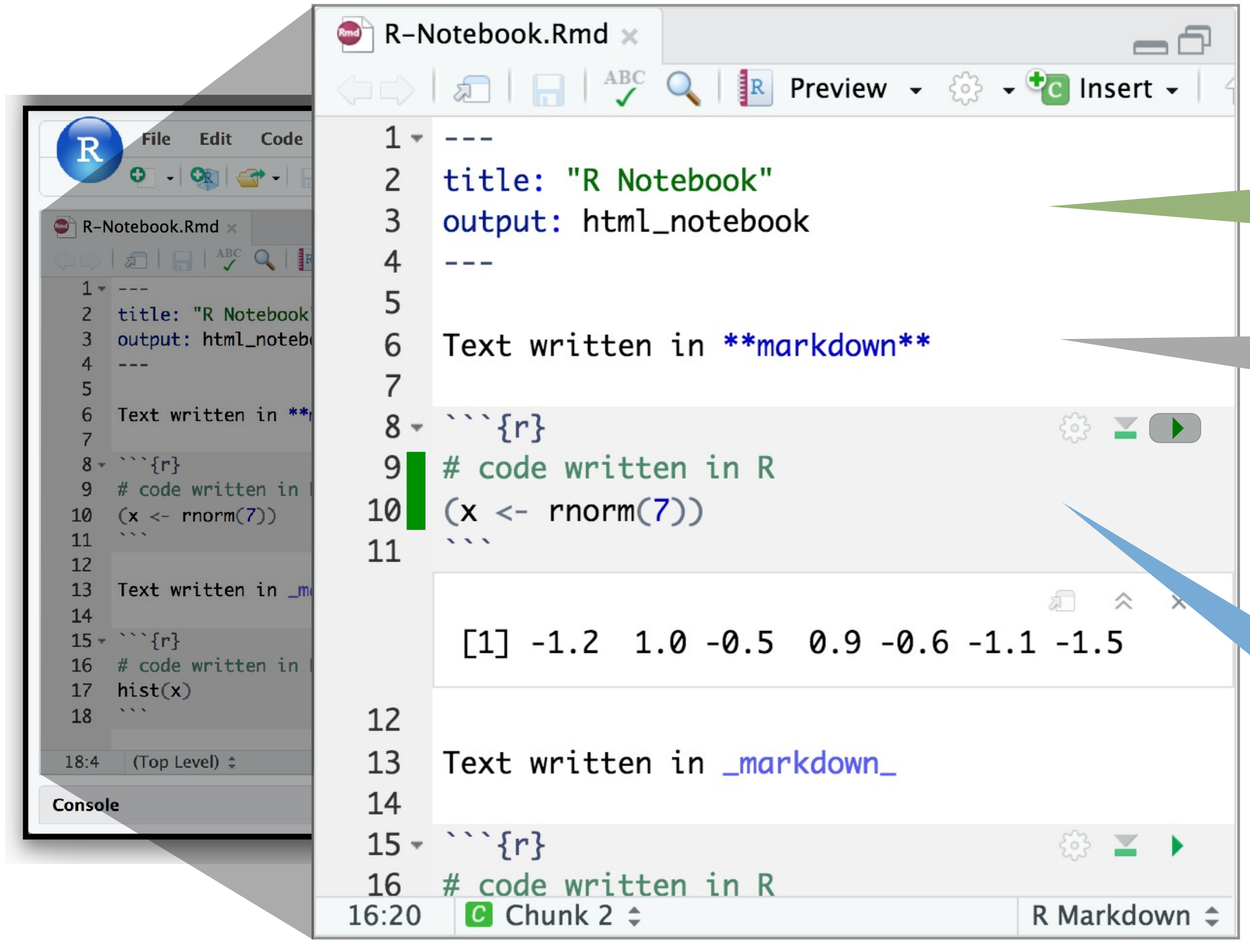
- There is a geom_smooth() argument commented out in the code i have given you. What will this line do if added? Remove the pound signs (don’t forget the plus sign above) and re-run the code.
- What is revealed by the slopes here?

rmarkdown


rmarkdown
www.rstudio.com

R Markdown

Plain text file with 3 types of content:



A header surrounded by — — —

Text in markdown

Code chunks surrounded by ```

Markdown language

```
Heading
=======

## Sub-heading

Paragraphs are separated
by a blank line.

Two spaces at the end of a
line
produces a line break.

Text attributes _italic_,
**bold**, `monospace`.

Horizontal rule:

---

Bullet list:

  * apples
  * oranges
  * pears

Numbered list:

  1. lather
  2. rinse
  3. repeat
```

```
<h1>Heading</h1>

<h2>Sub-heading</h2>

<p>Paragraphs are separated
by a blank line.</p>

<p>Two spaces at the end of a line<br
/>
produces a line break.</p>

<p>Text attributes <em>italic</em>,
<strong>bold</strong>,
<code>monospace</code>.</p>

<p>Horizontal rule:</p>

<hr />

<p>Bullet list:</p>

<ul>
<li>apples</li>
<li>oranges</li>
<li>pears</li>
</ul>

<p>Numbered list:</p>

<ol>
<li>lather</li>
```

Heading

Sub-heading

Paragraphs are separated by a blank line.

Two spaces at the end of a line
produces a line break.

Text attributes *italic*, **bold**, `monospace`.

Horizontal rule:

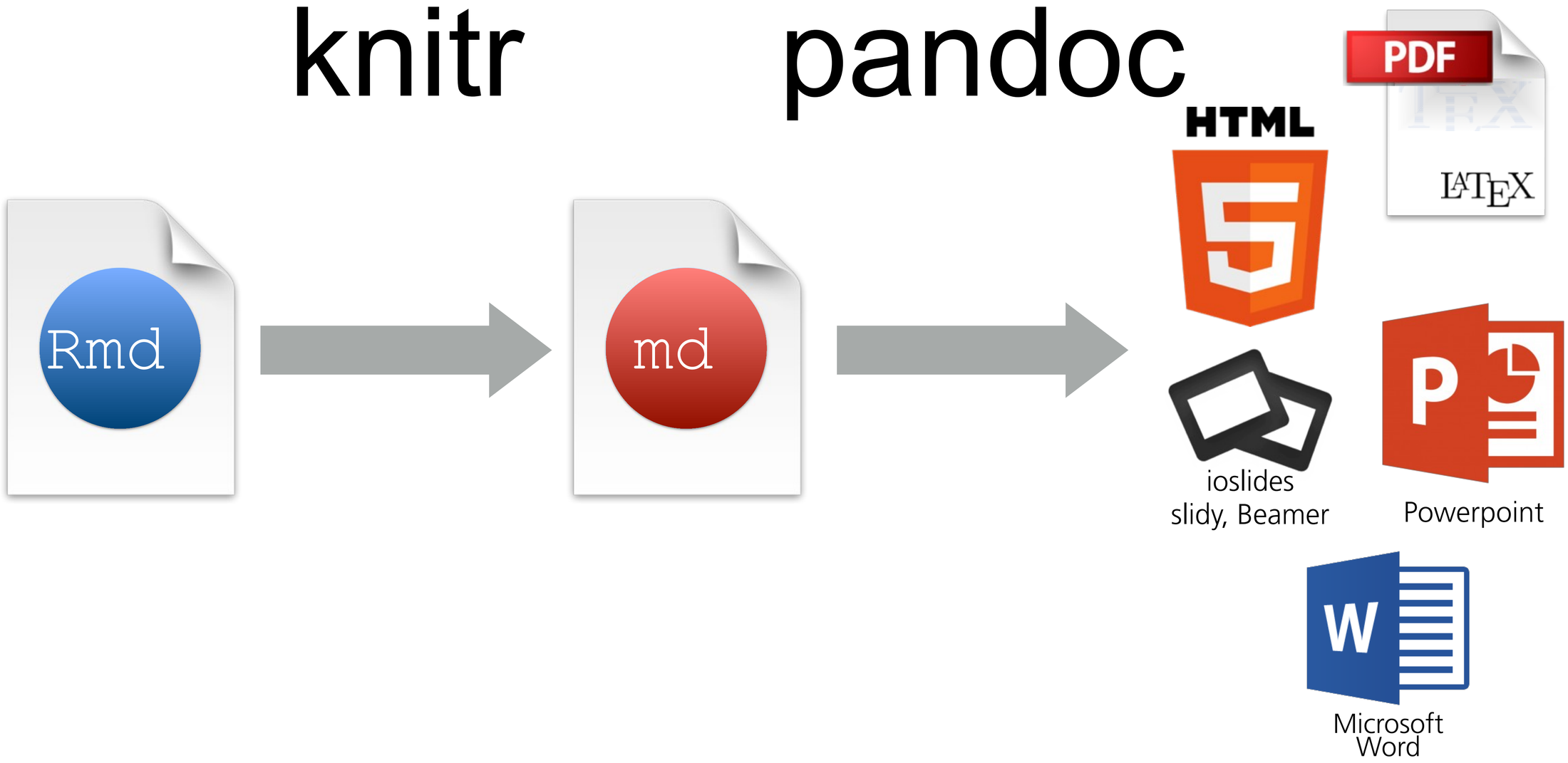
Bullet list:

- apples
- oranges
- pears

Numbered list:

1. lather
2. rinse
3. repeat

An example.


knitr
pandoc
Rmd
md
HTML
PDF
LaTeX
ioslides
slidy, Beamer
Powerpoint
Microsoft
Word

Logistics

1. Knitr runs the document in a fresh R session, which means you need to load the libraries that the document uses *in the document*

CC BY-SA RStudio

Logistics

1. Knitr runs the document in a fresh R session, which means you need to load the libraries that the document uses *in the document*

2. Objects made in one code chunk will be available to code in later code chunks.

CC BY-SA RStudio

knitr is multi-lingual!

- ☑ SAS
- ☑ python
- ☑ more

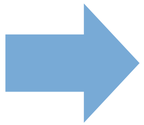
CC BY-SA RStudio

python

To embed non-R code, change the chunk label from r to the language to use.

````
Some python code,

```{python}
x = 'hello, python world!'
print(x)
print(x.split(' '))
```
````



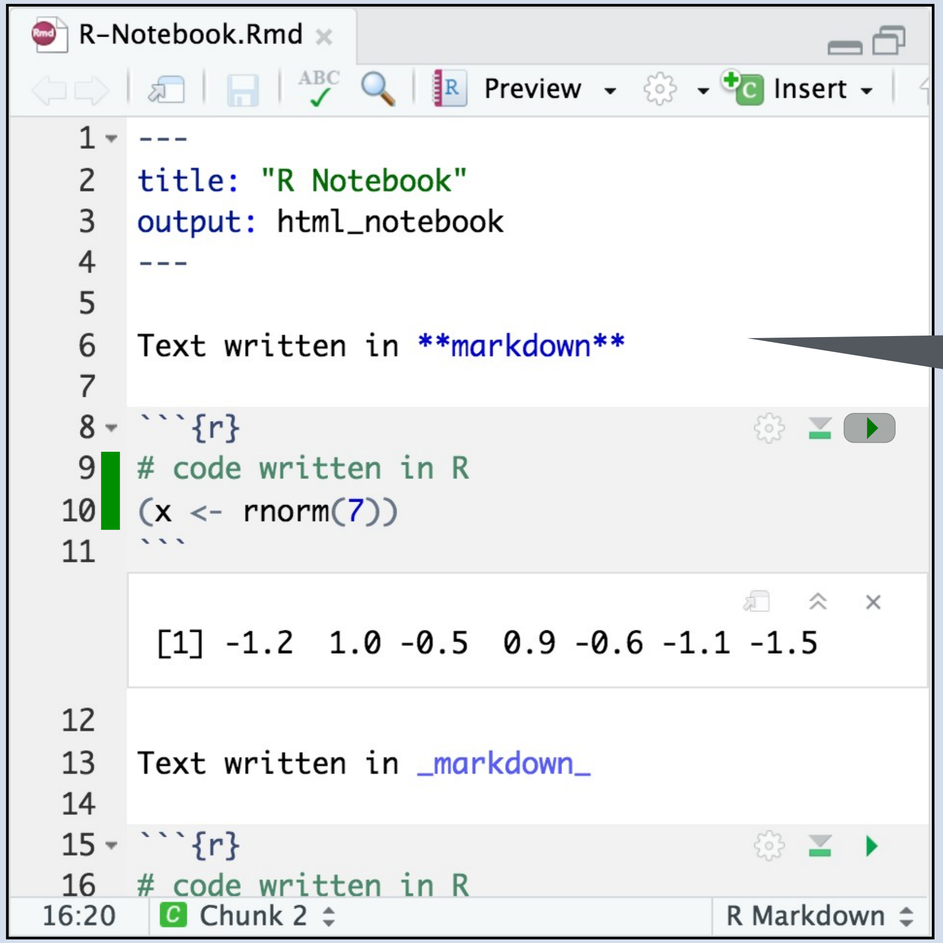
Some python code:

```
x = 'hello, python world!'
print(x)
print(x.split(' '))
```

```
## hello, python world!
## ['hello,', 'python', 'world!']
```

CC BY-SA RStudio

Markdown

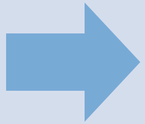


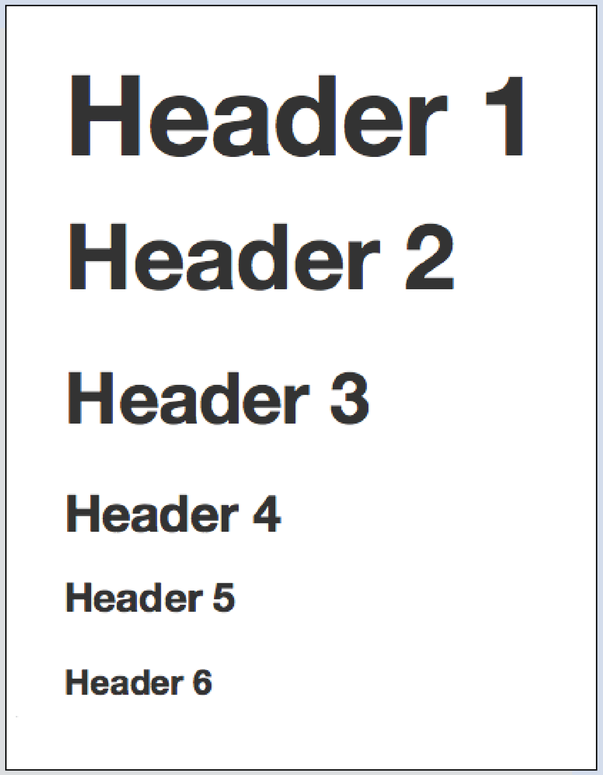
Text in markdown

Headers

Use # to create headers.
Multiple #’s create lower level headers.

```
# Header 1
## Header 2
### Header 3
#### Header 4
##### Header 5
###### Header 6
```



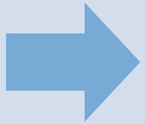


Text

Add two spaces
at the end of a line
to start a new line

Text is rendered as plain text. Surround text with _, **, or ` to format it.

```
Text
_italics_
**bold**
`code`
```



Text

italics

bold

`code`

Lists

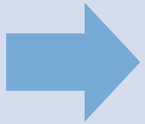
Use asterisks to make bullet points.
Use numbers to make numbered lists.

```
Bullets

* bullet 1
* bullet 2

Numbered list

1. item 1
2. item 2
```



Bullets

- bullet 1
- bullet 2

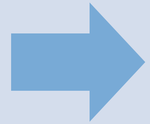
Numbered list

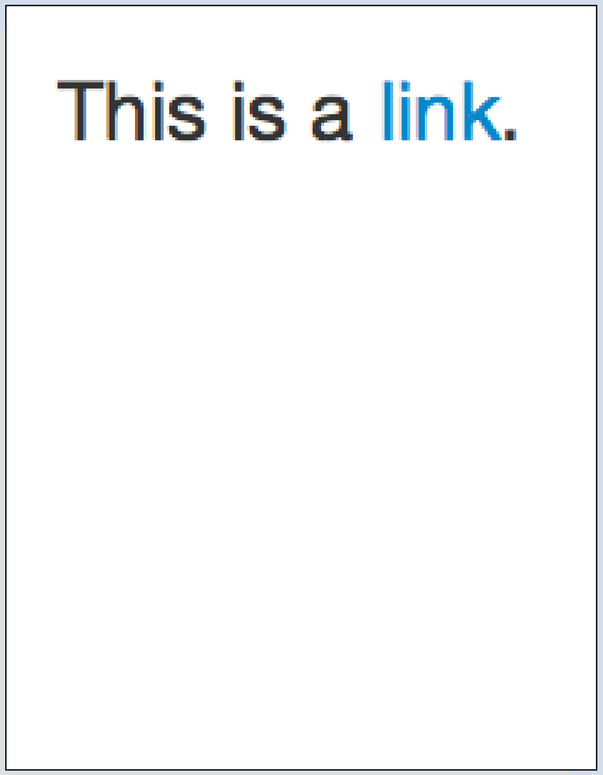
1. item 1
2. item 2

Hyperlinks

Use brackets to denote a link.
Place the URL in parentheses.

```
This is a
[link](www.git.com).
```





Images

Use a link preceded by an ! to insert an image.

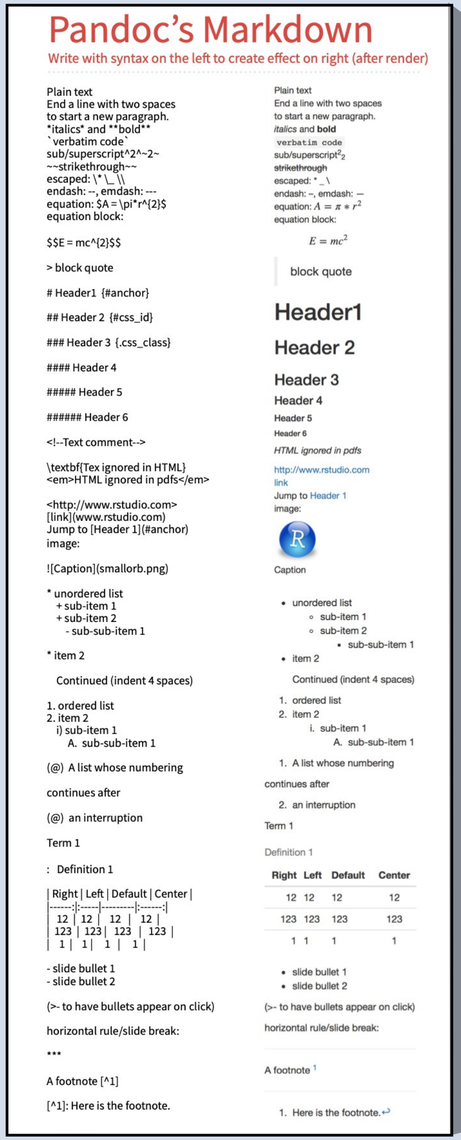
The link text should be a URL (if the image is hosted online), or a file path (if the image is saved as a file)

```
![](images/1.png)
The RStudio logo.
```



Markdown

Dictionary of formatting cues.



Pandoc's Markdown

Write with syntax on the left to create effect on right (after render)

```
Plain text
End a line with two spaces
to start a new paragraph.
*italics* and **bold**
`verbatim code`
sub/superscript^2^~2~
~~strikethrough~~
escaped: \* \_ \\
endash: --, emdash: ---
equation: $A = \pi*r^{2}$
equation block:

$$E = mc^{2}$$

> block quote

# Header1  {#anchor}

## Header 2  {#css_id}

### Header 3  {.css_class}

#### Header 4

##### Header 5

###### Header 6

<!--Text comment-->

\textbf{Tex ignored in HTML}
<em>HTML ignored in pdfs</em>

<http://www.rstudio.com>
[link](www.rstudio.com)
Jump to [Header 1](#anchor)
image:

![Caption](smallorb.png)

* unordered list
  + sub-item 1
  + sub-item 2
    - sub-sub-item 1

* item 2

    Continued (indent 4 spaces)

1. ordered list
2. item 2
   i) sub-item 1
      A.  sub-sub-item 1

(@)  A list whose numbering

continues after

(@)  an interruption

Term 1

:   Definition 1

| Right | Left | Default | Center |
|------:|:-----|---------|:------:|
|  12 |  12 |  12  |  12 |
| 123 | 123 | 123  | 123 |
|   1 |   1 |   1  |   1 |

- slide bullet 1
- slide bullet 2

(>- to have bullets appear on click)

horizontal rule/slide break:

***

A footnote [^1]

[^1]: Here is the footnote.
```

Plain text
End a line with two spaces
to start a new paragraph.
italics and **bold**
`verbatim code`
sub/superscript^2~2~
~~strikethrough~~
escaped: * _ \
endash: –, emdash: —
equation: $A = \pi * r^2$
equation block:

$$E = mc^2$$

> block quote

Header1

Header 2

Header 3

Header 4

Header 5

Header 6

HTML ignored in pdfs

http://www.rstudio.com
link
Jump to Header 1
image:

Caption

- unordered list
  - sub-item 1
  - sub-item 2
    - sub-sub-item 1
- item 2

    Continued (indent 4 spaces)

1. ordered list
2. item 2
   i. sub-item 1
      A. sub-sub-item 1

1. A list whose numbering

continues after

2. an interruption

Term 1

Definition 1

Right	Left	Default	Center
12	12	12	12
123	123	123	123
1	1	1	1

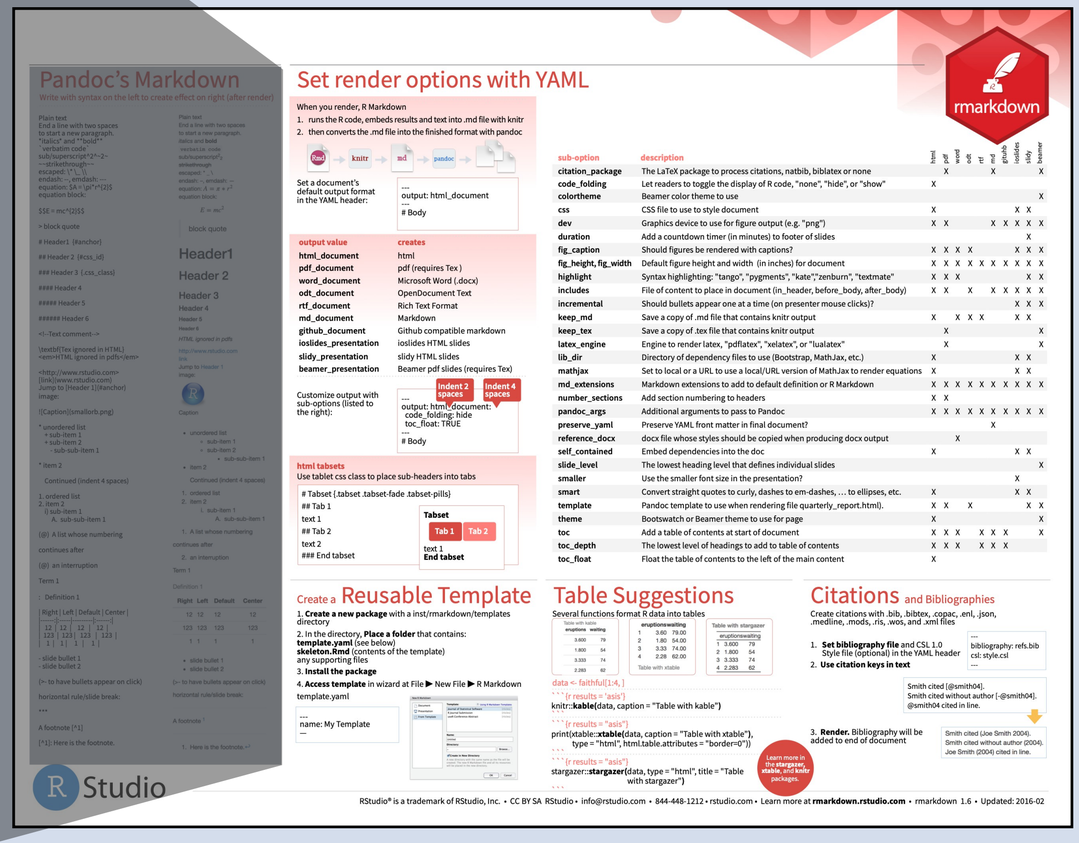
- slide bullet 1
- slide bullet 2

(>- to have bullets appear on click)

horizontal rule/slide break:

A footnote [1]

1. Here is the footnote. ↩



Pandoc's Markdown

Write with syntax on the left to create effect on right (after render)

Set render options with YAML

When you render, R Markdown
1. runs the R code, embeds results and text into .md file with knitr
2. then converts the .md file into the finished format with pandoc

Rmd → knitr → md → pandoc →

Set a document's default output format in the YAML header:

```
---
output: html_document
---
# Body
```

output value	creates
html_document	html
pdf_document	pdf (requires Tex)
word_document	Microsoft Word (.docx)
odt_document	OpenDocument Text
rtf_document	Rich Text Format
md_document	Markdown
github_document	Github compatible markdown
ioslides_presentation	ioslides HTML slides
slidy_presentation	slidy HTML slides
beamer_presentation	Beamer pdf slides (requires Tex)

Customize output with sub-options (listed to the right):

```
---
output: html_document:
  code_folding: hide
  toc_float: TRUE
---
# Body
```

Indent 2 spaces · Indent 4 spaces

html tabsets
Use tablet css class to place sub-headers into tabs

```
# Tabset {.tabset .tabset-fade .tabset-pills}
## Tab 1
text 1
## Tab 2
text 2
### End tabset
```

Tabset
Tab 1 | Tab 2
text 1
End tabset

rmarkdown

sub-option	description	html	pdf	word	odt	rtf	md	github	ioslides	slidy	beamer
citation_package	The LaTeX package to process citations, natbib, biblatex or none		X				X				X
code_folding	Let readers to toggle the display of R code, "none", "hide", or "show"	X									
colortheme	Beamer color theme to use										X
css	CSS file to use to style document	X								X	X
dev	Graphics device to use for figure output (e.g. "png")	X	X				X	X	X	X	X
duration	Add a countdown timer (in minutes) to footer of slides									X	
fig_caption	Should figures be rendered with captions?	X	X	X	X					X	X
fig_height, fig_width	Default figure height and width (in inches) for document	X	X	X	X	X	X	X	X	X	X
highlight	Syntax highlighting: "tango", "pygments", "kate","zenburn", "textmate"	X	X	X						X	X
includes	File of content to place in document (in_header, before_body, after_body)	X	X		X		X	X	X	X	X
incremental	Should bullets appear one at a time (on presenter mouse clicks)?									X	X
keep_md	Save a copy of .md file that contains knitr output	X		X	X					X	X
keep_tex	Save a copy of .tex file that contains knitr output		X								X
latex_engine	Engine to render latex, "pdflatex", "xelatex", or "lualatex"		X								X
lib_dir	Directory of dependency files to use (Bootstrap, MathJax, etc.)	X								X	X
mathjax	Set to local or a URL to use a local/URL version of MathJax to render equations	X								X	X
md_extensions	Markdown extensions to add to default definition or R Markdown	X	X	X	X	X	X	X	X	X	X
number_sections	Add section numbering to headers	X	X								
pandoc_args	Additional arguments to pass to Pandoc	X	X	X	X	X	X	X	X	X	X
preserve_yaml	Preserve YAML front matter in final document?						X				
reference_docx	docx file whose styles should be copied when producing docx output			X							
self_contained	Embed dependencies into the doc	X							X	X	
slide_level	The lowest heading level that defines individual slides										X
smaller	Use the smaller font size in the presentation?								X		
smart	Convert straight quotes to curly, dashes to em-dashes, … to ellipses, etc.	X							X	X	
template	Pandoc template to use when rendering file quarterly_report.html).	X	X		X					X	X
theme	Bootswatch or Beamer theme to use for page	X									X
toc	Add a table of contents at start of document	X	X	X		X	X	X			X
toc_depth	The lowest level of headings to add to table of contents	X	X	X		X	X	X			
toc_float	Float the table of contents to the left of the main content	X									

Create a Reusable Template

1. **Create a new package** with a inst/rmarkdown/templates directory
2. In the directory, **Place a folder** that contains:
   template.yaml (see below)
   skeleton.Rmd (contents of the template)
   any supporting files
3. **Install the package**
4. **Access template** in wizard at File ▶ New File ▶ R Markdown

template.yaml

```
---
name: My Template
---
```

Table Suggestions

Several functions format R data into tables

Table with kable

	eruptions	waiting
1	3.600	79
2	1.800	54
3	3.333	74
4	2.283	62

Table with xtable

	eruptions	waiting
1	3.60	79.00
2	1.80	54.00
3	3.33	74.00
4	2.28	62.00

Table with stargazer

	eruptions	waiting
1	3.600	79
2	1.800	54
3	3.333	74
4	2.283	62

```
data <- faithful[1:4, ]
```
```
{r results = 'asis'}
knitr::kable(data, caption = "Table with kable")
```
```
{r results = "asis"}
print(xtable::xtable(data, caption = "Table with xtable"),
      type = "html", html.table.attributes = "border=0"))
```
```
{r results = "asis"}
stargazer::stargazer(data, type = "html", title = "Table with stargazer")
```

Learn more in the **stargazer, xtable,** and **knitr** packages.

Citations and Bibliographies

Create citations with .bib, .bibtex, .copac, .enl, .json, .medline, .mods, .ris, .wos, and .xml files

1. **Set bibliography file** and CSL 1.0 Style file (optional) in the YAML header
2. **Use citation keys in text**

```
---
bibliography: refs.bib
csl: style.csl
---
```

Smith cited [@smith04].
Smith cited without author [-@smith04].
@smith04 cited in line.

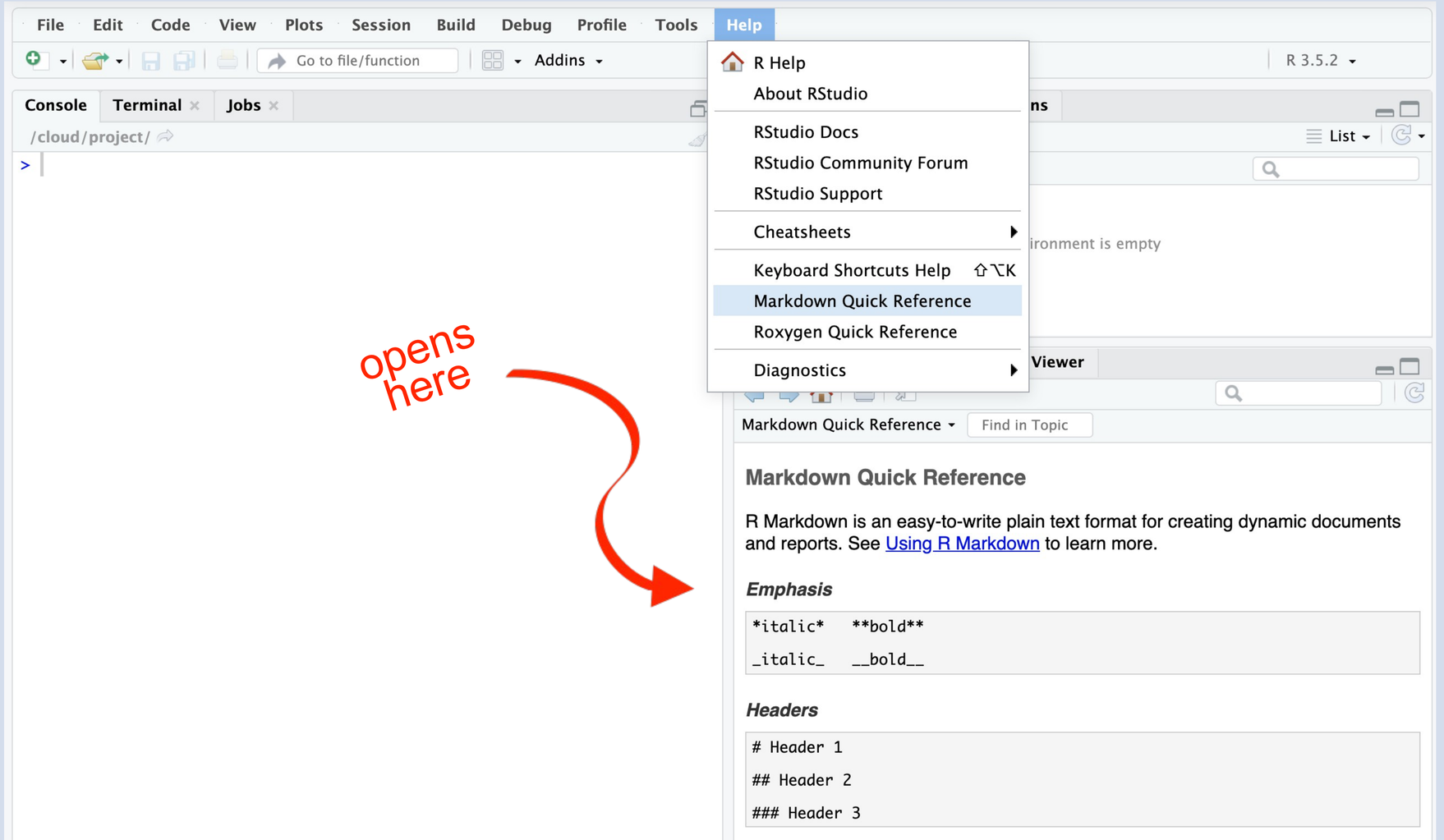
Smith cited (Joe Smith 2004).
Smith cited without author (2004).
Joe Smith (2004) cited in line.

3. **Render.** Bibliography will be added to end of document

RStudio® is a trademark of RStudio, Inc. • CC BY SA RStudio • info@rstudio.com • 844-448-1212 • rstudio.com • Learn more at **rmarkdown.rstudio.com** • rmarkdown 1.6 • Updated: 2016-02

IDE Reference

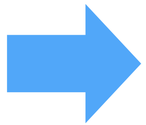
Go to **Help > Markdown Quick Reference**



Inline code

Place code in a sentence with **`r <code>`**. R Markdown will replace the code with its results.

```
Today is
`r Sys.Date()`.
```



Today is 2015-04-16.

CC BY-SA RStudio

Inline code

Code whose results are inserted into the text.

```
Today is `r Sys.Date()`.
```

Surround with `r  `

Code to run. Only the result will be included.

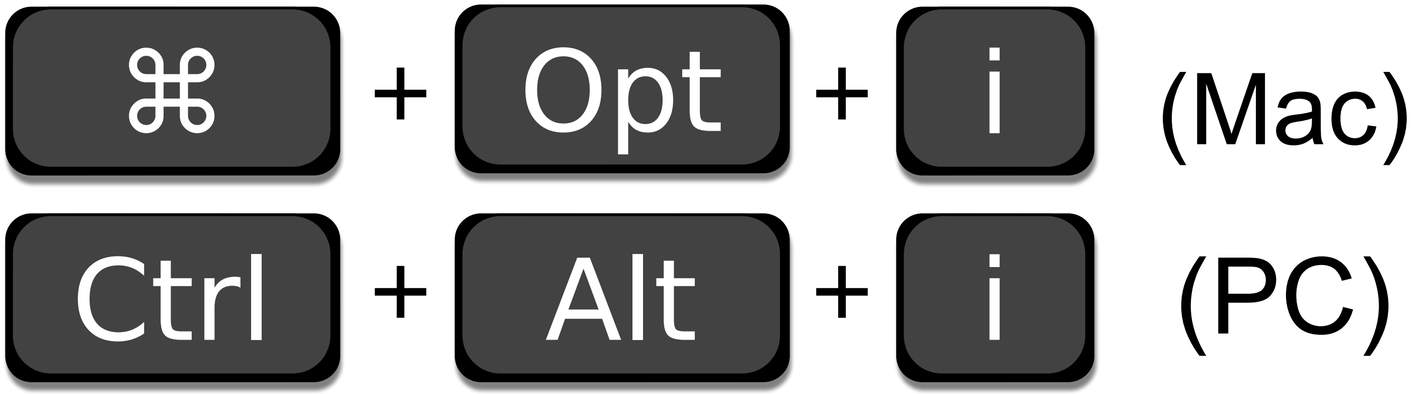
CC BY-SA RStudio

The sample includes `r nrow(flfp)` countries. The average FLFP rate is `r round(mean(flfp$wvs_flfp), 2)`.

Code chunks

Insert a chunk of R code with

```
```{r}
# some code
```
```



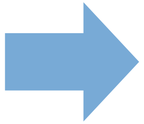
CC BY-SA RStudio

chunk options

By default, R Markdown includes both the code and its results

````
Here's some code

```{r}
dim(iris)
```
````



Here's some code

```
dim(iris)
```

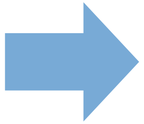
```
## [1] 150   5
```

CC BY-SA RStudio

echo

Add options in the brackets after r.
echo = FALSE hides the code.

````
Here's some code
```{r echo=FALSE}
dim(iris)
```
````



Here's some code

```
## [1] 150   5
```

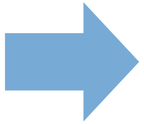
CC BY-SA RStudio

eval

eval = FALSE prevents the code from being run. As a result, no results will be displayed with the code.

````
Here's some code

```{r eval=FALSE}
dim(iris)
```
````



Here's some code

```
dim(iris)
```

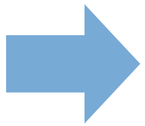
CC BY-SA RStudio

include

include = FALSE runs the code, but prevents both the code and the results from appearing (e.g. to setup).

````
Here's some code

```{r include=FALSE}
dim(iris)
```
````



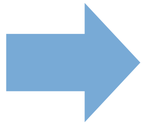
Here's some code

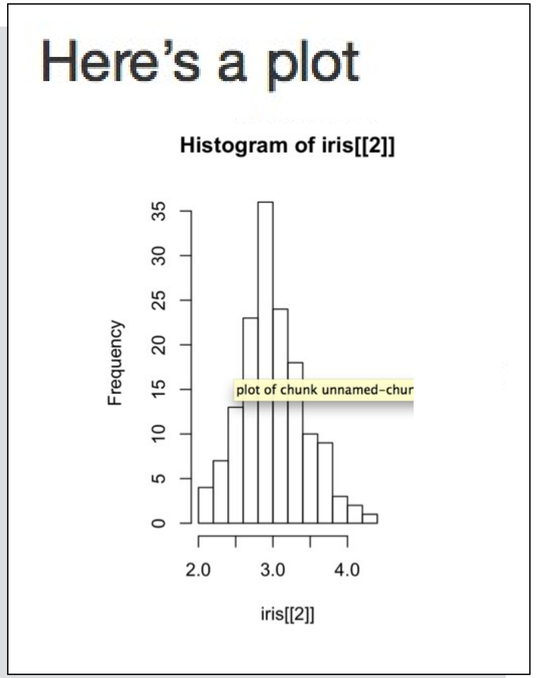
CC BY-SA RStudio

fig.height, fig.width

Specify the dimension of plots (in inches) with fig.width and fig.height. Separate multiple arguments with commas.

````
Here's a plot
```{r echo=FALSE, fig.width=3, fig.height=5}
hist(iris[[2]])
```
````





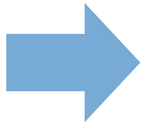
CC BY-SA RStudio

warning

Suppresses warnings. Works similarly with messages.

````
Here's some code

```{r warning=FALSE}
warning("Something troublesome I do not want you to see")
```
````



Here's some code

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global chunk options

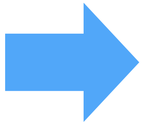
Hides the code in every chunk unless over-rewritten within {r}.

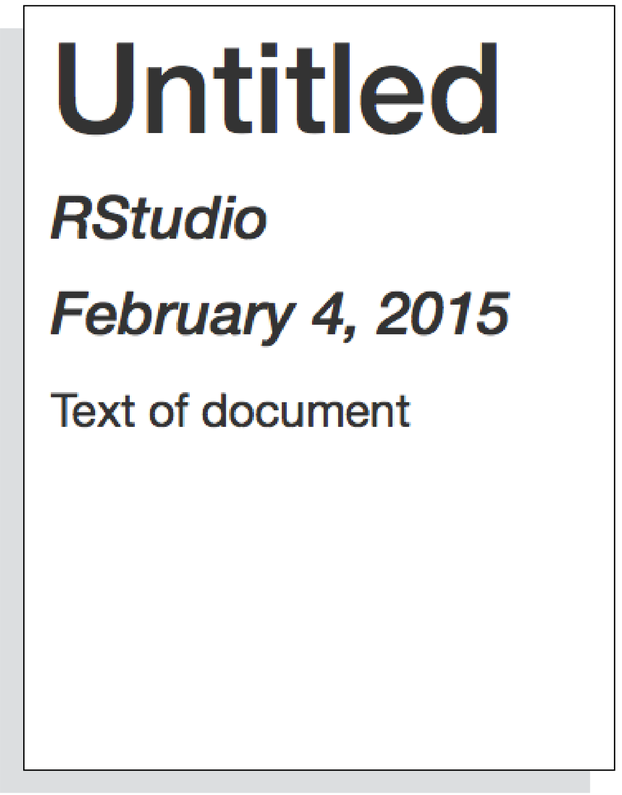
````
```{r setup}
knitr::opts_chunk$set(echo = FALSE)
```
````

Header

A section of *key:value* pairs
separated by dashed lines ———

```
---
title: "Untitled"
author: "RStudio"
date: "February 4, 2015"
output: html_document
---
Text of document
```

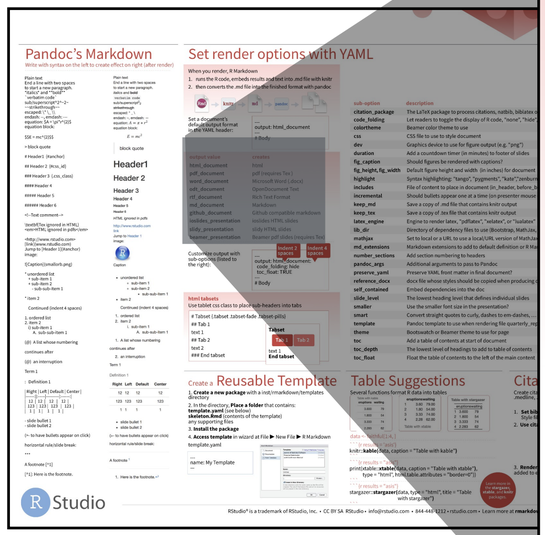




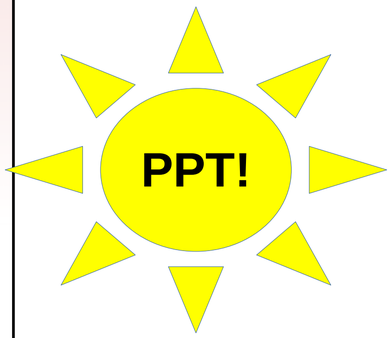
CC BY-SA RStudio

output

The output: field sets the format of the final report



output value	creates
html_document	html
pdf_document	pdf (requires Tex)
word_document	Microsoft Word (.docx)
odt_document	OpenDocument Text
rtf_document	Rich Text Format
md_document	Markdown
github_document	Github compatible markdown
ioslides_presentation	ioslides HTML slides
slidy_presentation	slidy HTML slides
beamer_presentation	Beamer pdf slides (requires Tex)



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Be sure to..

- If you have not done so you, look at the utility of rmarkdown by using rmarkdown to generate a powerpoint presentation about… rmarkdown.
- You can do that by opening up and pushing ”knit” above in Rstudio on the the script called ”rmarkdown introduction.R”